SMOOTHING CALABI–YAU THREEFOLDS WITH NODES AND DEL PEZZO CONE POINTS

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Abstract. We give a necessary and sufficient smoothing criterion for projective Calabi–Yau threefolds with nodes and exact anticanonical cones over smooth del Pezzo surfaces of degrees five, six and seven, and over $\mathbb{P}^1 \times \mathbb{P}^1$. Choose a local deformation component at each cone point. The criterion is a relation among the corresponding local homology classes in a crepant resolution: every one-dimensional local summand must have nonzero coefficient; on a two-dimensional degree-six component, three specified linear functionals must be nonzero. At each degree-five point, all five pairings with the conic-pencil classes must be nonzero. Their kernels form the local discriminant in a smooth four-dimensional base. If the relation space has nonzero projection at every degree-six point, the deformation base with those local components selected is smooth, and every tangent integrates convergently. The proof uses crepant partial resolutions, local Hodge comparisons, generalized T^1 lifting, and analytic blow-down. Simultaneous partial resolutions and boundary periods prove necessity along arbitrary smoothing arcs.

A threefold with thirty nodes and two degree-six cone points satisfies the weaker nonzero-vector condition for two choices of local components but has no smoothing. Its full deformation ring is $\mathbb{C}\{z_1, \dots, z_{32}, x, y, u, v, b\}/(xy, xb, uv, ub)$, with four smooth components. A toric hypersurface with fourteen nodes and one degree-seven cone point is smoothable and has a smooth thirty-dimensional deformation base, although every primitive single-degree ambient construction considered here fails to smooth it. An explicit ambient smoothing completes a classification of 77 reflexive four-polytopes with at most nine vertices and a pentagonal face, restricted to hypersurfaces with isolated nodes and degree-six or degree-seven cone points and one singular point over each singular two-face: one is smoothable and 76 are not.

1. Introduction

A projective Calabi–Yau threefold can have only locally smoothable singularities and nevertheless admit no smoothing. For ordinary double points, the obstruction is expressed by the exceptional curves of a small resolution: a smoothing exists precisely when their classes satisfy a relation

$$\sum_j \lambda_j [C_j] = 0, \qquad \lambda_j \neq 0 \quad \text{for every } j. \tag{1}$$

This is Friedman's criterion, together with the unobstructedness and necessity results discussed in [Fri86, Nam94, RT09]; [RT09, Thm. 1.3] states the smoothing criterion itself. We prove an analogous criterion when nodes occur together with anticanonical cones over smooth del Pezzo surfaces of degrees five, six and seven, and over $\mathbb{P}^1 \times \mathbb{P}^1$, of degree eight. The degree-six cone has a two-dimensional smoothing component, and a nonzero vector on that component need not be a smoothing direction. Three separate nonvanishing conditions are

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necessary. An explicit nonsmoothable threefold shows that they cannot be replaced by a single nonzero-vector condition, even if the local components may be chosen anew.

1.1. **The smoothing criterion.** Let X be a connected normal projective complex threefold with $\omega_X \cong \mathcal{O}_X$ and $H^1(X, \mathcal{O}_X) = 0$. Suppose every singularity is an ordinary double point or is analytically isomorphic to the anticanonical cone over a smooth del Pezzo surface of degree five, six or seven, or over $\mathbb{P}^1 \times \mathbb{P}^1$. The assumption concerns these exact analytic germs. Let $Y_0 \to X$ be the smooth crepant analytic resolution obtained from chosen small resolutions of the nodes and the standard divisorial resolutions of the cones. It is compact and Moishezon.

A *branch profile* S chooses a reduced local deformation component at each singular point. At a node or a $\mathbb{P}^1 \times \mathbb{P}^1$ cone the base is a line; at a degree-five point it is a smooth four-dimensional germ; at a degree-seven point there is one reduced line; at a degree-six point there are a line and a plane, denoted A_1 and A_2. These names agree with the root subspaces in the del Pezzo lattice. For the link L_p of a cone point, the circle-bundle projection identifies

$$H_2(L_p; \mathbb{Q}) \cong K_{E_p}^{\perp} \subset H_2(E_p; \mathbb{Q}).$$

Write R_{p,S_p} for the nodal line or the selected root subspace, and set

$$K_S = \ker \left\{ \bigoplus_p R_{p,S_p} \longrightarrow H_2(Y_0; \mathbb{Q}) \right\}. \tag{2}$$

At a $\mathbb{P}^1 \times \mathbb{P}^1$ cone the selected subspace is the entire link-homology line $\mathbb{Q}(f_1 - f_2)$, where f_1, f_2 are the ruling classes on the exceptional surface. At a degree-six point, mark $E_p = \mathrm{Bl}_3\mathbb{P}^2$ by the line class h and exceptional classes e_1, e_2, e_3. Its A_2 subspace is

$$R_{p,A_2} = \{\mu_1 e_1 + \mu_2 e_2 + \mu_3 e_3 : \mu_1 + \mu_2 + \mu_3 = 0\}.$$

At a degree-five point mark $E_p = \mathrm{Bl}_4 \mathbb{P}^2$ by $h, e_1, \ldots, e_4$. The selected subspace is the entire $K_{E_p}^{\perp}$, and its five conic-pencil classes are

$$f_i = h - e_i \ (1 \le i \le 4), \qquad f_5 = 2h - e_1 - e_2 - e_3 - e_4.$$

Put $\lambda_i(\kappa_p) = \kappa_p \cdot f_i$. Their sum is zero because $\sum_i f_i = -2K_{E_p}$.

Theorem A (smoothing criterion)**.** *The threefold X has a convergent one-parameter smoothing inducing S if and only if K_S contains a vector nonzero at every rank-one summand and satisfying $\mu_1\mu_2\mu_3 \neq 0$ at every A_2 summand and $\prod_{i=1}^5 \lambda_i \neq 0$ at every degree-five point.*

Equivalently, none of the stated scalar functionals vanishes identically on K_S. Thus the criterion is a finite linear-algebra test for each of the finitely many profiles. When neither degree-five nor degree-six points occur, its form is the same as the nodal criterion. At an A_2 point the three excluded lines are precisely the local discriminant. Necessity holds for arbitrary smoothing arcs, including arcs tangent to that discriminant to higher order. At degree five, the five excluded hyperplanes are likewise the exact local discriminant, and necessity holds for arbitrary arcs.

The integrability statement is stronger than the existence assertion. Fix analytic local classifying maps for a semiuniversal deformation of X, and let $\mathrm{Def}^S(X)$ be the scheme-theoretic inverse image of the chosen local branches. Denote the tangent sheaf by Θ_X.

Theorem B (smooth selected deformation spaces)**.** *If $K_S \to R_{p,S_p}$ is nonzero at every degree-six point, then $\mathrm{Def}^S(X)$ is smooth, of dimension*

$$h^1(X, \Theta_X) + \dim_{\mathbb{Q}} K_S.$$

Every tangent vector to this space is the tangent of a convergent one-parameter deformation on the selected branches.

No nonzero-projection hypothesis is required at nodes, degree-five or degree-seven points, or $\mathbb{P}^1 \times \mathbb{P}^1$ cones in this theorem. The hypothesis is automatic when the smoothing criterion holds. Smoothness of a selected base does not, by itself, assert the existence of a smooth fibre.

1.2. **Scope among del Pezzo cones.** The degree-seven and degree-six cases arise from the pentagonal and hexagonal faces in the toric examples. Both associated cones are smoothable and are not complete intersections. Their local deformation spaces exhibit the two phenomena treated here: distinct smoothing components in degree six, and a nonreduced structure transverse to the smoothing line in degree seven. Theorems A and B also allow the degree-five cone and the anticanonical cone over $\mathbb{P}^1 \times \mathbb{P}^1$, of degree eight. These hypotheses specify the local comparisons established below; smoothable del Pezzo cones also occur in the other degrees.

For a smooth del Pezzo surface E, the anticanonical cone means

$$\operatorname{Spec} \bigoplus_{m\geq 0} H^0(E, -mK_E).$$

This convention includes the weighted anticanonical models in degrees one and two. The classical local facts in Table 1 are recalled in [Gro97r, §§3.3, 5.5 and Example 2.19]; the branch structures in degrees six and seven follow from [Alt97]. Proposition 3.1 proves the full local-base and discriminant statements in degree five.

The degree-eight cone over $\mathbb{P}^1 \times \mathbb{P}^1$ already occurs as the partial resolution of a degree-seven point. Lemma 3.5 proves smoothness of its full local base and identifies its tangent line with the difference of the ruling classes. When such points occur on the original threefold, the partial modification is the identity near them. Their Milnor fibres retract to $\mathbb{RP}^3$; the descended nodal period supplies the necessity argument in Lemma 4.6. Thus they require one nonzero coefficient on the ruling difference, with no additional hypothesis in the selected-base smoothness theorem.

The degree-five surface is not toric. Totaro's Bott vanishing theorem [Tot20] supplies its local Hodge comparison. The four-parameter family of Grassmannian sections and its boundary periods identify the five conic-pencil conditions and prove necessity at every order. Degree-five points also remain unchanged on the partial model. Thus the present criterion covers all locally smoothable anticanonical cones over smooth del Pezzo surfaces that are not complete intersections. For degrees one to four, the existing complete-intersection theory does not by itself give the stated criterion for mixtures with the non-complete-intersection cones considered in this paper.

Table 1. Anticanonical cones over smooth del Pezzo surfaces and the scope of the present results. The local assertions alone do not imply smoothability of a compact threefold containing these germs.

Degree / surface	Local deformation behavior	Role in the present results
1–3	Hypersurfaces; smoothable, with smooth local bases.	Classical complete-intersection cases; outside the stated mixed criterion.
4	Intersection of two quadrics; smoothable, with smooth local base.	As in degrees one to three.
5	Smooth four-dimensional base; five discriminant hyperplanes.	Covered by Theorems A and B; five nonzero conic-pencil pairings.
6	A smoothing line and a smoothing plane; three discriminant lines in the plane.	Both components covered by Theorems A and B.
7	One reduced smoothing line, with an obstructed transverse tangent.	Covered by Theorems A and B.
8, $\mathbb{P}^1 \times \mathbb{P}^1$	Smooth one-dimensional local base; noncentral fibres smooth.	Covered by Theorems A and B; coefficient on the ruling difference.
8, $\mathbb{F}_1$	No smoothing; local base $\operatorname{Spec} \mathbb{C}[t]/(t^2)$.	Excluded by local nonsmoothability.
9, $\mathbb{P}^2$	Rigid.	Excluded by local nonsmoothability.

1.3. **A nonsmoothable example and its deformation space.** The distinction between a nonzero local vector and a smoothing direction cannot be removed by changing the branch profile.

Theorem C. *There is a projective Calabi–Yau threefold with thirty nodes and two degree-six cone points such that a relation in K_S is nonzero at every singular point for either mixed profile, but the threefold has no smoothing on any profile. Its semiuniversal analytic local ring is*

$$\mathbb{C}\{z_1, \dots, z_{32}, x, y, u, v, b\}/(xy, xb, uv, ub).$$

It is reduced, has tangent dimension 37*, and has four smooth irreducible components of dimensions* $34, 34, 34, 35$*. All their scheme-theoretic intersections are smooth.*

The example is a general anticanonical hypersurface in the toric fourfold associated with an explicit reflexive polytope with 25 vertices. Three divisor identities prove nonsmoothability independently of the sufficiency argument. They force an old nodal coefficient to vanish on each uniform profile and one of the three A_2 coefficients to vanish on each mixed profile. The full deformation ring then follows from Theorem B, the local branch equations, and a convergent change of coordinates. A prescribed tangent integrates precisely when the four displayed quadratic expressions vanish.

1.4. **The geometric argument.** The first-order comparison identifies the image of the global tangent space in the sum of the local tangent spaces with the kernel of the link-homology map. Contraction with one global volume form, local cohomology with supports, and the $\partial\bar{\partial}$-lemma give this comparison for the full local tangent spaces, including the transverse infinitesimal direction at a degree-seven cone.

To integrate the selected directions, we replace the cone points by proper crepant partial resolutions. The degree-seven cone is replaced by the anticanonical cone over $\mathbb{P}^1 \times \mathbb{P}^1$; the A_1 and A_2 choices at a degree-six cone are replaced by two and three nodes, respectively. Degree-five and original $\mathbb{P}^1 \times \mathbb{P}^1$ cones remain unchanged. Each resulting singularity has a smooth full local deformation base. Friedman's generalized T^1 lifting theorem [Fri26, Thm. 3.10] therefore proves unobstructedness of the partial model. An analytic blow-down lemma and a lift of locally trivial global tangents transfer this smoothness to the selected deformation space of X. The partial model need not be projective; compact Moishezon resolutions supply the required Hodge theory.

For necessity at an A_2 point, an explicit simultaneous partial resolution of the rank-one matrix family is an isomorphism over every noncentral fibre. Pulling it back along an arbitrary smoothing arc replaces the central cone by three nodes without changing the smooth fibres. Nodal necessity then supplies three nonzero coefficients; intersection with the exceptional del Pezzo surface makes their sum zero. This proves the stronger A_2 condition at every order. At a $\mathbb{P}^1 \times \mathbb{P}^1$ cone, the vanishing cycle is the quotient of the nodal sphere by the antipodal involution; its nonzero period gives the corresponding rank-one necessity condition. At degree five, the four-parameter Grassmannian section family has five discriminant hyperplanes. Its boundary carries every absolute Milnor cycle. The resulting holomorphic boundary periods identify the five conditions and prove necessity along arbitrary arcs.

1.5. **Toric constructions and their limits.** The general theorems also distinguish abstract deformations from deformations induced by a toric ambient space. For Batyrev hypersurfaces [Bat94], we study the ambient families obtained from an admissible Minkowski decomposition at one primitive lattice degree, using Altmann's local theory and the global construction of Ilten–Vollmert [Alt97, IV12]. Definition 7.3 specifies these families.

The hypersurface X_{19} has fourteen nodes and one degree-seven cone point. Corollary 9.3 proves that its full deformation space is smooth of dimension 30 and that it is smoothable. Nevertheless, Theorem 9.2 excludes a smoothing by every single-degree ambient family through the relative anticanonical system. Its proof combines Altmann's grading calculation with a propagation rule on the edges of the two facets containing the pentagonal face. Definition 8.11 calls a facet *locked* when this rule forces all its edge-dilation parameters to agree. The obstruction is uniform over the entire degree lattice.

The complementary construction is X_9, with two nodes and one degree-seven cone point. Theorem 11.8 gives an explicit ambient smoothing at degree $(0, 1, -1, -1)$. Theorem 10.3 classifies all 77 admissible polytopes with at most nine vertices and a pentagonal face: 76 give nonsmoothable hypersurfaces, and the remaining one is X_9. The larger edge-propagation census describes the frequency of the ambient obstruction, without asserting that its separate degree-coverage condition holds throughout that census.

Sections 2–4 prove the general theorems, and Section 5 gives the counterexample and its full deformation ring. Sections 6–11 develop the toric applications. Appendix B records the exact computations and the data from which they can be reproduced. The results leave open smoothness of selected bases whose degree-six projection vanishes, and the structure of unrestricted deformation bases in general, including possible degree-seven nilpotents.

2. Local branches and global first-order deformations

2.1. **The analytic setting and the local lattices.** Throughout Sections 2–4, X satisfies the projective Calabi–Yau hypotheses of the introduction. All its singularities are isolated nodes or the exact anticanonical cones over $E = \mathrm{dP}_5$, dP_6, dP_7, or $\mathbb{P}^1 \times \mathbb{P}^1$. The degree-five, degree-six and degree-seven surfaces are unique up to isomorphism over $\mathbb{C}$. The degree-five surface is not toric; the other three surfaces are toric. Fix a nowhere-vanishing section Ω of ω_X.

At a node choose a small resolution; at a cone use the zero-section contraction $\mathrm{Tot}(K_E) \to \mathrm{Cone}(E)$. These proper analytic maps are isomorphisms off the singular points and glue to a smooth crepant analytic resolution $f_0 : Y_0 \to X$. The gluing is Hausdorff because two points over the same singular point lie in one proper local model; points with distinct images can be separated on X. Properness is local on the target. Thus Y_0 is compact, and it is Moishezon because it is bimeromorphic to the projective variety X. The $\partial\bar{\partial}$-lemma holds on Y_0 [DGMS75, Thm. 5.22 and Cor. 5.23].

Let L_p be a small link of a singularity. At a cone point the circle-bundle Gysin sequence identifies

$$H_2(L_p; \mathbb{Q}) = K_{E_p}^{\perp} \subset H_2(E_p; \mathbb{Q}).$$

For $E = \mathrm{Bl}_r \mathbb{P}^2$, use the usual marking $h, e_1, \ldots, e_r$. Its intersection form and canonical class are

$$\mathrm{diag}(1, -1, \ldots, -1), \qquad K_E = -3h + \sum_i e_i.$$

At degree seven the selected line is $\mathbb{Q}(e_1 - e_2)$. For $E = \mathbb{P}^1 \times \mathbb{P}^1$, the ruling classes satisfy $f_1^2 = f_2^2 = 0$, $f_1 \cdot f_2 = 1$, and $K_E = -2f_1 - 2f_2$; the selected line is the entire space $K_E^{\perp} = \mathbb{Q}(f_1 - f_2)$. At degree six set

$$\ell = -h + e_1 + e_2 + e_3, \qquad A = -e_1 + e_3, \qquad B = e_2 - e_3. \tag{3}$$

Then $K_E^{\perp} = \mathbb{Q}\ell \oplus \langle A, B \rangle_{\mathbb{Q}}$, the orthogonal decomposition into the A_1 and A_2 root subspaces. In the plane,

$$aA + bB = -ae_1 + be_2 + (a - b)e_3. \tag{4}$$

At a node the chosen resolution identifies its link-homology line with the span of the exceptional curve. Changing the small resolution changes its generator by a sign and does not change any nonvanishing test.

At degree five the selected subspace is all of $K_E^{\perp}$. For $E = \mathrm{Bl}_4 \mathbb{P}^2$ its five conic-pencil classes are

$$f_i = h - e_i \quad (1 \leq i \leq 4), \qquad f_5 = 2h - e_1 - e_2 - e_3 - e_4, \qquad \sum_{i=1}^{5} f_i = -2K_E. \tag{5}$$

For $\gamma = xh + \sum_i y_i e_i \in K_E^{\perp}$, define

$$\lambda_i(\gamma) = \gamma \cdot f_i = x + y_i \quad (i \leq 4), \qquad \lambda_5(\gamma) = -x, \qquad \sum_i \lambda_i = 0. \tag{6}$$

These coordinates identify the integral perpendicular lattice with the negative A_4 lattice: the inverse is $x = -\lambda_5$, $y_i = \lambda_i + \lambda_5$, and $\gamma^2 = -\sum_i \lambda_i^2$.

Definition 2.1. A branch profile S selects the entire line at each node and each $\mathbb{P}^1 \times \mathbb{P}^1$ cone, the entire four-dimensional base at each degree-five point, the unique reduced degree-seven branch, and one of the two reduced branches at every degree-six point. Write R_{p,S_p} for the corresponding subspace of $H_2(L_p;\mathbb{Q})$ described above, and K_S for (2). We also write

$$K = \ker\left\{\bigoplus_p H_2(L_p;\mathbb{Q}) \longrightarrow H_2(Y_0;\mathbb{Q})\right\}. \tag{7}$$

These subspaces are also $R_{p,S_p} = \ker\{H_2(L_p;\mathbb{Q}) \to H_2(\mathcal{M}_{p,S_p};\mathbb{Q})\}$, where $\mathcal{M}_{p,S_p}$ is a smooth Milnor fibre on the chosen component [Wue26b, Prop. 2.1 and Lem. 4.2]. For the $\mathbb{P}^1\times\mathbb{P}^1$ and degree-five cones the same assertion follows from $H_2(\mathcal{M}_{p,S_p};\mathbb{Q}) = 0$, proved in Lemmas 3.6 and 3.3. When a profile is fixed, we occasionally abbreviate R_{p,S_p} to $\mathcal{R}_p$.

The scheme-theoretic local bases are, in analytic coordinates,

(8)

germ	local base ring
node	$\mathbb{C}\{t\}$
$\mathrm{Cone}(\mathbb{P}^1\times\mathbb{P}^1)$	$\mathbb{C}\{t\}$
$\mathrm{Cone}(\mathrm{dP}_5)$	$\mathbb{C}\{b_1,b_2,b_3,b_4\}$
$\mathrm{Cone}(\mathrm{dP}_7)$	$\mathbb{C}\{s,w\}/(w^2,sw)$
$\mathrm{Cone}(\mathrm{dP}_6)$	$\mathbb{C}\{s,u,v\}/(su,sv)$.

The degree-five entry is proved in Proposition 3.1; the other cone entries are Altmann's calculations. At degree six the A_1 branch is $u=v=0$ and the A_2 branch is $s=0$ [Alt97, §§5–8]. Fix analytic classifying maps from a semiuniversal base $\mathrm{Def}(X)$ to these local bases. Their scheme-theoretic inverse image of the selected components is denoted $\mathrm{Def}^S(X)$; the chosen maps form part of this notation. Write $V_S = T_0\,\mathrm{Def}^S(X)$. The line and plane are smooth, but the plane's smoothing locus excludes the three lines $ab(a-b)=0$ in the marking (4). In particular, the condition of being a nonzero tangent in the plane is weaker than being a first-order smoothing direction. At degree five the smoothing locus excludes the five hyperplanes $\lambda_i = 0$, as proved in Lemma 3.2.

2.2. **The full local Hodge comparison.** Write $\mathbb{T}^i_X = \mathrm{Ext}^i(L_X,\mathcal{O}_X)$, where L_X is the cotangent complex. The following comparison permits the use of Kähler differentials in the low-degree local-to-global sequence.

Lemma 2.2. *Let X be a normal Gorenstein threefold whose non-lci locus is finite. Then*

$$\mathrm{Ext}^i(\Omega^1_X,\mathcal{O}_X) \;\cong\; \mathrm{Ext}^i(L_X,\mathcal{O}_X) \qquad (i\le 2),$$

both as sheaves and globally, compatibly with the local-to-global spectral sequence.

Proof. The cohomology sheaves $\mathcal{H}^{-q}(L_X)$ for $q\ge 1$ vanish where X is a local complete intersection, so they are supported on a finite set and have finite length. A finite-length module on a Cohen–Macaulay threefold has grade 3, so $\mathcal{E}xt^p(-,\mathcal{O}_X)$ and $\mathrm{Ext}^p(-,\mathcal{O}_X)$ vanish on it for $p<3$. In the spectral sequence $E_2^{p,q} = \mathrm{Ext}^p(\mathcal{H}^{-q}(L_X),\mathcal{O}_X) \Rightarrow \mathrm{Ext}^{p+q}(L_X,\mathcal{O}_X)$ a term with $q\ge 1$ contributing in total degree ≤ 2 has $p\le 1<3$, hence vanishes. Only $q=0$ survives in degrees ≤ 2, and $\mathcal{H}^0(L_X) = \Omega^1_X$. □

The local-to-global sequence begins

$$0 \longrightarrow H^1(X,\Theta_X) \longrightarrow \mathbb{T}^1_X \xrightarrow{\alpha} \bigoplus_p T^1_{X,p} \xrightarrow{\mathrm{ob}} H^2(X,\Theta_X).$$

We use contraction with the restriction of the fixed global volume form Ω in every local comparison below. Thus numerical local parameters are Hodge-normalized; independently chosen equation coordinates may differ by nonzero scalar factors.

Lemma 2.3 (local Hodge comparison). *Let (Y,p) be a three-dimensional ordinary double point or the anticanonical cone over $E = \mathrm{dP}_5$, dP_6, dP_7, or $\mathbb{P}^1\times\mathbb{P}^1$. Let $\pi_p\colon \widehat{Y}_p \to Y$ be the small resolution at a node and $\widehat{Y}_p = \mathrm{Tot}(K_E)$ at a cone point, let Z_p be its exceptional locus, and put $U_p = \widehat{Y}_p \setminus Z_p$. Then the connecting homomorphism gives an isomorphism*

$$T^1_{Y,p} \xrightarrow{\ \sim\ } K_p^{\mathrm{cr}} := \ker\Big\{ H^2_{Z_p}(\widehat{Y}_p;\Omega^2_{\widehat{Y}_p}) \longrightarrow H^2(\widehat{Y}_p;\Omega^2_{\widehat{Y}_p})\Big\}. \tag{9}$$

There is also a natural isomorphism

$$H_2(L_p;\mathbb{C}) \cong H^3(L_p;\mathbb{C}) \xrightarrow{\ \sim\ } K_p^{\mathrm{cr}}, \tag{10}$$

and the two isomorphisms are compatible with the local Gysin map to any global crepant resolution containing $\widehat{Y}_p$.

Proof. The identifications $T^1_{Y,p} \cong H^1(U_p;\Theta_{U_p}) \cong H^1(U_p;\Omega^2_{U_p})$ are Schlessinger's comparison and contraction with the prescribed crepant volume form. The comparison applies to the non-lci cones: the hypothesis in [FL25, Lem. 1.17 and Lem. 4.1] is weakened in [FL25, Rem. 4.2] to an isolated singularity of depth at least 3.

For a node, $\widehat{Y}_p = \mathrm{Tot}_{\mathbb{P}^1}(\mathcal{O}(-1)\oplus\mathcal{O}(-1))$. The projection is affine, and the tangent sequence together with $H^1(\mathbb{P}^1,\mathcal{O}(m)) = 0$ for $m \geq -1$ gives $H^1(\widehat{Y}_p;\Theta_{\widehat{Y}_p}) = 0$. The local cohomology sequence therefore identifies the one-dimensional space $T^1_{Y,p}$ with K_p^{cr}. The residue calculation for the node identifies this class with the generator of $H^3(L_p)$; equivalently, this is the local Gysin identification in Friedman's construction [Fri86].

Now suppose $\widehat{Y}_p = \mathrm{Tot}(K_E)$, with projection $\rho\colon \widehat{Y}_p \to E$ and zero section again denoted E. Contraction with the crepant volume form identifies

$$\Omega^2_{\widehat{Y}_p}(\log E)(-E) \cong \Theta_{\widehat{Y}_p}(-\log E),$$

and the logarithmic tangent sequence is

$$0 \longrightarrow \mathcal{O}_{\widehat{Y}_p} \longrightarrow \Theta_{\widehat{Y}_p}(-\log E) \longrightarrow \rho^*\Theta_E \longrightarrow 0.$$

Since ρ is affine and $\rho_*\mathcal{O}_{\widehat{Y}_p} = \bigoplus_{k\geq 0}\mathcal{O}_E(-kK_E)$, Kodaira vanishing and Bott vanishing give

$$H^1(\widehat{Y}_p;\mathcal{O}_{\widehat{Y}_p}) = \bigoplus_{k\geq 0} H^1(E;\mathcal{O}_E(-kK_E)) = 0,$$
$$H^1(\widehat{Y}_p;\rho^*\Theta_E) = \bigoplus_{k\geq 0} H^1\big(E;\Omega^1_E\otimes\mathcal{O}_E(-(k+1)K_E)\big) = 0.$$

For the degree-five surface, the Bott vanishing used here is Totaro's theorem [Tot20, Theorem 2.1]; the twist $-(k+1)K_E$ is ample for every $k \geq 0$. For the other surfaces it is the usual toric Bott vanishing. Thus $H^1(\widehat{Y}_p;\Omega^2(\log E)(-E)) = 0$. The exact sequence of [FL25, Thm. 2.1(v)], whose theorem explicitly allows every three-dimensional isolated canonical Gorenstein germ, now gives (9). Moreover [FL25, Thm. 2.1(vi)] gives a natural inclusion

$$H^3(L_p;\mathbb{C}) = \mathrm{Gr}^2_F H^3(L_p;\mathbb{C}) \hookrightarrow K_p^{\mathrm{cr}}$$

and a decomposition of K_p^{cr} with this as one summand. If E is toric and its boundary polygon has n vertices, Altmann's calculation gives $\dim T^1_{Y,p} = n-3$. The circle-bundle Gysin sequence gives $\dim H^3(L_p) = 9 - \deg E = n-3$. Hence the other summand is zero and the displayed inclusion is an isomorphism. At degree five the same dimension comparison uses $\dim T^1 = 4$ [Gro97r, §5.5] and $\dim H^3(L_p) = 9-5 = 4$. Its construction in local cohomology is compatible with the Gysin map, which proves the final assertion. □

Lemma 2.4 (the local branches)**.** *Under the isomorphism $T^1_{Y,p} \cong H_2(L_p;\mathbb{C})$ of Lemma 2.3, the tangent space to each reduced local component is exactly the corresponding subspace $\mathcal{R}_p$: the whole line at a node or a $\mathbb{P}^1 \times \mathbb{P}^1$ cone, the entire $K_E^\perp$ at a degree-five cone, the unique root line at a dP_7 cone, and respectively the A_1 line and the A_2 plane at a dP_6 cone.*

Proof. At $E = \mathbb{P}^1 \times \mathbb{P}^1$ both spaces are one-dimensional, and $K_E^\perp = \mathbb{C}(f_1 - f_2)$ gives the assertion. At degree five the assertion follows from the full four-dimensional comparison and the smooth irreducible base of Proposition 3.1. For the degree-six and degree-seven points, consider a unit-edge polygon $P = (v_0, \ldots, v_{n-1})$, put $e_i = v_{i+1} - v_i$. Altmann identifies the degree of the versal tangent space used here with

$$\mathcal{M}(P) = \left\{ (t_i) \in \mathbb{C}^n : \sum_i t_i e_i = 0 \right\} / \mathbb{C}(1, \ldots, 1), \tag{11}$$

and a lattice Minkowski decomposition gives the class of its edge-dilation vector (t_i). The Hodge isomorphism of Lemma 2.3 is natural under lattice automorphisms of P, with one necessary character: contraction with the crepant volume form twists equivariance by the determinant of the induced automorphism of the three-dimensional cone lattice. This equivariance is computed with the standard toric volume form. The restriction of Ω is g times that form, for an analytic unit g. Altmann's concentration of T^1 in the single Gorenstein degree implies $\mathfrak{m}_p T^1_{X,p} = 0$. Multiplication by g therefore acts on $T^1_{X,p}$ through the scalar $g(p)$. The comparison determined by Ω differs only by this nonzero scalar, so the invariant subspaces identified below are unchanged.

There is one unit-edge reflexive pentagon up to $\mathrm{GL}_2(\mathbb{Z})$. Take

$$P_7 = \big((-1,-1), (0,-1), (1,0), (0,1), (-1,0)\big).$$

Its nontrivial decomposition has dilation vector $t = (0,1,0,1,0)$, up to replacing t by $1-t$, and spans the reduced versal base. The reflection $s(x,y) = (y,x)$ fixes $[t] \in \mathcal{M}(P_7)$ and has determinant -1. On $H_2(L_p) = K_E^\perp$ it acts by -1 on the unique root line and by $+1$ on the complementary line. The determinant twist therefore forces the Hodge image of $[t]$ to be the root line.

Likewise there is one unit-edge reflexive hexagon. Take

$$P_6 = \big((1,0), (1,1), (0,1), (-1,0), (-1,-1), (0,-1)\big).$$

The determinant-one rotation $r(x,y) = (x-y, x)$ cyclically permutes its vertices. On $\mathcal{M}(P_6)$ its characteristic polynomial is $(\lambda+1)(\lambda^2+\lambda+1)$. The (-1)-eigenspace is spanned by the alternating dilation $(0,1,0,1,0,1)$, the triangle-plus-triangle decomposition which gives Altmann's one-dimensional smoothing component; the complementary plane is spanned by the coarsenings of the three-segment decomposition and is the two-dimensional component. On $K_E^\perp \cong A_1 \oplus A_2$, the same rotation is -1 on A_1 and a Coxeter element, with characteristic polynomial $\lambda^2+\lambda+1$, on A_2. Equivariance now identifies the line with A_1 and the plane with A_2. The boundary classes and edge-dilation vectors above give finite intersection and representation calculations; Appendix B records their exact verification. □

2.3. **The image of the global tangent space.** The use of the full local spaces in Lemma 2.3 is essential: the comparison also accounts for tangent directions that do not belong to a reduced local component.

Theorem 2.5. *Under the Hodge identifications determined by* Ω*, localization gives exact sequences*

$$0 \longrightarrow H^1(X, \Theta_X) \longrightarrow \mathbb{T}^1_X \longrightarrow K \otimes_{\mathbb{Q}} \mathbb{C} \longrightarrow 0, \tag{12}$$

$$0 \longrightarrow H^1(X, \Theta_X) \longrightarrow V_S \longrightarrow K_S \otimes_{\mathbb{Q}} \mathbb{C} \longrightarrow 0. \tag{13}$$

Proof. Let $E = \operatorname{Exc}(f_0)$ and $U = Y_0 \setminus E = X_{\text{reg}}$. Excision and the local cohomology sequence give

$$H^1(U; \Omega^2_U) \longrightarrow \bigoplus_p H^2_{E_p}(\widehat{X}_p; \Omega^2_{\widehat{X}_p}) \longrightarrow H^2(Y_0; \Omega^2_{Y_0}). \tag{14}$$

Here $\widehat{X}_p$ denotes the restriction of Y_0 over a sufficiently small neighborhood of p. The algebraic cone calculations in Lemma 2.3 compute these analytic local groups as well: the vanishing statements are equivalent to the corresponding vanishing of completed higher direct images, by formal functions. The analytic local-support comparison itself is that of [FL25, Thm. 2.1].

Under Lemma 2.3, the map from $\bigoplus_p K^{\text{cr}}_p$ to $H^2(Y_0; \Omega^2_{Y_0})$, followed by the de Rham map, is the topological link map followed by Poincaré duality. In particular the square

$$\begin{array}{ccc} \bigoplus_p H_2(L_p; \mathbb{C}) & \longrightarrow & H_2(Y_0; \mathbb{C}) \\ \downarrow\cong & & \downarrow \text{PD} \\ \bigoplus_p K^{\text{cr}}_p & \longrightarrow & H^4(Y_0; \mathbb{C}) \end{array}$$

commutes. The $\partial\bar{\partial}$-lemma makes $H^2(Y_0; \Omega^2_{Y_0}) \to H^4(Y_0; \mathbb{C})$ injective. Hence an element of the sum of local tangent spaces has zero image in $H^2(Y_0; \Omega^2_{Y_0})$ if and only if its link class lies in $K \otimes_{\mathbb{Q}} \mathbb{C}$.

Contraction with Ω and Schlessinger's depth-three comparison identify

$$H^1(U; \Omega^2_U) \cong H^1(U; \Theta_U) \cong \mathbb{T}^1_X;$$

see [FL25, Rem. 4.2]. Exactness of (14) therefore gives every element of $K \otimes \mathbb{C}$ as the localization of a global tangent. Conversely, the localization of any global tangent has zero supported Gysin image, hence lies in that kernel. The kernel of localization is $H^1(X, \Theta_X)$ by the local-to-global sequence. This proves (12). Taking the inverse image of the chosen local tangent subspaces and applying Lemma 2.4 proves (13). ☐

At an A_2 point, (13) can supply a nonzero local vector on a discriminant line. It asserts the existence of a global first-order deformation with those local components; it makes no smoothing assertion for such a vector. The next two sections determine which directions integrate and which profiles contain smooth fibres.

3. Local partial resolutions and simultaneous families

We construct the local modifications used to compare deformation spaces. Write C_d for the anticanonical cone over dP_d, and write Q for the anticanonical cone over $\mathbb{P}^1 \times \mathbb{P}^1$. The latter is different from the ordinary double point: its polarization is $\mathcal{O}(2,2)$, rather than $\mathcal{O}(1,1)$. All morphisms below are proper over a sufficiently small representative of the cone germ. The constructions are algebraic, so they also define modifications of analytic germs.

Fix a nowhere vanishing volume form on the regular locus of the global threefold, and use its pullback on every crepant modification. All identifications of tangent classes with homology classes in this section use this single form and Lemma 2.3. In particular, coordinates introduced in polynomial models are initially algebraic coordinates. Their conversion into Hodge-normalized coordinates will be stated separately.

3.1. **The degree-five discriminant and boundary periods.** The degree-five cone has a smooth local deformation base, so it will remain unchanged on the global partial resolution. Its discriminant requires five separate nonvanishing conditions.

Proposition 3.1. *The full semiuniversal deformation base of C_5 is a smooth four-dimensional germ. It admits linear coordinates in which the reduced discriminant is the union of five distinct hyperplanes. Every noncentral singular fibre has only ordinary double points, one for each discriminant hyperplane containing its parameter.*

Proof. Let $V = \mathbb{C}^5$, $A = \bigwedge^2 V$, and

$$D = \{p \in A : p \wedge p = 0\} = \mathrm{Cone}(\mathrm{Gr}(2, V)).$$

Choose a general six-dimensional subspace $W \subset A$. The surface $E = \mathrm{Gr}(2, V) \cap \mathbb{P}(W)$ is smooth by Bertini. Adjunction gives $K_E = -H$, and its degree is five. All smooth degree-five del Pezzo surfaces over $\mathbb{C}$ are isomorphic, so its anticanonical cone is the required germ. Put $U = W^{\perp} \subset A^*$ and $B = A/W = U^*$. The quotient map

$$L : D \longrightarrow B \tag{15}$$

varies the four cutting linear forms by constants and has central fibre C_5.

The Grassmannian cone is Cohen–Macaulay of dimension seven. For example, its five quadratic Pfaffians have the graded resolution

$$0 \longrightarrow S(-5) \longrightarrow S(-3)^5 \longrightarrow S(-2)^5 \longrightarrow S \longrightarrow \mathcal{O}_D \longrightarrow 0, \qquad S = \mathrm{Sym}(A^*).$$

The four components of L cut a three-dimensional central fibre and hence form a regular sequence at the vertex. Miracle flatness over the smooth four-dimensional base proves flatness there. Homogeneity then proves flatness everywhere: scaling transports each point into this neighborhood and acts compatibly on B.

We check the full Kodaira–Spencer map. A constant vector $c \in A$ gives the perturbation $p \wedge c$ of the Pfaffian equations on W, up to the common factor two. If this abstract first-order class is zero, the cotangent presentation of the embedded germ shows that its homogeneous degree--1 part comes from a constant translation $w \in W$. The quadratic ideal has no linear elements, so

$$p \wedge (c - w) = 0 \quad \text{for every } p \in W.$$

Write $a = c - w$. If a has rank four, the map $a \wedge - : \bigwedge^2 V \to \bigwedge^4 V$ has rank five; its kernel cannot contain the six-dimensional space W. If a has rank two with support P, its kernel is $P \wedge V$, of dimension seven. The decomposable bivectors in $\mathbb{P}(P \wedge V)$ form the four-dimensional Schubert variety of two-planes meeting P: choose a line in P and then a second direction modulo that line. A hyperplane $\mathbb{P}(W)$ in this $\mathbb{P}^6$ would meet the Schubert variety in dimension at least three, contradicting $\dim E = 2$. Thus $a = 0$. The Kodaira–Spencer map $B \to T^1_{C_5}$ is injective and is an isomorphism because $\dim T^1_{C_5} = 4$ [Gro97r, §5.5]. This also shows that all of $T^1_{C_5}$ has degree -1.

Embed a semiuniversal base minimally with ring $\mathbb{C}\{z_1, \ldots, z_4\}/I$, $I \subset (z_1, \ldots, z_4)^2$. The classifying map of (15) pulls the z_i back to four functions with independent linear terms. They are analytic coordinates on B. Hence the map from the ambient convergent power-series ring is injective, forcing $I = 0$. The family is therefore semiuniversal for the full germ, including its scheme structure; local prorepresentability is not required.

For a nonzero decomposable p with support P_p, one has $T_pD = P_p \wedge V$. An alternating form $\alpha \in A^*$ annihilates this tangent space precisely when $P_p \subset \ker \alpha$. Such a nonzero form has rank two. Thus the dual Grassmannian is the rank-two locus in $\mathbb{P}(A^*)$. It meets $\mathbb{P}(U)$ transversely in five points $[\alpha_1], \dots, [\alpha_5]$. Indeed, for a rank-two $\alpha \in U$, failure of transversality is equivalent to

$$W \cap \bigwedge^2 \ker \alpha \neq 0.$$

Every nonzero bivector in this intersection is decomposable and would be a nonvertex singular point of C_5, contrary to the smoothness of E. The same argument excludes a positive-dimensional intersection. Its length is the degree five of the Grassmannian.

Set $K_i = \ker \alpha_i$ and $H_i = \ker(\alpha_i : B \to \mathbb{C})$. The critical locus of L away from the vertex is the union of the punctured three-dimensional vector spaces $\bigwedge^2 K_i$. The maps

$$L : \bigwedge^2 K_i \longrightarrow H_i$$

are isomorphisms: their kernels are the intersections just excluded, and both dimensions are three. Thus a point $b \in H_i \setminus 0$ has one associated critical point $p_i(b)$, linear in b. Two such points cannot coincide for $b \neq 0$. Otherwise every form in the pencil spanned by α_i, α_j would vanish on the same two-plane and hence have rank at most two, giving a line in the finite dual intersection. The complete reduced discriminant is consequently $\bigcup_i H_i$.

Finally, near a nonzero $p_i(b)$ choose a Grassmannian chart in which $\alpha_i = e_4^* \wedge e_5^*$ and the varying plane is spanned by

$$e_1 + x_3e_3 + x_4e_4 + x_5e_5, \qquad e_2 + y_3e_3 + y_4e_4 + y_5e_5.$$

With the nonzero cone coordinate r, the function α_i is $r(x_4y_5 - x_5y_4)$. Three other components of L restrict to coordinates on its critical locus, by the preceding isomorphism. Eliminate them. The remaining equation has nondegenerate quadratic part in x_4, x_5, y_4, y_5, so the holomorphic Morse lemma gives an ordinary double point with transverse smoothing parameter $\alpha_i(b)$, up to a unit. ∎

Lemma 3.2. *Under the local Hodge comparison and the conic-pencil coordinates* (6)*, the discriminant of Proposition 3.1 is*

$$\prod_{i=1}^{5} \lambda_i = 0, \qquad \sum_{i=1}^{5} \lambda_i = 0. \tag{16}$$

Proof. Permuting the four blown-up points gives the S_4 action on the first four conic pencils. The quadratic Cremona involution at the first three points lifts to an automorphism of E and interchanges the fourth and fifth pencils, fixing the first three. These automorphisms generate S_5. On $\mathrm{Tot}(K_E)$ their canonical lifts preserve the tautological two-form η and the volume form $d\eta$. The local Hodge comparison is therefore equivariant and realizes $T^1_{C_5}$ as the standard sum-zero representation on the five coordinates λ_i.

The tangent cone of the discriminant is intrinsic to the deformation functor. An automorphism of the central germ induces a classifying map on a semiuniversal base; its invertible derivative is the natural action on T^1. It preserves the discriminant because it preserves the isomorphism classes of the fibres. Hence S_5 permutes the five hyperplanes in Proposition 3.1. This permutation action is faithful. A nontrivial kernel would contain A_5 and make each normal line invariant under A_5. As A_5 has no nontrivial characters and no invariants in the standard four-dimensional representation, this is impossible.

The image of this faithful action has order 120, so it is the full symmetric group on the five hyperplanes. The stabilizer of one hyperplane has order 24. In its action on the five conic pencils it must fix a pencil: without a fixed point its orbits would have sizes $2+3$, giving order at most $2!3! = 12$, while a size-five orbit cannot divide 24. It is thus the S_4 stabilizing one pencil. The restriction of the standard representation to this subgroup is the sum of a trivial line and its three-dimensional standard representation. The unique invariant normal line is generated by the corresponding λ_i. This identifies all five hyperplanes.

For a global volume form $g\,d\eta$, multiplication by the analytic unit g acts on $T^1_{C_5}$ by $g(0)$: its degree--1 concentration gives $\mathfrak{m}_0 T^1_{C_5} = 0$. Thus the fixed global normalization changes the comparison by one nonzero scalar and preserves the five hyperplanes. ☐

Lemma 3.3. *For a smooth Milnor fibre M of C_5 with boundary link L,*

$$H_2(M;\mathbb{Q}) = 0, \qquad \dim H_3(M;\mathbb{Q}) = 4.$$

Both maps

$$H_3(L;\mathbb{Q}) \longrightarrow H_3(M;\mathbb{Q}), \qquad H_3(M,L;\mathbb{Q}) \longrightarrow H_2(L;\mathbb{Q}) \tag{17}$$

are isomorphisms.

Proof. For b outside the discriminant the affine fibre compactifies as

$$V_b = \mathrm{Gr}(2,V) \cap \mathbb{P}(L^{-1}(\mathbb{C}b)), \qquad X_b = V_b \setminus E.$$

This is a smooth three-hyperplane section of the Grassmannian. It is smooth at E because the four-hyperplane section E is smooth, and away from E by Proposition 3.1. Near the compact smooth divisor E, the squared affine norm is $F/|s|^2$, where s is a local equation for E and F is smooth and positive. Its normal derivative does not vanish sufficiently close to E. Thus every sufficiently large Euclidean truncation of X_b removes only a collar at infinity and has its homotopy type. Scaling by ε identifies $X_b \cap \{\|p\| \le R\}$ with $X_{\varepsilon b} \cap \{\|p\| \le \varepsilon R\}$. Fix the latter radius and take R large. This gives a small Milnor fibre; transport in the smooth local family gives the assertion for every nearby smooth parameter.

Lefschetz gives $b_1(V_b) = 0$ and $b_2(V_b) = 1$. For completeness the middle Betti number follows from the Chern calculation. Write $H = c_1(\mathcal{S}^*)$ and $P = c_2(\mathcal{S}^*)$ for the tautological bundle on the Grassmannian. Its tangent bundle is $\mathcal{S}^* \otimes \mathcal{Q}$, so

$$c_1 = 5H, \qquad c_2 = 11H^2 + P, \qquad c_3 = 15H^3.$$

Dividing by $(1+H)^3$ gives $c_3(TV_b) = 2H^3 - 3HP$. Pieri's rule gives $\int_{V_b} H^3 = 5$ and $\int_{V_b} HP = 2$; hence $\chi(V_b) = 4$ and $b_3(V_b) = 0$. The Gysin sequence for $E \subset V_b$ gives $H^2(X_b;\mathbb{Q}) = 0$ and

$$H^3(X_b;\mathbb{Q}) \cong \ker\{H^2(E;\mathbb{Q}) \longrightarrow H^4(V_b;\mathbb{Q})\},$$

of dimension four. The link circle-bundle sequence likewise gives $b_2(L) = b_3(L) = 4$. Lefschetz duality and the pair sequence make the second map in (17) a surjection between four-dimensional spaces. It is an isomorphism. Exactness then makes the first map surjective and hence an isomorphism as well. ☐

Lemma 3.4 (degree-five boundary periods)**.** *There are rational classes $\beta_1, \dots, \beta_5 \in H_3(L;\mathbb{Q})$ whose pairings with the local Hodge classes are nonzero multiples of the five functionals λ_i. For any convergent smoothing arc of C_5, equipped with a relative volume form generating its dualizing sheaf, the periods of the images of all five classes in the smooth fibre are nonzero for every sufficiently small noncentral parameter.*

Proof. Use the semiuniversal family (15). The boundary family of a small representative is a smooth proper submersion over the entire small base B, including 0; its homology local system is therefore trivial. By (17), every absolute Milnor cycle has a unique boundary class. Near a general point of the discriminant hyperplane H_i, choose the class β_i whose image is the nodal vanishing sphere, and transport it in the boundary family over B.

The Pfaffian resolution above shows that the Grassmannian cone has a homogeneous dualizing generator of weight five. Dividing by the four base differentials gives a nowhere vanishing relative generator Ω_b^0 of weight one. Integration on transported compact boundary cycles gives holomorphic functions on the whole base,

$$P_i(b) = \int_{\beta_{i,b}} \Omega_b^0.$$

Holomorphic dependence uses only the smooth locus near the boundary, so it holds also at parameters with a singular interior. Scaling changes the boundary radius; compare the two boundaries through their annular collars, whose central radial isotopy fixes the homology marking. The connected scaling action then preserves transported classes and gives $P_i(cb) = cP_i(b)$. Thus P_i is linear. Near a general point of H_i, it equals the nodal vanishing period and has a simple zero along H_i. Consequently

$$P_i(b) = c_i\alpha_i(b), \qquad c_i \neq 0. \tag{18}$$

We verify that its derivative is the claimed Hodge functional. Choose first-order product trivializations on the punctured central germ, with Kodaira–Spencer cocycle θ_{jk}. Differentiating the relative three-form gives

$$\dot{\Omega}_k - \dot{\Omega}_j = d(\iota_{\theta_{jk}}\Omega_0^0).$$

The Čech–de Rham representative for the derivative of a period therefore has overlap term $\iota_{\theta_{jk}}\Omega_0^0$, precisely the contraction cocycle defining Lemma 2.3. In the logarithmic de Rham model of the circle bundle, this is the natural comparison underlying [FL25, Theorem 2.1(vi)]. The link has $H^3(L;\mathbb{C})$ of pure Hodge type $(2,2)$, so its F^3 is zero and this comparison determines the entire class. Thus $dP_i|_0$ pairs β_i with the local Hodge class, up to the one common convention for Gysin maps and orientation. There is no ambiguity from the chosen extension of the volume form: a difference with the same overlap cocycle is a global form $a\Omega_0^0$. Normality extends a across the vertex, and contraction with the Euler vector field makes this form exact, since all its homogeneous weights are strictly positive. The same identity applies to a convergent series after shrinking. Lemma 3.2 and (18) identify its kernel with $\lambda_i = 0$.

Finally let $b = b(t)$ be an arbitrary smoothing arc. Its actual relative volume form is $g(p,t)\Omega^0$, where g is an analytic unit on the pulled-back family. Lift g to a holomorphic function of the ambient coordinates (p,t); after shrinking it is a unit and defines the same multiplier over $B \times \Delta$. The boundary periods are holomorphic functions $P_i^g(b,t)$. They vanish on $H_i \times \Delta$, by the nodal vanishing-cycle calculation at its general points and analytic continuation. Hence

$$P_i^g(b,t) = \alpha_i(b)u_i(b,t).$$

To compute its linear term without expanding g term by term, fix a smooth direction b_* and a compact cycle δ representing β_i in X_{b_*}. Scaling this cycle gives

$$P_i^g(sb_*,t) = s\int_\delta g(sp,t)\Omega_{b_*}^0 = s\,g(0,t)c_i\alpha_i(b_*) + O(s^2).$$

The estimate is uniform on the fixed compact cycle. Smooth directions form a dense open set, so $d_b P_i^g(0,t) = g(0,t)c_i\alpha_i$. Thus $u_i(0,0) = g(0,0)c_i \neq 0$. Restricting to $b(t)$ proves nonvanishing, whatever its contact order with H_i. $\square$

3.2. **The quadric surface cone.**

Lemma 3.5. *The full semiuniversal deformation base of Q is a smooth one-dimensional germ. Its general fibre is smooth. On the crepant resolution $\mathrm{Tot}(K_{\mathbb{P}^1\times\mathbb{P}^1})$, the local Hodge class of its tangent line is the difference of the two ruling classes.*

Proof. The height-one polygon of Q, translated to its centre, has vertices

$$(1,0), \quad (0,1), \quad (-1,0), \quad (0,-1).$$

Its oriented edges are $u, v, -u, -v$, where u, v are linearly independent over $\mathbb{Q}$. The polygon has primitive edges, and the associated threefold singularity is isolated. Altmann's equations for the versal base are

$$\sum_{i=1}^{4} t_i^k d_i = 0 \quad (k \geq 1), \qquad (d_1, d_2, d_3, d_4) = (u, v, -u, -v),$$

modulo common translation of the t_i. The equations with $k = 1$ are $t_1 = t_3$ and $t_2 = t_4$. All higher equations belong to the ideal generated by these two linear equations. Consequently the quotient base is a smooth affine line, scheme-theoretically. The full versality theorem, including the obstruction calculation in [Alt97, Sections 6–7], applies here; the assertion concerns the entire base, including its possible nilpotent structure.

A smoothing can also be seen directly as

$$\{xy - zw = s\}/\boldsymbol{\mu}_2 \longrightarrow \mathbb{A}^1_s, \tag{19}$$

where the involution negates all four coordinates and fixes s. Taking invariants is exact in characteristic zero, so this quotient family is flat. Its central coordinate ring is the second Veronese subring of the ordinary quadric cone, hence is the section ring of $\mathcal{O}_{\mathbb{P}^1\times\mathbb{P}^1}(2,2)$. For $s \neq 0$ the quadric is smooth and the involution acts freely. Thus the family is a smoothing.

This family is semiuniversal, including its parameter normalization. Translate the displayed polygon by $(1,0)$ to obtain $[0,(1,1)] + [0,(1,-1)]$. Its Cayley cone has rays

$$(0,0,1,0), \quad (1,1,1,0), \quad (0,0,0,1), \quad (1,-1,0,1).$$

They span an index-two sublattice of $\mathbb{Z}^4$, and half their sum is integral. The corresponding toric fourfold is therefore $\mathbb{A}^4/\boldsymbol{\mu}_2$, with the involution negating all four coordinates. The two projection characters pull back to the products of the first two and the last two coordinates. Altmann's component family [Alt97, §8.1] is their difference, precisely (19) after relabeling. Its Kodaira–Spencer class is the nonzero difference of the two segment-dilation parameters. Since the full base is a smooth line, every sufficiently small analytic deformation of Q is locally a pullback of this family.

Finally, the proof of Lemma 2.3 applies without change to $E = \mathbb{P}^1 \times \mathbb{P}^1$: the same toric vanishing proves the local cohomology comparison. The circle-bundle sequence identifies its link homology with

$$K_E^{\perp} \subset H_2(E;\mathbb{C}) = \mathbb{C}f_1 \oplus \mathbb{C}f_2, \qquad K_E = -2f_1 - 2f_2.$$

This perpendicular space is $\mathbb{C}(f_1 - f_2)$ and has dimension one, which equals the tangent dimension just calculated. The comparison therefore identifies the tangent line with this ruling difference. $\square$

Lemma 3.6. *Let M be a smooth Milnor fibre of Q, with boundary L. Then M is homotopy equivalent to $\mathbb{RP}^3$ and $\pi_1(L) = \mathbb{Z}/2$. In particular,*

$$H_2(M;\mathbb{Q}) = 0, \qquad H_3(M;\mathbb{Q}) = \mathbb{Q}, \qquad H^1(L;\mathbb{Z}) = 0.$$

The boundary map $H_3(M, L;\mathbb{Q}) \to H_2(L;\mathbb{Q})$ is an isomorphism; under the circle-bundle identification its image is $\mathbb{Q}(f_1 - f_2)$.

Proof. After a linear change of coordinates the double cover of a noncentral fibre of (19) is $\sum_{i=1}^4 z_i^2 = s$. Its Milnor fibre retracts equivariantly onto the vanishing sphere $\sqrt{s}\,S^3$, and the involution acts antipodally. Passing to the free quotient gives the asserted retraction onto $\mathbb{RP}^3$ and its rational homology. The link is the circle bundle with Euler class $K_{\mathbb{P}^1\times\mathbb{P}^1} = -2f_1 - 2f_2$. In its homotopy sequence the image of $\pi_2(\mathbb{P}^1\times\mathbb{P}^1) \to \mathbb{Z}$ is $2\mathbb{Z}$, so $\pi_1(L) = \mathbb{Z}/2$ and $H^1(L;\mathbb{Z}) = 0$. The rational circle-bundle sequence gives $H_2(L;\mathbb{Q}) = \mathbb{Q}(f_1 - f_2)$. Lefschetz duality gives $\dim H_3(M, L;\mathbb{Q}) = \dim H^3(M;\mathbb{Q}) = 1$; the pair sequence and $H_2(M;\mathbb{Q}) = 0$ make its boundary map surjective onto the one-dimensional link group. It is therefore an isomorphism. □

Smoothness in Lemma 3.5 does not assert vanishing of cotangent cohomology in degree two. In fact the local calculation gives $\dim T^1_Q = 1$ and $\dim T^2_Q = 6$. The latter value, reproduced in `certificates/d19_quadric_partial_resolution.py`, is not needed in the proof: the explicit equations and full versality show directly that the obstruction maps vanish. In particular no complete-intersection hypothesis on Q is being used.

3.3. **Central fibres and their curve classes.** For the degree-six surface use the marking

$$E = \mathrm{Bl}_{p_1,p_2,p_3}\mathbb{P}^2, \qquad h^2 = 1, \quad e_i^2 = -1, \quad h\cdot e_i = e_i\cdot e_j = 0\ (i \neq j),$$

with the three points noncollinear. Its toric boundary, in cyclic order, is

$$e_1, \quad h - e_1 - e_2, \quad e_2, \quad h - e_2 - e_3, \quad e_3, \quad h - e_1 - e_3. \tag{20}$$

We retain the root conventions

$$\ell = -h + e_1 + e_2 + e_3, \qquad A = -e_1 + e_3, \qquad B = e_2 - e_3. \tag{21}$$

Thus the A_1 summand is $\mathbb{Q}\ell$, and the A_2 summand is $\mathbb{Q}A \oplus \mathbb{Q}B$.

Proposition 3.7. *There are proper crepant partial resolutions with the following properties.*

1. *A small morphism $W_7 \to C_7$ has precisely one singular point, of type Q. A smooth crepant resolution through W_7 differs from $\mathrm{Tot}(K_{\mathrm{dP}_7})$ by the flop of $h - e_1 - e_2$. Under this comparison the ruling difference of Q is the degree-seven root, up to the choice of its sign.*
2. *A morphism $W_{6,1} \to C_6$ is obtained from $\mathrm{Tot}(K_E)$ by contracting $C_0 = e_1$ and $C_3 = h - e_2 - e_3$. It has exactly two ordinary double points. The image of E is $\mathbb{P}^1 \times \mathbb{P}^1$.*
3. *A morphism $W_{6,2} \to C_6$ is obtained by contracting e_1, e_2, e_3 in $\mathrm{Tot}(K_E)$. It has exactly three ordinary double points. The image of E is $\mathbb{P}^2$.*

The last two morphisms to C_6 are divisorial. With Hodge-normalized nodal coefficients, the maps into the selected root subspaces are

$$\lambda\ell = \lambda[C_0] - \lambda[C_3], \qquad aA + bB = -ae_1 + be_2 + (a - b)e_3. \tag{22}$$

In particular the two coefficients over an A_1 point are $(\lambda, -\lambda)$, and the three coefficients over an A_2 point are $(\mu_1, \mu_2, \mu_3) = (-a, b, a - b)$.

Proof. For degree seven use the height-one pentagon

$$(0,0),(1,0),(2,1),(1,2),(0,1).$$

Subdivide it into the triangle with vertices $(0,0),(1,0),(0,1)$ and the quadrilateral with vertices $(1,0),(2,1),(1,2),(0,1)$. The triangle is unimodular. Translation of the quadrilateral by $(-1,-1)$ gives the polygon in Lemma 3.5. The subdivision preserves the support and introduces no rays; the induced toric morphism is therefore proper and small. Every ray has height one, so the morphism is crepant. The two maximal cones give one smooth chart and one Q chart, proving the assertion about its singularities.

Inserting the interior lattice point of the quadrilateral gives a smooth resolution through this model. Its triangulation and the star triangulation of the pentagon differ by one diagonal change. The corresponding curve on the original degree-seven surface is $c = h - e_1 - e_2$. It is a (-1)-curve and $K_E \cdot c = -1$; its normal bundle in $\mathrm{Tot}(K_E)$ is $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$. The diagonal change is its ordinary flop. The surface after the flop is the contraction of c, namely $\mathbb{P}^1 \times \mathbb{P}^1$. The classes $h - e_1$ and $h - e_2$ have intersection zero with c, square zero, and mutual intersection one. They descend to its two rulings. Their difference is $e_2 - e_1$, the degree-seven root. This also identifies their classes on the common complement of the flopping curves. Indeed representatives of these two pencils can be chosen disjoint from c; their differences give the same class under the homology comparison across the flop.

For degree six use the polygon

$$v_0 = (1,0),\ v_1 = (1,1),\ v_2 = (0,1),\ v_3 = (-1,0),\ v_4 = (-1,-1),\ v_5 = (0,-1)$$

and its centre $o = (0,0)$. The six triangles $[o, v_i, v_{i+1}]$ are unimodular; the star fan gives $\mathrm{Tot}(K_E)$ with boundary marking (20). To contract the boundary curve at v_i, merge its two adjacent triangles into $[o, v_{i-1}, v_i, v_{i+1}]$. For $i = 0, 3$, these mergers are disjoint in their interiors and leave two unimodular triangles. For $i = 0, 2, 4$ they give three quadrilaterals covering the polygon. Each merged cell is a unit parallelogram: its four lifted rays have the relation with coefficients $(1,-1,1,-1)$ in cyclic order, and three adjacent lifted rays have determinant ± 1. Its affine toric chart is consequently $xy - zw = 0$, with no finite quotient. No boundary cone changes, and the subdivisions cover the original cone, proving properness, crepancy, and the claimed singularity counts.

The same contractions can be checked intrinsically. Every curve in (20) has self-intersection and canonical degree -1. Its normal sequence in the total canonical bundle splits, since its extension class belongs to $H^1(\mathbb{P}^1, \mathcal{O}) = 0$. The selected curves are disjoint, so their contractions are independent ordinary nodal contractions. Contracting e_1, e_2, e_3 on E gives $\mathbb{P}^2$. Contracting $e_1, h - e_2 - e_3$ gives the smooth toric degree-eight surface with four boundary curves of square zero, namely $\mathbb{P}^1 \times \mathbb{P}^1$. These surfaces are still contracted by the subsequent morphism to C_6, explaining its divisorial nature.

Finally, (22) follows by substitution in (21). These are also the deformation-theoretic transfer maps: inclusion of the nodal exceptional supports in the full exceptional surface sends a tuple of local classes to the sum of their curve classes. Naturality of the connecting maps in local cohomology and contraction with the same volume form make this map commute with the Hodge comparison. In the degree-seven case the comparison is made on the common regular open set across the flop. There is thus no independent scalar choice at each newly introduced node in (22). □

3.4. **A simultaneous modification of the one-dimensional branch.** The two-node model extends over the entire A_1 branch. Here it is useful to give the extension explicitly, since its central-fibre description alone would not justify lifting a family. Altmann's triangle-plus-triangle family is the affine cone over the Segre embedding of $(\mathbb{P}^1)^3$, equipped with the function

$$u_0v_0w_0 - u_1v_1w_1 = s; \tag{23}$$

see [Alt97, Section 8.2(i)]. The products in this equation are tensor coordinates on the cone, rather than functions on the product of projective spaces.

Retain the first two projective factors. The resulting incidence space is

$$V = \mathrm{Tot}_{\mathbb{P}^1\times\mathbb{P}^1}\big(\mathcal{O}(-1,-1)\oplus\mathcal{O}(-1,-1)\big).$$

Its morphism to the tensor cone sends a pair of fibre vectors to the tensor $u\otimes v\otimes w$. It is proper, since the incidence description is closed in the product of the cone with $\mathbb{P}^1\times\mathbb{P}^1$. It is an isomorphism away from the vertex: a nonzero decomposable tensor determines its first two factor lines uniquely. On V, equation (23) defines a morphism to the s-line and hence a simultaneous modification of that family.

There are exactly two critical points in the central fibre. To see this, write the fibre coordinates locally as A, B. Differentiating with respect to them forces $u_0v_0 = u_1v_1 = 0$. The two possible base points are $([1:0],[0:1])$ and $([0:1],[1:0])$. On their respective affine charts the equations are exactly

$$\beta A - \alpha B = s, \qquad \gamma C - \delta D = s. \tag{24}$$

Their differentials vanish only at the origin of each four-dimensional chart, and their quadratic forms are nondegenerate. On the other two projective charts one coefficient of a fibre coordinate is a unit, so the map is smooth there. These equations also prove flatness: near a critical point the morphism is the standard nodal deformation, and elsewhere it is smooth. Equivalently, the local rings are torsion-free over the regular one-dimensional base.

The zero section belongs only to $s = 0$. Thus the modification is an isomorphism over every $s \neq 0$ and away from the vertex of the central fibre. Its central fibre is $W_{6,1}$: resolving the two nodes on the side which blows up the corresponding points of the zero section gives the marked degree-six surface and the two curves C_0, C_3. This can equally be read from the first degree-six subdivision in Proposition 3.7.

The two displayed equations have the same algebraic parameter s. Their Hodge-normalized parameters have opposite signs when their exceptional curves are oriented by $[C_0], [C_3]$. More precisely, the image is $(\lambda, -\lambda)$ with $\lambda = c_1 s$ to first order for a nonzero constant c_1. This follows either from the local-cohomology comparison above or by pairing with E: $E\cdot C_0 = E\cdot C_3 = -1$, while the branch class lies in $K_E^{\perp}$. Neither nodal derivative vanishes, by (24). The normalized branch coordinate is chosen using this c_1, rather than by equating two independently chosen residue normalizations.

3.5. **The two-dimensional branch and its discriminant.**

Proposition 3.8. *The A_2 branch of C_6 admits a proper simultaneous partial resolution whose central fibre is $W_{6,2}$. The morphism is an isomorphism over every nonzero point of the two-dimensional base, including its discriminant. In linear coordinates (a,b) compatible with the Hodge normalization, the three nodal tangent parameters are*

$$(\mu_1,\mu_2,\mu_3) = (-a, b, a-b).$$

The discriminant is the union of the three lines $ab(a-b) = 0$. At a nonzero point of any one of these lines the fibre has precisely one ordinary double point; off these lines it is smooth.

Proof. We first work in algebraic parameters before imposing the Hodge normalization. Write $D = \mathbb{C}^3/\mathbb{C}(1,1,1)$ and represent its points by (d_1, d_2, d_3). The branch family is

$$\mathcal{M}_d = \{M : \operatorname{rank} M \le 1, \quad M_{ii} = z + d_i \ (1 \le i \le 3)\}. \tag{25}$$

Common translation of the d_i is absorbed in z. This is the actual restricted versal family, not just a first-order model. For an explicit comparison with [Alt97, Section 5.6], take edge parameters (a, b, c, a, b, c) and the matrix

$$\begin{pmatrix} z+c & z_1 & z_2 \\ z_4 & z+b & z_3 \\ z_5 & z_6 & z+a \end{pmatrix}.$$

Its nine 2×2 minors are the six equations $z_i t_i - z_{i-1} z_{i+1} = 0$ (cyclic indices), with $(t_1, \ldots, t_6) = (z+a, z+b, z+c, z+a, z+b, z+c)$, and the three equations $t_2 t_3 - z_1 z_4 = t_3 t_4 - z_2 z_5 = t_1 t_2 - z_3 z_6 = 0$. They generate exactly the branch ideal. This also follows from [Alt97, Section 8.2(ii)], which describes its total space as the cone over $\mathbb{P}^2 \times \mathbb{P}^2$ with the two diagonal-difference functions.

Resolve that total space by retaining its column line:

$$\widetilde{\mathcal{M}} = \{(M, L) : \operatorname{im} M \subseteq L\} = \operatorname{Tot}_{\mathbb{P}^2}(\mathcal{O}(-1)^{\oplus 3}) \longrightarrow \{M : \operatorname{rank} M \le 1\}. \tag{26}$$

Write $M = xy^{\mathsf{T}}$, where $x \neq 0$ and $(x, y) \sim (tx, t^{-1}y)$. The incidence relation proves properness. A nonzero rank-one matrix has a unique column line, and on an open set $M_{k\ell} \neq 0$ this line is represented by its ℓth column. These formulas give regular inverses, proving that (26) is an isomorphism away from the zero matrix.

Compose (26) with the diagonal-difference map to D. On $x_k = 1$, let i, j be the two other indices. Elimination of y_k, z gives the single equation

$$x_i y_i - x_j y_j = d_i - d_j, \qquad y_k = x_i y_i - d_i + d_k. \tag{27}$$

The two independent base coordinates can be taken as $d_i - d_j$ and $d_k - d_i$. Thus each chart is the product of the standard nodal deformation with a smooth one-dimensional base. This proves flatness over the full plane, including its origin and discriminant. In the central fibre its only singular point in this chart has all four coordinates zero. The three resulting points lie over the three distinct coordinate points of $\mathbb{P}^2$ and give exactly three nodes. The zero section is the exceptional $\mathbb{P}^2$. Resolving the nodes by blowing up their points on this surface recovers $\mathrm{Bl}_3 \mathbb{P}^2$ and the curves e_1, e_2, e_3. The star fan is the second subdivision of Proposition 3.7; in particular the central morphism is crepant and is $W_{6,2} \to C_6$.

The zero matrix has all diagonal differences zero. It therefore occurs only over $0 \in D$. The isomorphism away from the zero matrix consequently restricts to an isomorphism on every fibre with $d \neq 0$, with no exception along a discriminant line. The chart equations show that the three singularity parameters are the cyclic differences

$$d_2 - d_3, \quad d_3 - d_1, \quad d_1 - d_2.$$

Their sum is zero. A nonzero base point lies on at most one of their zero lines, and the corresponding chart has one node by its nondegenerate quadratic equation. All remaining charts are smooth. This proves both the discriminant statement and the assertion about its noncentral fibres.

To identify parameters with the fixed marking, label the three coordinate points, and choose signs in the nodal equations, so that their differences in the order e_1, e_2, e_3 are $d_3 - d_1, d_2 - d_3, d_1 - d_2$. Taking $(d_1, d_2, d_3) = (a, b, 0)$ gives $(-a, b, a - b)$ algebraically. The determinant-twisted equivariance in Lemma 2.4 identifies the two-dimensional parameter representation with the A_2 root representation, uniquely up to a nonzero scalar: rotation and reflection act as the usual irreducible two-dimensional representation of the triangle symmetry group. Thus, after matching this marking, the Hodge comparison multiplies these three algebraic differences by one common nonzero scalar. Rescale a, b simultaneously by that scalar. Naturality under inclusion of supports now gives exactly

$$aA + bB = -ae_1 + be_2 + (a-b)e_3$$

with the prescribed global volume form. The linear rescaling leaves the three discriminant lines unchanged. This proves the asserted normalized parameter map. $\square$

The constants relating algebraic parameters to Hodge-normalized parameters may differ between singular points, and between the A_1 and A_2 summands at one degree-six point. They are fixed by the one global volume form, not by unrelated choices of local coordinates. Equations (22) are identities after this normalization. Their vanishing conditions are independent of the remaining common nonzero factors.

The simultaneous constructions also distinguish a nonzero tangent on a smoothing component from a tangent transverse to its discriminant. Every nonzero A_1 parameter smooths the germ, whereas a nonzero vector in the A_2 plane may leave one of the three nodes unsmoothed. Because the simultaneous modification is an isomorphism on every noncentral fibre, this distinction can be applied to an arbitrary analytic classifying arc, including an arc tangent to a discriminant line to high order. It is this property that permits the necessity argument in Section 4.

4. The smoothing criterion

Throughout this section, X is a connected normal projective complex threefold with $\omega_X \cong \mathcal{O}_X$ and $H^1(X, \mathcal{O}_X) = 0$, whose singularities are isolated ordinary double points and the specified analytic anticanonical cones over dP_5, dP_6, dP_7, and $\mathbb{P}^1 \times \mathbb{P}^1$. Fix a local component S_p at every cone point, and write $S = (S_p)$. We retain the entire nodal base and the entire degree-five and $\mathbb{P}^1 \times \mathbb{P}^1$ cone bases, and use the reduced component at a degree-seven point. The smooth crepant resolution $Y_0 \to X$, the link subspaces R_{p,S_p}, and their relation space

$$K_S = \ker\left(\bigoplus_p R_{p,S_p} \longrightarrow H_2(Y_0; \mathbb{Q})\right) \tag{28}$$

are as in Section 2. All tangent identifications below use contraction with one fixed generator of $H^0(X, \omega_X)$. In particular, coefficients in (28) are the resulting homological coordinates; they are not independently normalized local equation parameters.

Choose an analytic semiuniversal deformation of X, analytic local semiuniversal deformations of its singularities, and analytic classifying maps from the first base to the latter bases. We denote by $\mathrm{Def}^S(X)$ the scheme-theoretic inverse image of the product of the selected local components. These choices specify the analytic subgerm and its completed local ring. We use $\mathbf{Def}_X$ for the functor of marked deformations over local Artin $\mathbb{C}$-algebras, and $\mathrm{Def}(X)$ for an analytic semiuniversal base.

4.1. Global partial resolutions.

Proposition 4.1. *The local morphisms of Proposition 3.7, together with the identity near degree-five and original $\mathbb{P}^1 \times \mathbb{P}^1$ cones, determine a proper crepant analytic modification $\pi : Z \to X$. Its singularities are the original nodes, two or three nodes over each degree-six point* *according to its selected component, and one anticanonical $\mathbb{P}^1\times\mathbb{P}^1$ cone over each degree-seven point, together with the original degree-five and $\mathbb{P}^1\times\mathbb{P}^1$ cones. Moreover,*

$$R\pi_*\mathcal{O}_Z \simeq \mathcal{O}_X, \qquad \omega_Z \cong \mathcal{O}_Z, \qquad H^1(Z,\mathcal{O}_Z)=0. \tag{29}$$

Every smooth resolution of Z is Moishezon and satisfies the $\partial\bar\partial$-lemma.

Proof. Choose mutually disjoint sufficiently small neighborhoods U_p of the degree-six and degree-seven cone points, and restrict the local morphisms to proper morphisms $W_p \to U_p$. Glue the W_p to the unchanged complement of these degree-six and degree-seven points using their specified isomorphisms over $U_p \setminus \{p\}$. This complement includes every original degree-five and $\mathbb{P}^1\times\mathbb{P}^1$ cone. The images of these charts are open. Two points with different images in X are separated by inverse images of disjoint neighborhoods in X; two distinct points over the same cone point are separated in the Hausdorff space W_p. Thus the glued space is Hausdorff. The map is proper over every U_p and over the complement of these points, and properness is local on the target. This proves properness globally, and hence compactness of Z.

The local models are Gorenstein and canonical. In particular, X and Z have rational singularities. If $g:\widetilde{Z}\to Z$ is a resolution, then πg is a resolution of X and $Rg_*\mathcal{O}_{\widetilde{Z}} \simeq \mathcal{O}_Z$, $R(\pi g)_*\mathcal{O}_{\widetilde{Z}} \simeq \mathcal{O}_X$. Composition gives the first assertion of (29), and Leray gives its last assertion. Local crepancy means that the pullback of the chosen volume form extends to a generator of the dualizing sheaf on every local model. These generators agree on the overlaps, giving $\omega_Z \cong \mathcal{O}_Z$.

A smooth resolution is compact and bimeromorphic to projective X, hence Moishezon. A smooth projective modification of it exists, and the $\partial\bar\partial$-lemma descends under a proper bimeromorphic map between compact complex manifolds [DGMS75, Theorem 5.22 and Corollary 5.23]. □

Let $Y \to Z$ be the smooth crepant resolution obtained by resolving the new nodes on the sides specified in Proposition 3.7 and resolving the degree-five and quadric cones by their vertex blowups. At a degree-six point this recovers the original neighborhood in $\operatorname{Tot} K_E$. At a degree-seven point it differs from Y_0 by the described flop. Near a degree-five or original $\mathbb{P}^1\times\mathbb{P}^1$ cone the two resolutions agree. Removing a smooth curve from a smooth threefold does not change H_2: the relative groups in degrees two and three vanish by the real codimension-four Thom isomorphism. The common complements therefore identify $H_2(Y_0;\mathbb{Q})$ and $H_2(Y;\mathbb{Q})$. The degree-seven root is represented on that complement by the difference of the two ruling classes, so the identifications preserve the local substitutions used in (28).

4.2. **Blowing down Artin deformations analytically.** The following lemma supplies a morphism of deformation functors even when Z is not algebraic. Its sheaf-theoretic part is the construction in [San14, Proposition 2.12 and Corollary 2.13]; we include the analytic realization.

Lemma 4.2 (analytic blow-down)**.** *Let $f: Z\to X$ be a proper holomorphic morphism of complex spaces with $f_*\mathcal{O}_Z=\mathcal{O}_X$ and $R^1f_*\mathcal{O}_Z=0$. A flat deformation Z_A over a local Artin $\mathbb{C}$-algebra A determines a flat analytic deformation*

$$X_A = (|X|, f_*\mathcal{O}_{Z_A})$$

and a proper morphism $f_A: Z_A\to X_A$ deforming f. This construction is natural in marked deformations and commutes with every Artin base change.

Proof. The marking identifies the underlying topological space of Z_A with $|Z|$, so the displayed pushforward is defined as a sheaf of A-algebras. Let $U \subset X$ be a sufficiently small Stein open subset and put $V = f^{-1}(U)$, $\mathcal{F} = \mathcal{O}_{Z_A}|_V$. Leray and Cartan's theorem B give $H^1(V, \mathcal{O}_Z) = 0$. For every finite A-module M we claim

$$H^1(V, \mathcal{F} \otimes_A M) = 0. \tag{30}$$

Indeed, a composition series of M has successive quotients $\mathbb{C}$. Flatness of $\mathcal{F}$ preserves its short exact sequences under tensor product. The long cohomology sequences, starting from $\mathcal{F} \otimes_A \mathbb{C} = \mathcal{O}_Z|_V$, prove (30) by induction.

It follows that $M \mapsto H^0(V, \mathcal{F} \otimes_A M)$ is exact on finite A-modules. Applying it to a finite presentation of M by free modules gives the natural isomorphism

$$H^0(V, \mathcal{F}) \otimes_A M \xrightarrow{\sim} H^0(V, \mathcal{F} \otimes_A M). \tag{31}$$

Consequently $H^0(V, \mathcal{F})$ is A-flat: exactness on finite modules suffices over the Noetherian ring A. Passing to stalks on X proves flatness of $f_*\mathcal{O}_{Z_A}$ and the corresponding sheaf version of (31). In particular its reduction modulo the maximal ideal $\mathfrak{m}_A$ is $\mathcal{O}_X$.

To prove analyticity, take U small enough to be a closed analytic subspace of a polydisc D. Equation (31), with $M = \mathbb{C}$, lifts its embedding coordinates to global sections of $\mathcal{F}$. Together with the structure morphism to $\mathrm{Spec}_{\mathrm{an}}\, A$ these sections define a holomorphic map

$$g_A : V_A \longrightarrow D \times \mathrm{Spec}_{\mathrm{an}}\, A.$$

Its underlying continuous map is the proper composite $V \to U \hookrightarrow D$, hence g_A is proper. Grauert's direct image theorem for complex spaces, including nonreduced spaces, implies that $g_{A*}\mathcal{O}_{V_A}$ is coherent on the polydisc thickening [Gra60, §6, Hauptsatz I]. The natural algebra map

$$\mathcal{O}_{D \times \mathrm{Spec}_{\mathrm{an}} A} \longrightarrow g_{A*}\mathcal{O}_{V_A}$$

is surjective modulo $\mathfrak{m}_A$, since its reduction is $\mathcal{O}_D \to \mathcal{O}_U$. Its coherent cokernel $\mathcal{C}$ therefore satisfies $\mathcal{C} = \mathfrak{m}_A\mathcal{C}$, and nilpotence gives $\mathcal{C} = 0$. Its kernel is a coherent ideal. The resulting closed analytic subspace has underlying ringed space $(U, f_*\mathcal{O}_{Z_A}|_U)$.

This ringed space is independent of the lifted coordinates. Thus the local analytic realizations glue, and the adjunction counit induces f_A. Locally this is the holomorphic map to the closed subspace just constructed; its central restriction is f and its underlying map is proper. Finally, if $A \to A'$ is any homomorphism of local Artin $\mathbb{C}$-algebras, then A' is a finite A-module. The sheaf version of (31) identifies the blow-down of $Z_A \times_A A'$ with $X_A \times_A A'$. These natural tensor identifications respect composition. An isomorphism of marked source deformations pushes forward to an isomorphism of the constructed target deformations, proving the functorial assertion. □

4.3. **Unobstructedness of the partial resolution.**

Lemma 4.3. *The functor* $\mathbf{Def}_Z$ *is unobstructed. Both* $\mathbf{Def}_Z$ *and* $\mathbf{Def}_X$ *are prorepresentable, and Lemma 4.2 induces a morphism of their formal bases*

$$\beta : \widehat{\mathrm{Def}}(Z) \longrightarrow \widehat{\mathrm{Def}}(X). \tag{32}$$

Proof. By Proposition 4.1, Z is compact, normal and Gorenstein with trivial dualizing sheaf, $H^1(Z, \mathcal{O}_Z) = 0$, and isolated rational, hence Du Bois, singularities. A resolution satisfies the $\partial\bar{\partial}$-lemma. Friedman's theorem [Fri26, Theorem 3.10] gives the generalized T^1 lifting property. We explain why here it gives ordinary T^1 lifting.

Set $A_k = \mathbb{C}[t]/(t^{k+1})$ and $B_k = A_k[\epsilon]/(\epsilon^2)$. For a deformation Z_k/A_k let Z_{k-1} denote its restriction. Relative first-order deformations, with an identification over $\epsilon = 0$, form $T^1(Z_k/A_k)$. There is an exact sequence and its local counterpart

$$T^1(Z_k/A_k) \longrightarrow T^1(Z_{k-1}/A_{k-1}) \xrightarrow{\delta} \mathbb{T}^2_Z,$$
$$T^1_{\mathrm{loc}}(Z_k/A_k) \longrightarrow T^1_{\mathrm{loc}}(Z_{k-1}/A_{k-1}) \xrightarrow{\delta_{\mathrm{loc}}} H^0(Z, \mathcal{T}^2_Z).$$

Here $\mathbb{T}^2_Z = \mathrm{Ext}^2(\Omega^1_Z, \mathcal{O}_Z)$ and $\mathcal{T}^2_Z = \mathcal{E}xt^2(\Omega^1_Z, \mathcal{O}_Z)$ are the global and local obstruction spaces used in the lifting theorem. Localization $\ell : \mathbb{T}^2_Z \to H^0(Z, \mathcal{T}^2_Z)$ commutes with the connecting maps. Generalized lifting means that ℓ is injective on $\operatorname{im}\delta$ for every k and every Z_k.

The full local deformation bases of Z are smooth: this is the usual nodal calculation, Proposition 3.1, and Lemma 3.5. This implies surjectivity of the displayed local T^1 map with its markings, as follows. For a fixed local deformation over A_k and a marked relative deformation over B_{k-1}, use the specified identification over A_{k-1} to glue them over

$$D_k = A_k \times_{A_{k-1}} B_{k-1}.$$

The deformation category of analytic spaces has the Rim–Schlessinger gluing property: the local structure sheaves are formed by the fibre product over the common restriction, and the result comes with compatible identifications on both factors. This is gluing with a specified identification, not merely a choice of isomorphism classes; see also the categorical formulation in [Gro97r, §1]. The natural map $B_k \to D_k$ is surjective with kernel (ϵt^k). Smoothness of the local deformation functor lifts the glued object to B_k. Choose an isomorphism from its restriction to D_k to that object, and restrict this same isomorphism to the two factors. The resulting identifications agree over A_{k-1} and give exactly the required relative first-order lift. Infinitesimal automorphisms can change these choices but do not prevent their existence; local prorepresentability is not used.

Thus $\delta_{\mathrm{loc}} = 0$, so $\ell\delta = 0$ and injectivity on $\operatorname{im}\delta$ gives $\delta = 0$. Ordinary T^1 lifting proves unobstructedness [Fri26, Theorem 3.3]. In particular this argument concerns the full base, including its nilpotent structure; it does not infer smoothness from a statement about reduced supports.

For completeness, the vanishing of infinitesimal automorphisms follows for both $T = X$ and $T = Z$ from the same argument. Choose an equivariant resolution $h : \widetilde{T} \to T$. Canonical singularities allow the volume form to pull back holomorphically, and contraction gives an injection $\Theta_{\widetilde{T}} \to \Omega^2_{\widetilde{T}}$. Rationality, Hodge symmetry on the Moishezon resolution, and Serre duality give

$$h^0(\widetilde{T}, \Omega^2_{\widetilde{T}}) = h^2(\widetilde{T}, \mathcal{O}_{\widetilde{T}}) = h^2(T, \mathcal{O}_T) = h^1(T, \mathcal{O}_T) = 0.$$

Since $h_*\Theta_{\widetilde{T}} = \Theta_T$, this proves $H^0(T, \Theta_T) = 0$, and the deformation functors are prorepresentable; compare [Fri26, Lemma 3.2]. The natural Artin transformation supplied by Lemma 4.2 therefore gives (32). □

4.4. **Lifting the entire selected tangent space.**

Lemma 4.4. *Put $V_S = T_0 \operatorname{Def}^S(X)$. There is an exact sequence*

$$0 \longrightarrow H^1(X, \Theta_X) \longrightarrow V_S \longrightarrow K_S \otimes_{\mathbb{Q}} \mathbb{C} \longrightarrow 0, \tag{33}$$

and every vector in V_S belongs to the image of $d\beta$.

Proof. The exact sequence is the restriction of Theorem 2.5 to the selected local tangent subspaces. We construct lifts through $d\beta$, including lifts of its locally trivial kernel.

Let $\kappa \in K_S \otimes \mathbb{C}$. At a degree-six A_1 point set $C_0 = e_1$, $C_3 = h - e_2 - e_3$. Since the convention of Section 2 is $\ell = C_0 - C_3$, replace the coefficient $\lambda\ell$ by nodal coefficients $(\lambda, -\lambda)$ at (C_0, C_3). At an A_2 point replace $\sum_i \mu_i e_i$, $\sum_i \mu_i = 0$, by the three nodal coefficients μ_i. At a degree-seven point use the same multiple of the quadric ruling difference. Retain the old nodal coefficients and the ruling-difference coefficients at original $\mathbb{P}^1 \times \mathbb{P}^1$ cones, and retain the whole local vector at each degree-five point. The curve identifications of Proposition 3.7 and the common complements identify the sum of this tuple ν with the sum of κ in $H_2(Y;\mathbb{C})$; it is therefore zero.

Write F for the exceptional locus of $Y \to Z$, with local pieces $F_q \subset Y_q$. The local Hodge comparisons identify the components of ν with local tangent classes and with their images in $H^2_{F_q}(Y_q, \Omega^2_{Y_q})$. In the support sequence

$$H^1(Y \setminus F, \Omega^2) \longrightarrow \bigoplus_q H^2_{F_q}(Y_q, \Omega^2) \longrightarrow H^2(Y, \Omega^2_Y), \tag{34}$$

their image in the last group is the class of the corresponding curves. Under the injection $H^2(Y, \Omega^2_Y) \hookrightarrow H^4(Y;\mathbb{C})$ supplied by the $\partial\bar\partial$-lemma it is the Poincaré dual of their homological sum, hence zero. Exactness gives a class on $Y \setminus F$. Contraction with the volume form and the depth-three comparison of deformations with the regular locus give a tangent $\eta \in \mathbb{T}^1_Z$ with local tuple ν, as in the proof of Theorem 2.5.

Let E' be the full exceptional set of $Y \to X$. Restriction from $Y \setminus F$ to $Y \setminus E'$ commutes with the boundary maps in (34) and the inclusion of supports $H^2_F(Y, \Omega^2_Y) \to H^2_{E'}(Y, \Omega^2_Y)$. At a degree-six point this sends the nodal tuple to its sum of curve classes in E; at a degree-seven point it sends the ruling difference to the original root; at a degree-five or original $\mathbb{P}^1 \times \mathbb{P}^1$ cone it is the identity. These are exactly the local components of κ. The construction of blow-down agrees with restriction on the common regular locus. Thus $d\beta(\eta)$ has the prescribed local tangent on X, including at points where its coefficient is zero.

Given any desired vector $v \in V_S$ over κ, its difference from $d\beta(\eta)$ belongs to $H^1(X, \Theta_X)$. Choose a Stein cover $\{U_i\}$ with exactly one member containing each singular point and all other members avoiding the singular set. Every overlap lies in the locus where π is an isomorphism. Represent the difference by a Čech cocycle $\{\xi_{ij}\}$ of vector fields. Glue the trivial deformations of $\pi^{-1}(U_i)$ by the automorphisms $1 + \epsilon\xi_{ij}$ on the overlaps. The additive cocycle condition gives the gluing identity modulo ϵ^2. Pushing forward these trivial local deformations and their transitions recovers the original cocycle on X. Hence this locally trivial global tangent also lifts, and adding its lift to η gives v. □

4.5. **Smoothness of a selected deformation base.**

Theorem 4.5. *Suppose the projection $K_S \to R_{p,S_p}$ is nonzero at every degree-six point p. Then $\operatorname{Def}^S(X)$ is smooth, of dimension*

$$h^1(X, \Theta_X) + \dim_{\mathbb{Q}} K_S.$$

Every tangent in $T_0 \operatorname{Def}^S(X)$ is realized by a convergent curve in this germ. No nonzero-projection assumption is required at nodes, degree-five or degree-seven points, or $\mathbb{P}^1 \times \mathbb{P}^1$ cones.

Proof. Let B be the completed local ring of $\operatorname{Def}(Z)$. By Lemma 4.3 it is a formal power-series ring, in particular a domain. At a degree-seven point the local base is $\mathbb{C}[[s, v]]/(v^2, sv)$, so the pullback of v to B vanishes. At a degree-six point the local base is $\mathbb{C}[[s, u, v]]/(su, sv)$, with A_1 component $u = v = 0$ and A_2 component $s = 0$. If A_1 was selected, the nonzero projection hypothesis and Lemma 4.4 imply that the pullback of s has a nonzero linear term. It is nonzero in B, and the two equations force the pullbacks of u, v to vanish. For a selected A_2 component, one of u, v has a nonzero linear term, forcing the pullback of s to vanish. Thus β factors through the full selected formal subspace. Nodes, degree-five cones, and original $\mathbb{P}^1 \times \mathbb{P}^1$ cones impose no branch equations, since their entire local bases are retained. Its differential onto that subspace is surjective by Lemma 4.4.

Write its completed target ring minimally as $A = \mathbb{C}[[x_1, \dots, x_r]]/I$, with $I \subset (x_1, \dots, x_r)^2$. Surjectivity of the differential means that the images of the x_i in B have independent linear terms. Extend them to a formal coordinate system of B. The composite $\mathbb{C}[[x_1, \dots, x_r]] \to B$ is injective, whereas every element of I maps to zero. Hence $I = 0$. Equation (33) gives the dimension. An analytic local ring is regular if its completion is regular, so the analytic subgerm $\mathrm{Def}^S(X)$ is smooth. Analytic coordinates on that germ realize every tangent by a convergent curve. This last step does not require convergence of the formal map β. $\square$

4.6. **Necessity from local periods.** We recall the form of the period argument that will be used after a simultaneous partial resolution. Its hypotheses require compactness; the intermediate space need not be projective.

Lemma 4.6 (period necessity)**.** *Let T be a connected compact normal Gorenstein threefold with $\omega_T \cong \mathcal{O}_T$, $H^1(T, \mathcal{O}_T) = 0$, and a smooth crepant resolution $\widetilde{T} \to T$. Suppose its singularities are nodes and anticanonical cones over dP_5, dP_6, dP_7, or $\mathbb{P}^1 \times \mathbb{P}^1$, and that it has a convergent proper smoothing which uses the A_1 component at every degree-six point. Then the relation space of its selected local subspaces in $H_2(\widetilde{T}; \mathbb{Q})$ contains a vector nonzero on every rank-one summand and satisfying $\prod_{i=1}^5 \lambda_i \neq 0$ at every degree-five point.*

Proof. We give the argument in a form applicable to the possibly nonprojective space T; the local period computations and link markings are those of [Wue26b, §§3–4], with the $\mathbb{P}^1 \times \mathbb{P}^1$ case supplied below and the degree-five case supplied by Lemma 3.4. Write T_t for a smooth nearby fibre. Remove disjoint small singular neighborhoods and let W be the common exterior with boundary $\bigsqcup_p L_p$. Smooth local triviality along W gives decompositions

$$T_t = W \cup \bigsqcup_p M_p, \qquad \widetilde{T} = W \cup \bigsqcup_p N_p,$$

where M_p is the local Milnor fibre and N_p a resolution neighborhood, with the canonical common link identifications.

The smoothing has a relative holomorphic volume form after shrinking the disc. Here are the details needed without a polarization. The relative dualizing sheaf is invertible because the family is flat with Gorenstein central fibre, and is compatible with base change. Its first Chern class restricts to zero on W, by smooth triviality and $\omega_T \cong \mathcal{O}_T$, and on each M_p, by a local trivialization near the central singularity. The links satisfy $H^1(L_p; \mathbb{Z}) = 0$: for nodes this follows from $L_p \cong S^2 \times S^3$, and for the cone links it follows from the homotopy sequence of the circle bundle with nonzero Euler class K_E over the simply connected del Pezzo surface. Mayer–Vietoris therefore injects $H^2(T_t; \mathbb{Z})$ into the sum of the corresponding groups of the pieces, so $c_1(\omega_{T_t}) = 0$. Semicontinuity gives $H^1(T_t, \mathcal{O}_{T_t}) = 0$; the exponential sequence implies $\omega_{T_t} \cong \mathcal{O}_{T_t}$. Connectedness and Grauert base change now give a relative section Ω generating the canonical sheaf on every fibre. Its restriction to a smooth fibre is closed.

At a node with local equation $xy - zw = b(t)$, the period of the residue form on the vanishing sphere is $c\, b(t)$, where $c \neq 0$. Indeed, after writing the quadratic form as $\sum_i z_i^2$, the sphere is $\sqrt{b}\,S^3$ and the residue form scales by b; its period at $b=1$ is π^2 up to orientation and the linear change of coordinates. The global form is a holomorphic unit times this residue form. Uniformly on the shrinking sphere its period is therefore

$$b(t)\big(c\,g(p,0)+o(1)\big),\qquad g(p,0)\neq 0,$$

which is nonzero for every sufficiently small $t\neq 0$, regardless of the vanishing order of $b(t)$.

At a $\mathbb{P}^1\times\mathbb{P}^1$ cone, Lemma 3.5 identifies the local smoothing with a pullback of (19), say with parameter $b(t)$. The nodal residue form on its double cover is invariant under the involution. The total quotient $\mathbb{A}^4/\boldsymbol{\mu}_2$ is Gorenstein because $-I\in\mathrm{SL}_4$; adjunction for its parameter function shows that the descended residue generates the relative dualizing sheaf, including at the central vertex. This remains true after pullback along $b(t)$. On the noncentral fibre the covering is free, and the antipodal involution preserves the orientation of the vanishing 3-sphere. Thus the period on the quotient cycle $\sqrt{b(t)}\,S^3/\boldsymbol{\mu}_2$ is half its period on the sphere. The global volume form differs from the descended residue form by a holomorphic unit g, and the cycle contracts to the central vertex. Consequently its period is

$$b(t)\big(\tfrac{1}{2}c\,g(p,0)+o(1)\big),\qquad c\neq 0.$$

It is nonzero for all sufficiently small $t\neq 0$, for any vanishing order of $b(t)$. Lemma 3.6 identifies this cycle with a generator of $H_3(M_p;\mathbb{Q})$ and identifies the local boundary map with the ruling-difference line. The link has $H^1(L_p;\mathbb{Z})=0$ even though its fundamental group is $\mathbb{Z}/2$, as required in the preceding canonical-bundle argument.

For the degree-six and degree-seven branches in the lemma, the smooth affine fibre is $M=V\setminus E$, with $V=(\mathbb{P}^1)^3$ for degree six and $V=\mathrm{Bl}_{\mathrm{pt}}\,\mathbb{P}^3$ for degree seven, and $E\in|-\frac{1}{2}K_V|$. The comparison with a compact local Milnor fibre, including its boundary marking, is [Wue26b, Lem. 4.2]. Weak Lefschetz makes $H^2(V;\mathbb{Q})\to H^2(E;\mathbb{Q})$ injective; the respective Picard ranks are $(3,4)$ and $(2,3)$. Since $H^3(V;\mathbb{Q})=0$, the pair sequence gives $b_3(M)=1$. More explicitly, take the character $w=\chi^m$ with $m=(1,1,1)$ on $(\mathbb{P}^1)^3$, and $m=(1,1,-1)$ on the blowup fan with rays $e_1,e_2,e_3,-e_1-e_2-e_3,e_1+e_2+e_3$. The closure of $w=1$ is E, and

$$\Omega_M=\frac{w}{(w-1)^2}\frac{dx}{x}\wedge\frac{dy}{y}\wedge\frac{dz}{z}$$

has order zero at each toric boundary divisor: the order is $\langle m,v\rangle-2\min(\langle m,v\rangle,0)-1=0$, since $\langle m,v\rangle=\pm1$. Its only pole is the double pole at E, so it generates ω_M. In adjacent toric charts it is $\pm dp\wedge dq\wedge dz/(q-pz)^2$. The transitions are

$$(p',q',z')=(pz^c,qz^c,z^{-1}),\qquad c=0\text{ or }-1$$

in the two respective cases. Fix $\varepsilon>0$ and $R>1$. A closed three-cycle consists of the two caps

$$\{p=0,\ q=\varepsilon Re^{i\psi},\ |z|\le R\},\qquad\{p'=\varepsilon R^ce^{i\psi},\ q'=0,\ |z'|\le R^{-1}\}$$

and the intervening chain

$$p=r\varepsilon e^{i\theta},\quad q=-(1-r)\varepsilon Re^{i(\theta+\varphi)},\quad z=Re^{i\varphi},\qquad 0\le r\le 1.$$

The boundaries agree under the stated transition, with opposite orientations. On the chain $q-pz=-\varepsilon Re^{i(\theta+\varphi)}$, so it misses E; the caps also miss E. The form vanishes on the caps and pulls back on the chain to $\pm dr\wedge d\theta\wedge d\varphi$. Its period is consequently $\pm4\pi^2$, in particular nonzero.

The one-dimensional local smoothing is equivariant with positive weights on the cone coordinates and weight one on its parameter b. Graded Nakayama gives a homogeneous generator of its relative canonical sheaf near the vertex. Its nonvanishing locus is invariant and contains the vertex, hence contains the entire fibre $b = 1$: every orbit in that fibre contracts to the vertex. Its restriction is therefore a constant multiple of the preceding form. The multiplier is constant because an algebraic unit on $V \setminus E$ has divisor supported on E, and $\mathrm{Pic}(V)$ is torsion-free with $[E] \neq 0$. Scaling the fixed compact cycle into the fibre over $b(t)$ contracts it to the vertex. If the homogeneous form has weight a, its global period is

$$b(t)^a\big(c\,g(p,0)+o(1)\big), \qquad c \neq 0.$$

It is again nonzero for small $t \neq 0$. At a degree-five point, Lemma 3.4 provides five classes $\nu_{p,i}$, the images of its boundary classes β_i, with nonzero periods. Their dual link functionals are the five λ_i. Every one of these specified local cycles survives in $H_3(T_t;\mathbb{Q})$: otherwise it would bound a rational chain and have zero period against the closed form Ω_t.

By Poincaré duality choose, for each specified local cycle, a rational class $\gamma \in H_3(T_t;\mathbb{Q})$ pairing nontrivially with it. Its Mayer–Vietoris boundary belongs to

$$\ker\left\{\bigoplus_q H_2(L_q;\mathbb{Q}) \longrightarrow H_2(W;\mathbb{Q}) \oplus \bigoplus_q H_2(M_q;\mathbb{Q})\right\}.$$

It therefore gives a relation in $H_2(\widetilde{T};\mathbb{Q})$ and belongs at each point to the selected local subspace. Lefschetz duality identifies the local restriction of γ with its pairing against $H_3(M_p;\mathbb{Q})$. At a rank-one germ, the boundary map $H_3(M_p, L_p;\mathbb{Q}) \to H_2(L_p;\mathbb{Q})$ is an isomorphism onto the root line: its image is that line by the local Milnor-fibre calculation, and its domain has dimension one. The corresponding coefficient is therefore nonzero. At a degree-five point, (17) is an isomorphism onto the full link space. Compatibility of the boundary with the Lefschetz pairing identifies its pairing with β_i with the preceding nonzero pairing against $\nu_{p,i}$. By Lemma 3.4 this is a nonzero multiple of λ_i on the resulting local relation vector. Thus each required scalar functional is nonzero on some rational global relation. Avoiding the finitely many proper kernels gives one relation on which all of them are nonzero. ☐

4.7. **The exact criterion.**

Theorem 4.7. *The threefold X has a convergent one-parameter smoothing inducing S if and only if K_S contains a vector nonzero on every rank-one summand, satisfying $\prod_{i=1}^5 \lambda_i(\kappa_p) \neq 0$ at every degree-five point, and, at every A_2-assigned point, with*

$$\kappa_p = \mu_1 e_1 + \mu_2 e_2 + \mu_3 e_3, \qquad \mu_1+\mu_2+\mu_3 = 0, \qquad \mu_1\mu_2\mu_3 \neq 0.$$

Equivalently, none of these individual scalar functionals vanishes identically on K_S.

Proof. Suppose first that such a relation exists. It gives a vector of V_S by (33) and implies the nonzero-projection hypothesis of Theorem 4.5. Realize that tangent by a convergent curve in $\mathrm{Def}^S(X)$. At each rank-one point the local smoothing parameter has nonzero derivative. At each A_2 point all three local discriminant factors have nonzero derivatives, by the normalized identifications of Proposition 3.7. At each degree-five point the five discriminant factors have nonzero derivatives by Lemma 3.2. Thus the local fibres are smooth for every sufficiently small nonzero parameter. Properness and openness of smoothness exclude additional singularities outside neighborhoods of the finitely many original singularities. The curve therefore gives a smoothing of X.

Conversely, let $\mathcal{X} \to \Delta$ be any convergent proper smoothing inducing S. At each A_2 point pull back the simultaneous incidence modification of Proposition 3.8 along its actual analytic classifying arc. Since this arc is a smoothing, it meets the central point of the local base only at $t = 0$, after shrinking. The pulled-back modification is proper and flat over Δ, is an isomorphism off the central cone point, and is an isomorphism over every noncentral fibre. Glue these morphisms to the identity on the complement of those central points in $\mathcal{X}$. The Hausdorffness and properness argument of Proposition 4.1 applies to this cover of the total space; flatness is local on the source. We obtain a proper flat family $\mathcal{T} \to \Delta$ with $T_t \cong X_t$ for $t \neq 0$.

Its central fibre T replaces each A_2 point by three nodes and leaves all other germs unchanged. Its smooth crepant resolution is Y_0: the three new exceptional curves are e_1, e_2, e_3 in the original del Pezzo surface. Rationality and crepancy, as in (29), give $\omega_T \cong \mathcal{O}_T$ and $H^1(T, \mathcal{O}_T) = 0$. The remaining degree-six points use A_1. Lemma 4.6 therefore supplies a relation nonzero at every node and every remaining rank-one cone subspace, and with all five λ_i nonzero at each degree-five point. In particular, the three new nodes over any one A_2 point have nonzero coefficients μ_1, μ_2, μ_3.

Pair this relation with the original exceptional divisor E in Y_0. Every term belonging to another original singular point has zero intersection with E, whereas $E \cdot e_i = K_E \cdot e_i = -1$. Hence $\mu_1 + \mu_2 + \mu_3 = 0$. Replacing the three nodal terms by $\sum_i \mu_i e_i$ reconstructs a vector in K_S with all the required nonvanishing conditions. This argument uses the entire classifying arc; it imposes no assumption on its first nonzero order and excludes an escape from a discriminant line at any higher order. Finally, a finite-dimensional rational vector space is not a union of finitely many proper linear subspaces, which proves the last equivalence. □

In particular, when no degree-six points occur, every selected base is smooth, even if a coordinate at a node, a degree-seven point, or a $\mathbb{P}^1 \times \mathbb{P}^1$ cone vanishes identically on its tangent image. Smoothness of a deformation base and existence of a smooth fibre are distinct conclusions. For a degree-six point assigned A_2, a nonzero local vector alone does not ensure a smoothing: each of its three scalar coefficients is required by Theorem 4.7.

If the singularities are only nodes and anticanonical cones over dP_5 or $\mathbb{P}^1 \times \mathbb{P}^1$, the selected local bases are the entire local bases. Thus $\mathrm{Def}^S(X) = \mathrm{Def}(X)$ is smooth, of dimension $h^1(X, \Theta_X) + \dim_{\mathbb{Q}} K$. Existence of a smoothing still requires the nonzero relation of Theorem 4.7.

5. A nonsmoothable threefold and its deformation space

The three separate nonvanishing conditions on an A_2 summand cannot be replaced by nonvanishing of the vector in that summand. The example below also shows that changing the choices of local components does not repair this weaker condition.

Theorem 5.1. *There is a normal projective Calabi–Yau threefold X with exactly thirty ordinary double points and two anticanonical dP_6 cone points such that K_S contains a relation nonzero at every singular point for each of the two mixed profiles $S = LP, PL$, but X has no smoothing. A smooth projective crepant resolution Y_0 has*

$$(h^{1,1}(Y_0), h^{2,1}(Y_0)) = (23, 21).$$

Here L denotes the A_1 line and P the A_2 plane, in the order of the two cone points. We prove the nonsmoothability assertion directly from nodal necessity and the simultaneous partial resolution of the A_2 family. Its proof is independent of the sufficiency assertion in Theorem 4.7.

5.1. **The threefold and its relation matrix.** Let $P \subset \mathbb{Z}^4 \otimes \mathbb{R}$ be the convex hull of the following vertices. All indices in this section start at zero.

(35)

i	v_i	i	v_i	i	v_i
0	$(1,0,0,0)$	1	$(0,1,0,0)$	2	$(0,0,1,0)$
3	$(0,0,0,1)$	4	$(1,-1,1,-1)$	5	$(0,0,1,-1)$
6	$(1,-1,0,0)$	7	$(-1,1,0,0)$	8	$(0,0,-1,1)$
9	$(-1,1,-1,1)$	10	$(0,0,0,-1)$	11	$(0,0,-1,0)$
12	$(-1,1,-1,0)$	13	$(0,-1,1,0)$	14	$(-1,0,1,0)$
15	$(0,-1,0,1)$	16	$(-1,0,0,1)$	17	$(-1,0,1,-1)$
18	$(0,-1,1,-1)$	19	$(-1,0,-1,1)$	20	$(0,-1,-1,1)$
21	$(0,-1,-1,0)$	22	$(0,-1,0,-1)$	23	$(-1,0,0,-1)$
24	$(-1,0,-1,0)$				

This is the fan polytope: the ambient fan consists of the cones over its proper faces, and the anticanonical Newton polytope is P°. The polytope P is reflexive. It has 23 facets, 71 primitive edges, and 69 two-faces: 37 unimodular triangles, 30 unit parallelograms, and two unit-edge hexagons. The latter have cyclic orders

$$F_0 = (13, 15, 20, 21, 22, 18), \qquad F_1 = (14, 16, 19, 24, 23, 17) \tag{36}$$

and centres $(0,-1,0,0)$ and $(-1,0,0,0)$. These two centres and the origin are the only lattice points of P other than its vertices. Every parallelogram and hexagon has dual edge of lattice length one. The complete list of parallelograms, in the cyclic orders used below, is

(37)

j	node face	j	node face	j	node face
0	$(0,1,5,4)$	1	$(0,1,9,8)$	2	$(0,1,12,11)$
3	$(0,3,15,6)$	4	$(0,6,20,8)$	5	$(0,6,21,11)$
6	$(1,2,14,7)$	7	$(1,5,17,7)$	8	$(2,3,15,13)$
9	$(2,3,16,14)$	10	$(2,5,17,14)$	11	$(3,8,20,15)$
12	$(4,5,17,18)$	13	$(4,6,15,13)$	14	$(4,6,21,22)$
15	$(5,10,23,17)$	16	$(7,9,16,14)$	17	$(7,12,23,17)$
18	$(8,9,12,11)$	19	$(8,9,19,20)$	20	$(8,11,21,20)$
21	$(9,12,24,19)$	22	$(10,11,21,22)$	23	$(11,12,24,21)$
24	$(13,14,16,15)$	25	$(13,14,17,18)$	26	$(15,16,19,20)$
27	$(17,18,22,23)$	28	$(19,20,21,24)$	29	$(21,22,23,24)$

All these lattice assertions follow by computing the supporting hyperplanes and their face incidences from (35); the listed singular faces also specify the local lattice tests. In particular, each parallelogram has primitive three-dimensional cone lattice, and each triangle in the central subdivision of a hexagon is unimodular.

Let X be a general anticanonical hypersurface in the projective Gorenstein toric Fano fourfold defined by this fan. Nondegeneracy and the dual-edge lengths give one node for each face in (37) and one anticanonical dP_6 cone for each face in (36). There are no other singularities. The hypersurface avoids the torus fixed points, and

$$\omega_X \cong \mathcal{O}_X, \qquad H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0.$$

Insert the two centre rays and choose a projective crepant simplicial refinement retaining the central subdivisions of the hexagons. Its restriction gives a smooth projective crepant resolution $f : Y_0 \to X$; the remaining possible ambient singularities lie over torus fixed points and do not meet Y_0. These are the standard hypersurface and resolution constructions of [Bat94, §§4.1–4.2]. This choice selects a small resolution at each node; independent arbitrary choices of all thirty diagonals are not needed.

Write $E_i \cong \mathrm{Bl}_3\,\mathbb{P}^2$ for the exceptional divisor over F_i. In each of the cyclic orders (36), mark its boundary curves by

$$(C_0, \dots, C_5) = (e_1, h - e_1 - e_2, e_2, h - e_2 - e_3, e_3, h - e_1 - e_3). \tag{38}$$

The intersection form on (h, e_1, e_2, e_3) is $\mathrm{diag}(1, -1, -1, -1)$. With the root basis fixed in Section 2,

$$\ell = -h + e_1 + e_2 + e_3, \qquad A = -e_1 + e_3, \qquad B = e_2 - e_3,$$

the boundary intersection rows are

$$\begin{aligned} r_\ell &= (-1, 1, -1, 1, -1, 1), \\ r_A &= (1, -1, 0, 1, -1, 0), \\ r_B &= (0, 1, -1, 0, 1, -1). \end{aligned} \tag{39}$$

All three classes are perpendicular to K_{E_i}.

Define a 36×25 matrix M as follows. Its first thirty rows are indexed by (37): a cyclic square (i, j, k, l) gives entries $(1, -1, 1, -1)$ in those columns and zero elsewhere. These rows fix signed generators γ_j of the corresponding nodal link homology; replacing a small resolution reverses the relevant sign. The last six rows are r_ℓ, r_A, r_B on F_0, followed by r_ℓ, r_A, r_B on F_1, placed in the six vertex columns in the orders (36). Equations (37)–(39) specify every entry of M. The two additional columns for the centre rays are zero, because the root classes pair trivially with K_{E_i}.

The ambient divisor intersections detect precisely the homological relations. Indeed, P has 28 lattice points and no facet-interior lattice points. The only two-faces with interior points are the two hexagons, and their dual edges have no interior lattice points. Batyrev's divisor formula therefore has zero correction term and gives

$$h^{1,1}(Y_0) = 28 - 5 = 23. \tag{40}$$

The same construction of the divisor classes shows that the 27 ambient boundary divisors span $H^2(Y_0, \mathbb{Q})$ [Bat94, Theorem 4.4.2 and Corollary 4.5.1]. Thus, writing a tuple of local classes as

$$w = (n_0, \dots, n_{29}, \lambda_0, a_0, b_0, \lambda_1, a_1, b_1),$$

its total class

$$\sum_{j=0}^{29} n_j \gamma_j + \sum_{i=0}^{1} (\lambda_i \ell_i + a_i A_i + b_i B_i) \tag{41}$$

vanishes in $H_2(Y_0, \mathbb{Q})$ if and only if $wM = 0$. Consequently

$$K = \{w \in \mathbb{Q}^{36} : wM = 0\}$$

is the full homological relation space. Its restriction to a profile S is K_S. Whenever these coefficients are used as tangent coordinates, they are normalized by the fixed global volume form and Lemma 2.3; independent choices of raw local equation parameters are not implicit in M.

5.2. **A nonzero relation and the obstruction to smooth fibres.** For the profile LP, take

$$
\begin{aligned}
(\lambda_0, a_0, b_0, \lambda_1, a_1, b_1) &= (3, 0, 0, 0, 1, 0),\\
(n_0, \dots, n_{29}) &= (-2, -1, 2, 2, -2, 1, 1, -2, 2, 2,\\
&\quad -3, 6, -1, 1, -2, -4, -3, 2, 2, -4,\\
&\quad 5, -4, -4, 6, 3, 3, 1, 1, -1, 4).
\end{aligned} \tag{42}
$$

Multiplication by the explicitly defined matrix gives $wM = 0$. Every nodal coefficient is nonzero, and the two cone components are 3ℓ on E_0 and A on E_1, both nonzero. By (41), this is a homological relation, so the weaker nonvanishing condition holds. An explicit relation on PL has cone coordinates $(0, 1, 0, 3, 0, 0)$; its nodal coordinates agree with (42) for $0 \le j \le 24$ and are

$$(n_{25}, n_{26}, n_{27}, n_{28}, n_{29}) = (-1, 5, 2, -2, 7).$$

It too annihilates M and is nonzero at every singular point.

Let D_i be the ambient divisor associated with v_i, restricted to Y_0, and set

$$
\begin{aligned}
H_L &= -\sum_{i \in \{0,1,2,4,5,6,7,13,14,17,18\}} D_i,\\
H_M &= \sum_{i \in \{2,3,13,14,15,16\}} D_i,\\
H_P &= \sum_{i \in \{0,1,2,3,4,5,6,7,13,14,15,16,17,18\}} D_i.
\end{aligned} \tag{43}
$$

Pairing (41) with these three divisors gives, for every $w \in K$,

$$
\begin{aligned}
n_{13} - a_0 + b_0 - a_1 + b_1 &= 0,\\
b_0 + b_1 &= 0,\\
n_{16} + \lambda_0 + \lambda_1 &= 0.
\end{aligned} \tag{44}
$$

These equalities can be checked directly from the four-entry nodal rows and the six-entry rows (39). In particular they are identities on the full relation space, before any local component is selected. They imply the following restrictions:

$$n_{13}|_{K_{LL}} = 0, \qquad b_1|_{K_{LP}} = 0, \qquad b_0|_{K_{PL}} = 0, \qquad n_{16}|_{K_{PP}} = 0. \tag{45}$$

For a plane component the interpretation of b_i is intrinsic to the marking:

$$a_i A_i + b_i B_i = -a_i e_1 + b_i e_2 + (a_i - b_i) e_3. \tag{46}$$

It is the coefficient of the second of the three nodal curves in the simultaneous partial resolution.

Proof of the nonsmoothability assertion in Theorem 5.1. Suppose that X has a convergent one-parameter smoothing. At each cone point the classifying map from the disc factors through one of the two local components: in the domain $\mathbb{C}\{t\}$, the equations $\lambda_i a_i = \lambda_i b_i = 0$ force either $\lambda_i = 0$ or $a_i = b_i = 0$. Hence the smoothing has one of the four profiles.

We use the following form of nodal necessity. A smoothing and a chosen ordinary double point produce a homological relation on a crepant resolution whose coefficient at that node is nonzero; at the remaining cone points its components belong to the selected branch subspaces. To recall the argument, the local vanishing sphere has a nonzero period of the Calabi–Yau volume form, and so is nonzero in the homology of a smooth fibre. A dual global three-cycle has a boundary tuple on the links. The boundary exact sequence makes its total class zero on the resolution, and its component at the chosen node is nonzero. The boundary classes of the other local fillings give the stated branch restrictions. This is Friedman's nodal period argument [Fri86], with the cone boundary description of [Wue26b, Theorem 6.1]; it concerns the given smoothing, without a condition on its Kodaira–Spencer class.

For LL, apply this argument to node 13. The first identity in (44) forces its coefficient to be zero, a contradiction. For PP, apply it to node 16 and use the third identity.

For LP or PL, apply the simultaneous modification of Proposition 3.8 at the cone assigned to P. More explicitly, its local family consists of rank-at-most-one matrices with diagonal entries $z+d_1, z+d_2, z+d_3$. Remembering the column line and writing the matrix as $\xi\eta^{\mathsf{T}}$ gives, on $\xi_k = 1$,

$$\xi_i\eta_i - \xi_j\eta_j = d_i - d_j, \qquad \{i,j,k\} = \{1,2,3\}.$$

The modification is proper and flat over the two-dimensional base. Its central fibre replaces the cone by three nodes with small-resolution curves e_1, e_2, e_3. It is an isomorphism over every noncentral base point, since the zero matrix can occur only when all diagonal differences vanish. Pulling it back along the hypothetical smoothing and gluing to the identity off the central cone point therefore gives a proper flat family with the same smooth noncentral fibres. Its central threefold Z has the thirty old nodes, these three new nodes, and the remaining cone on its L branch. Crepancy and rationality give $\omega_Z \cong \mathcal{O}_Z$ and $H^1(Z, \mathcal{O}_Z) = 0$.

Apply nodal necessity to the new node represented by e_2. The resulting boundary relation on Y_0 has a nonzero coefficient at e_2. However, H_M pairs trivially with every old nodal curve and with the remaining A_1 root. On the modified exceptional surface,

$$H_M|_{E_i} = e_1 + (h - e_1 - e_2) = h - e_2, \qquad (H_M \cdot e_1, H_M \cdot e_2, H_M \cdot e_3) = (0,1,0).$$

Pairing the relation with H_M forces its e_2 coefficient to vanish. This contradiction excludes both mixed profiles. The argument applies also to smoothing arcs with arbitrarily high contact with the local discriminant. Together with (42), it proves the claimed failure of the weaker condition. The Hodge numbers are (40) and Lemma 5.2 below. □

5.3. **Tangent dimensions and the four selected deformation spaces.**

Lemma 5.2. *For this threefold,*

$$h^1(X, \Theta_X) = h^{2,1}(Y_0) = 21.$$

Proof. The polar P° has 26 lattice points and no facet-interior lattice points. Every edge of P is primitive, so the correction term in Batyrev's formula for $h^{2,1}$ vanishes. Thus

$$h^{2,1}(Y_0) = 26 - 5 = 21$$

by [Bat94, Corollary 4.5.1]. The counts may be obtained directly from $\langle m, v_i\rangle \geq -1$ using (35).

We verify the tangent-sheaf comparison needed for the first equality. A vector field at an isolated cone point vanishes at the vertex, and its local flow preserves the vertex ideal. It therefore lifts to the vertex blow-up $\mathrm{Tot}(K_E)$. At a node the local flow lifts to the blow-up and preserves each of the two ruling components of its exceptional quadric: a one-parameter flow cannot interchange these discrete choices. It consequently lifts to the chosen small resolution. Conversely a vector field on the resolution descends on the regular locus and extends across the point by normality, or equivalently by reflexivity of the tangent sheaf. Hence

$$f_*\Theta_{Y_0} = \Theta_X. \tag{47}$$

This is the equivariant-resolution comparison of [FL25, Definition 1.11 and Lemma 1.15] in these local models.

We also have $R^1 f_* \Theta_{Y_0} = 0$. At a degree-six point use the filtration by powers of the zero-section ideal. The tangent bundle restricts as $\Theta_E \oplus K_E$, so its graded terms are

$$\Theta_E \otimes \mathcal{O}_E(-kK_E), \qquad K_E \otimes \mathcal{O}_E(-kK_E), \qquad k \geq 0. \tag{48}$$

The second term has vanishing H^1: use Serre duality and rationality for $k = 0$, rationality for $k = 1$, and Kodaira vanishing for $k \geq 2$. For the first term, twist the toric Euler sequence

$$0 \longrightarrow \mathcal{O}_E^{\oplus 4} \longrightarrow \bigoplus_{j=0}^{5} \mathcal{O}_E(C_j) \longrightarrow \Theta_E \longrightarrow 0.$$

At $k = 0$, the sequence $0 \to \mathcal{O}_E \to \mathcal{O}_E(C_j) \to \mathcal{O}_{C_j}(-1) \to 0$ gives $H^1(E, \mathcal{O}_E(C_j)) = 0$. For $k \geq 1$, the divisor $C_j - kK_E$ is nef: its intersections with the six generators (38) of the effective cone of curves are nonnegative, since $C_j^2 = -1$ and $(-K_E) \cdot C_l = 1$ for every l. Toric vanishing gives $H^1(E, \mathcal{O}_E(C_j - kK_E)) = 0$. Finally $H^2(E, \mathcal{O}_E(-kK_E)) = 0$ for every $k \geq 0$, so the twisted Euler sequence proves the other H^1 vanishing in (48).

At a nodal small resolution the exceptional curve has normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$, and the corresponding graded tangent terms are sums of $\mathcal{O}_{\mathbb{P}^1}(k+2)$ and $\mathcal{O}_{\mathbb{P}^1}(k-1)$ for $k \geq 0$. Their H^1 groups also vanish. Induction on the infinitesimal neighbourhoods of the exceptional sets and the theorem on formal functions now give $R^1 f_* \Theta_{Y_0} = 0$; these ideal filtrations are cofinal with the pullbacks of the maximal-ideal filtrations at the singular points. Leray, (47), and contraction with the crepant volume form yield

$$H^1(X, \Theta_X) \cong H^1(Y_0, \Theta_{Y_0}) \cong H^1(Y_0, \Omega^2_{Y_0}),$$

which proves the assertion. □

Row reduction of the matrix specified above gives

$$\operatorname{rank} M = 20, \qquad \dim K = 16, \qquad \operatorname{rank} M_{\text{nodes}} = 19. \tag{49}$$

Here M_{nodes} consists of the first thirty rows. Its relation space has dimension 11, so projection of K to the six cone coordinates has dimension $16 - 11 = 5$. The second identity in (44) therefore describes its entire image:

$$\operatorname{im}(K \longrightarrow \mathbb{Q}^6) = \{(\lambda_0, a_0, b_0, \lambda_1, a_1, b_1) : b_0 + b_1 = 0\}. \tag{50}$$

For a profile S, let M_S consist of the nodal rows and the selected root rows, and let $\rho_i : K_S \to R_{i,S_i}$ be its projection to the ith cone summand. The ranks and resulting dimensions are In particular both cone projections are nonzero on every profile. Theorem 4.5 applies and

Table 2. Relation spaces and selected deformation spaces of X.

S	rows of M_S	rank M_S	$\dim K_S$	rank ρ_0	rank ρ_1	$\dim \operatorname{Def}^S(X)$
LL	32	19	13	1	1	34
LP	33	20	13	1	1	34
PL	33	20	13	1	1	34
PP	34	20	14	2	2	35

proves that all four selected deformation spaces are smooth, with the dimensions in the final column. These dimensions use Lemma 5.2 and the exact sequence of Theorem 2.5.

For the unrestricted deformation space the same theorem gives

$$0 \longrightarrow H^1(X,\Theta_X) \longrightarrow T_0 \operatorname{Def}(X) \longrightarrow K \otimes_{\mathbb{Q}} \mathbb{C} \longrightarrow 0. \tag{51}$$

Thus $\dim T_0 \operatorname{Def}(X) = 37$. The kernel of its projection to the two cone tangent spaces has dimension $21 + 11 = 32$. The second summand in this count consists of global tangent directions whose local components are supported at the old nodes.

5.4. The unrestricted analytic deformation germ.

Theorem 5.3. *For the threefold of Theorem 5.1 there is an isomorphism of convergent local rings*

$$\mathcal{O}_{\operatorname{Def}(X),0} \cong \frac{\mathbb{C}\{z_1,\dots,z_{32},x,y,u,v,b\}}{(xy,xb,uv,ub)}. \tag{52}$$

The completed local ring has the same presentation with formal power series. The four selected deformation spaces are precisely its irreducible components. They are smooth, and their ideals in the ambient smooth germ are

$$\begin{aligned} P_{LL} &= (y,v,b), \quad P_{LP} = (y,b,u),\\ P_{PL} &= (x,v,b), \quad P_{PP} = (x,u). \end{aligned} \tag{53}$$

In particular the full deformation germ is reduced, although its deformation functor is obstructed.

Proof. Choose coordinates on the two analytic local bases so that

$$\mathcal{O}_{\operatorname{Def}(X,p_i),0} = \mathbb{C}\{\lambda_i,a_i,b_i\}/(\lambda_i a_i, \lambda_i b_i) \tag{54}$$

and their differentials are the Hodge-normalized coordinates used in M. This normalization is possible by an invertible linear change on the line and on the plane separately; it preserves the displayed ideal. Fix the local classifying maps used to define $\operatorname{Def}^S(X)$.

By (50) and (51), the pulled-back functions

$$x = \lambda_0, \quad y = a_0, \quad u = \lambda_1, \quad v = a_1, \quad b = b_0$$

have independent differentials. Complete them to a minimal analytic embedding with 32 further coordinates $z_1,\dots,z_{32}$, and write

$$\mathcal{O}_{\operatorname{Def}(X),0} = R/I, \qquad R = \mathbb{C}\{z_1,\dots,z_{32},x,y,u,v,b\}, \qquad I \subseteq \mathfrak{m}^2,$$

where $\mathfrak{m}$ is the maximal ideal of R. Choose a representative $H \in R$ for the negative of the pullback of b_1. Equation (50) says exactly that $H = b + O(\mathfrak{m}^2)$. The local equations (54) imply

$$F := (xy, xb, uv, uH) \subseteq I. \tag{55}$$

We will prove equality and then remove the higher terms of H by a convergent change of coordinates.

Let $\operatorname{in}(I)$ denote the ideal of lowest nonzero homogeneous parts of elements of I in $\operatorname{gr}_{\mathfrak{m}} R = \mathbb{C}[z_1,\dots,z_{32},x,y,u,v,b]$. The four elements in (55) give

$$J := (xy,xb,uv,ub) \subseteq \operatorname{in}(I).$$

On the other hand, each of the four selected germs is a smooth closed subgerm of $\operatorname{Def}(X)$, by Table 2 and Theorem 4.5. Its tangent space has the corresponding ideal in (53); this follows from (51) and the linear relation $b_1 = -b_0$. If $g \in I$, substitute a regular local parametrization of one of these smooth germs into g. Since the result vanishes, the lowest homogeneous part of g vanishes on its tangent space. Therefore

$$\mathrm{in}(I) \subseteq P_{LL} \cap P_{LP} \cap P_{PL} \cap P_{PP}.$$

The elementary monomial-ideal identity

$$(y, v, b) \cap (y, b, u) \cap (x, v, b) \cap (x, u) = (xy, xb, uv, ub) \tag{56}$$

now proves

$$\mathrm{in}(I) = J. \tag{57}$$

For example, (56) can be checked monomial by monomial: membership in all four coordinate primes forces divisibility by at least one of xy, xb, uv, ub, and each of those four monomials belongs to every prime.

Since J is an intersection of prime ideals, the associated graded ring of R/I is reduced. Its maximal-ideal filtration is separated by Krull's intersection theorem. A nonzero nilpotent would have a nonzero initial form, and a power of that initial form would be zero, which is impossible. The same argument applies to the completion, whose associated graded ring is unchanged. Thus both local rings are reduced. This step uses the full equality (57); smoothness of the four selected germs by itself would not exclude an additional nilpotent structure.

Equation (57) also proves $I = F$ in the convergent ring. Indeed, for $g \in I$, express its lowest homogeneous part as a homogeneous polynomial combination of xy, xb, uv, ub. Subtract the same polynomial combination of xy, xb, uv, uH. The remainder is in I and has strictly higher order. For every integer N a finite iteration therefore gives $g \in F + \mathfrak{m}^N$. Krull intersection in the Noetherian local ring R/F implies $g \in F$. Hence

$$I = (xy, xb, uv, uH). \tag{58}$$

No convergence of an infinite sequence of correcting coefficients is needed for this ideal equality.

It remains to straighten H. The scheme-theoretic LL restriction imposes $y = b = v = H = 0$. By (58), its local ring is

$$R/(y, b, v, H) \cong \mathbb{C}\{z_1, \ldots, z_{32}, x, u\}/(H|_{y=b=v=0}).$$

This selected germ has dimension 34, equal to the dimension of the convergent power-series ring on the right. That ring is a domain, so the displayed restricting function must vanish identically. It follows that

$$H = cb + py + qv, \qquad c(0) = 1, \quad p(0) = q(0) = 0, \tag{59}$$

with convergent coefficient functions c, p, q. In particular c is a unit. Put $b' = cb+py = H-qv$. Its linear part is b, so the analytic inverse function theorem allows b' to replace b as an ambient coordinate, with the other 36 coordinates fixed. The identities

$$xb' = c(xb) + p(xy), \quad ub' = uH - q(uv),$$
$$c(xb) = xb' - p(xy), \qquad uH = ub' + q(uv)$$

give

$$(xy, xb, uv, uH) = (xy, xb', uv, ub').$$

Renaming b' as b proves (52) by a convergent coordinate change.

The change also identifies the scheme-theoretic profile restrictions. Their ideals before the change are

$$(y, b, v, H), \quad (y, b, u), \quad (x, v, H), \quad (x, u),$$

in the order LL, LP, PL, PP. Equation (59) and the invertibility of c send these ideals to those in (53). The four coordinate primes are pairwise incomparable, and (56) is their intersection. They are therefore exactly the irreducible components. Finally a tangent with $x = y = 1$ and $u = v = b = 0$ cannot extend to second order, because xy has a nonzero coefficient of t^2. Thus the full deformation functor is obstructed. □

The 32-dimensional smooth factor in (52) is not the locally trivial deformation space of X. Its tangent space is the kernel of localization at the two cone points; it contains the 21 locally trivial directions and 11 additional independent directions with local components at the old nodes. The final coordinate change has identity linear part, so this tangent interpretation is preserved.

Corollary 5.4. *All scheme-theoretic intersections of the components in Theorem 5.3 are smooth. The pairwise dimensions are*

pair	LL, LP	LL, PL	LL, PP	LP, PL	LP, PP	PL, PP
dimension	*33*	*33*	*32*	*32*	*33*	*33*

Every intersection of three or four distinct components is the same smooth 32*-dimensional germ* $x = y = u = v = b = 0$.

Proof. Add the corresponding coordinate ideals in (53). Each sum is generated by a subset of the five coordinates, giving the dimensions stated. No reduction of these intersection schemes is required. □

Corollary 5.5. *A global first-order deformation of* X *is the tangent of a convergent curve in* $\mathrm{Def}(X)$ *if and only if its five coordinates in* (52) *satisfy*

$$xy = xb = uv = ub = 0.$$

Equivalently, existence of a second-order extension of a tangent is sufficient for existence of a convergent deformation with that tangent. Each of the four components also contains a curve whose sufficiently small noncentral fibres have exactly one ordinary double point and no other singularities.

Proof. The quadratic equations give the necessity for a tangent by taking the coefficients of t^2. Conversely, a vector satisfying them belongs to one of the four linear components in (53). Its straight line in the convergent coordinates of (52) integrates it.

For the last assertion, on each K_S the coefficient in (45) is the only identically vanishing functional among the old nodal coefficients, the selected line coefficients, and the three separate plane coefficients (46). This follows by row reduction of M_S using the printed rows. Avoiding the kernels of the other finitely many functionals gives a tangent on the smooth selected germ with all these other coefficients nonzero. Integrate it there. Every other local singularity then smooths for sufficiently small nonzero parameter. On LL the remaining point is old node 13, and on PP it is old node 16. On LP or PL the remaining plane parameter is noncentral; its only possible singularity is the ordinary node on the corresponding discriminant line, as in Proposition 3.8. Were the final point also to smooth, the resulting curve would give a smoothing of X, contrary to Theorem 5.1. Compactness and openness of smoothness exclude additional singularities away from the original ones. □

The assertion about a prescribed tangent does not say that every prescribed higher jet extends. For example, $x = y = t^2$ and $u = v = b = z_1 = \cdots = z_{32} = 0$ gives a deformation over $\mathbb{C}[t]/(t^4)$ which has no extension with those coefficients over $\mathbb{C}[t]/(t^5)$, since $xy = t^4$. Its zero tangent nevertheless integrates. Likewise, the four minimal equations in (52) do not assert that $\mathrm{Ext}^2(\Omega^1_X, \mathcal{O}_X)$ has dimension four.

The finite calculations used in this section are the face and lattice counts for (35), the divisor intersections (44), and the ranks of the matrices specified by (37)–(39). Reproduction files and exact rational outputs are supplied with the paper.[1] The analytic ideal equality and coordinate change are proved above; they do not follow from a finite truncation of the deformation equations.

6. Germs from reflexive polytopes

We fix notation and recall the dictionary between 2-faces of a reflexive polytope and the singularities of its Calabi–Yau hypersurface. Everything in this section is standard; [Bat94] and [CLS11] are the references.

For the toric applications write $\widehat{X} = Y_0$ for the standard smooth crepant resolution and retain the link and branch notation of Section 2. We recall the earlier rank-one necessity statement because it suffices for the small-polytope classification. Theorem 4.7 supplies the stronger criterion at A_2 points.

Theorem 6.1 (mixed necessity; [Wue26b, Thm. 6.1])**.** *Let X be a compact complex threefold with $\omega_X \cong \mathcal{O}_X$ and $h^1(\mathcal{O}_X) = 0$ whose singular points are ordinary double points and anticanonical cones over dP_6 or dP_7, and suppose X admits a one-parameter smoothing. Set*

$$K(X) = \Big\{\kappa \in \ker\Big(\bigoplus_p H_2(L_p;\mathbb{Q}) \to H_2(\widehat{X};\mathbb{Q})\Big) \;:\; \kappa_p \in \mathcal{R}_p \text{ at every cone point } p\Big\}.$$

Then for every node q there is $\kappa \in K(X)$ with $\kappa_q \neq 0$; for every dP_7 point q there is $\kappa \in K(X)$ whose q-component is a nonzero element of the (-2)-line $\mathcal{R}_q$; and for every dP_6 point q smoothed through the one-parameter component there is $\kappa \in K(X)$ whose q-component is a nonzero element of the A_1 summand $\mathcal{R}_q$.

The step from "no one-parameter smoothing" to "not smoothable" is also the companion's, and we quote it because Theorem 10.3 rests on it.

Theorem 6.2 ([Wue26b, Cor. 6.7])**.** *Let X be as in Theorem 6.1 and admit no one-parameter smoothing. Then no fibre of a small representative of the miniversal deformation of X is smooth; consequently X is not smoothable over any base germ, and neither is any small deformation of X. Likewise no formal arc in the miniversal deformation meets all the local smoothing loci.*

The mechanism is that a smoothing over an arbitrary base germ would, by the curve selection lemma applied to the discriminant in the miniversal base, produce a one-parameter one; ruling out the one-parameter case therefore rules out all of them.

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[1] The geometric data and intersection calculations are in `a2_branch_selection_counterexample.py`; the Hodge and selected kernel calculations are in `branch_smoothness_inputs.py`; and the unrestricted kernel and ideal calculations are in `counterexample_deformation_germ.py`, all under `certificates/`. The independent standard-library calculation `a2_counterexample_rational_replay.py` reproduces the rational matrix identities and ranks.

We say the criterion *obstructs* X when some coordinate at which Theorem 6.1 asserts nonvanishing vanishes identically on $K(X)$, so that no such κ exists and X has no one-parameter smoothing; and that it is *silent* when none does. Theorem 4.7 strengthens this necessity statement to an equivalence by imposing all three scalar conditions at an A_2 point.

Convention 6.3. $N \cong \mathbb{Z}^4$ is a lattice, $M = \mathrm{Hom}(N, \mathbb{Z})$ its dual, $\langle \cdot, \cdot \rangle$ the pairing. $\Delta \subset N_{\mathbb{R}}$ is a *reflexive* polytope: a lattice polytope with the origin in its interior whose polar $\Delta^\circ = \{u \in M_{\mathbb{R}} : \langle u, v\rangle \geq -1 \ \forall v \in \Delta\}$ is again a lattice polytope. Σ is the face fan of Δ, whose cones are $\sigma_F = \mathrm{Cone}(F)$ for F a face, and $\mathbb{P}_\Delta = \mathrm{TV}(\Sigma)$ the associated toric fourfold, which is Gorenstein Fano. For a facet F we write $u_F \in M$ for the primitive inner normal, normalised by $\langle u_F, v\rangle = -1$ for $v \in F$; reflexivity is exactly the statement that u_F is a lattice point for every facet. $X \subset \mathbb{P}_\Delta$ denotes a Δ-regular anticanonical hypersurface in the sense of [Bat94, Def. 3.1.1]; it is a Calabi–Yau threefold with at worst Gorenstein canonical singularities.

6.1. **The dictionary.** Let G be a 2-face of Δ and $G^* \subset \Delta^\circ$ its dual edge. Write $\ell(G^*)$ for the lattice length of G^*. A Δ-regular X meets the closed orbit of σ_G, a curve in $\mathbb{P}_\Delta$, in exactly $\ell(G^*)$ points, and near each of them

$$(X, x) \ \cong \ \big(\mathrm{Cone}(S_G),\, 0\big), \tag{60}$$

the transverse affine toric threefold defined by σ_G in the saturated rank-three lattice $L_G = N \cap \mathrm{span}_{\mathbb{R}}(\sigma_G)$. The surface and its polarization are determined by this height-one cone. For a unit square this is the cone over $(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(1,1))$; for the pentagon and hexagon below it is the anticanonical cone over the indicated del Pezzo surface. Here the face polygon is a section of the ray cone, rather than the moment polygon of its polarization.

The germ (60) is a singular point of X precisely when G is not a unimodular triangle, and it is an *isolated* singular point precisely when G has no lattice points on the interiors of its edges, i.e. when every edge of G has lattice length one. We will only meet the following three:

G	S_G	germ
unit square	$\mathbb{P}^1 \times \mathbb{P}^1$	ordinary double point (node), $xy = zw$
unit-edge pentagon, one interior point	$\mathrm{dP}_7 = \mathrm{Bl}_2\mathbb{P}^2$	$\mathrm{Cone}(\mathrm{dP}_7)$
unit-edge hexagon, one interior point	$\mathrm{dP}_6 = \mathrm{Bl}_3\mathbb{P}^2$	$\mathrm{Cone}(\mathrm{dP}_6)$

A unit-edge pentagon with exactly one interior lattice point is reflexive after translating that point to the origin, and its face fan is the fan of dP_7; this is the sense in which the second row is a del Pezzo cone.

Definition 6.4. Δ lies in the *admissible framework* if every 2-face of Δ is either a unimodular triangle, or a unit square, or a pentagon or hexagon with one interior point; every edge of Δ has lattice length one; and every singular 2-face has $\ell(G^*) = 1$. For such a Δ, $\mathrm{Sing}(X)$ is a finite set with one point per singular 2-face, and every germ is a node or an anticanonical del Pezzo cone of degree 6 or 7.

This is the toric class studied in the remaining sections. Its germs are locally smoothable by the families described in Section 3 and [Alt97]. The general criterion of Theorem 4.7 also applies outside this toric framework.

6.2. **Nodes, coloops, and when the nodal mechanism is unavailable.** Let X have nodes $x_1, \ldots, x_k$, and let $\widehat{X} \to X$ be a small resolution of the nodes with exceptional curves $C_1, \ldots, C_k$. Following [Wue26b, §5.2] we call the matroid of the vectors $[C_j]$ the *node relation system*, and a node x_j a *coloop* when $[C_j]$ lies in no circuit; equivalently, when every relation $\sum \lambda_i [C_i] = 0$ has $\lambda_j = 0$.

Remark 6.5. If some node is a coloop, no relation of the form (1) exists, and Friedman's criterion is vacuous: the nodal mechanism cannot smooth X by itself. The threefolds of interest here are exactly those where this happens and a del Pezzo cone point is also present, so that a smoothing, if it exists, must move both kinds of germ at once. We call such an X *genuinely mixed*.

Both X_{19} and X_9 are genuinely mixed: the node system of X_{19} has rank 12 with two coloops, and that of X_9 has rank 2 with both nodes coloops.

7. Deformations of the ambient by Minkowski decomposition

We now recall the mechanism. The reader who knows [IV12] can skim to Definition 7.3, which fixes the object the two theorems quantify over.

7.1. **Altmann's construction, affine case.** Let $\widetilde{N}$ be the lattice of the germ, unrelated to the ambient N of Convention 6.3, and $\widetilde{M}$ its dual. Let $\delta \subset \widetilde{N}_{\mathbb{R}}$ be a pointed cone and $r \in \widetilde{M}$ a degree. Altmann [Alt95, Alt97] showed that Minkowski decompositions

$$\delta_r := \delta \cap [r = 1] = \Delta_0 + \cdots + \Delta_l$$

into polyhedra with common tail cone $\sigma := \delta \cap [r = 0]$ give l-parameter deformations of the affine toric variety $\mathrm{TV}(\delta)$, with deformation parameters in degree $-r$. For an isolated Gorenstein toric threefold germ, T^1 is concentrated in the single degree $-R^*$ and has dimension $\#(\text{vertices}) - 3$ [Alt95, Thm. 6.5]; for the pentagon this is 2, and the versal base is a line with an embedded component [Alt95, (7.1.4)]. The pentagon has, up to translation, exactly one nontrivial decomposition into lattice polyhedra, namely

$$\text{pentagon} \;=\; \text{primitive segment} \;+\; \text{unimodular triangle}, \tag{61}$$

and the corresponding one-parameter deformation smooths $\mathrm{Cone}(\mathrm{dP}_7)$, because the cones over a primitive segment and over a unimodular triangle are both smooth.

7.2. **The non-affine case: complexity-one T-varieties.** Our ambient $\mathbb{P}_\Delta$ is complete and, for the polytopes we need, not simplicial, so the affine statement does not apply directly. Ilten–Vollmert [IV12] give the required globalisation, in the language of polyhedral divisors of Altmann–Hausen [AH06, AHS08].

Fix a primitive degree $R \in M$ and *downgrade* the torus action of $\mathbb{P}_\Delta$ along R: restrict from the full four-torus to the three-dimensional subtorus $\ker R$. By [IV12, Rem. 1.8] the resulting complexity-one T-variety is described by a divisorial fan on $\mathbb{P}^1$ whose two nontrivial *slices* are the cross-sections

$$\mathcal{S}^+ := \Sigma \cap [R = 1], \qquad \mathcal{S}^- := \Sigma \cap [R = -1], \tag{62}$$

polyhedral complexes in the rank-three lattice $R^\perp \cap N$ together with the tail fan $\Sigma \cap [R = 0]$. The maximal cells of $\mathcal{S}^\pm$ are the cross-sections $\sigma_F \cap [R = \pm 1]$ of the maximal cones; a cell is *bounded* exactly when R is positive (resp. negative) on every ray of σ_F: a point of $\sigma_F \cap [R = 1]$ is $\sum_i \lambda_i v_i$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i \langle R, v_i \rangle = 1$, so each $\lambda_i \leq \langle R, v_i \rangle^{-1}$ when all $\langle R, v_i \rangle > 0$, and the cell is empty when all are negative. We call the union $\mathcal{S} := \mathcal{S}^+ \sqcup \mathcal{S}^-$ the *slice complex* at R.

Definition 7.1 ([IV12, Def. 2.1, 2.2, 4.1])**.** A *Minkowski decomposition of the slice complex* assigns, separately on each of the subdivisions $\mathcal{S}^+$ and $\mathcal{S}^-$, to every cell C a decomposition $C = C^0 + C^1$ into polyhedra with the same tail cone as C, subject to

(D1) both summands carry the full tail cone, and $C^0 + C^1 = C$;

(D2) *admissibility*: for each lattice character in the dual of the tail cone, at most one of the corresponding faces of the summands fails to contain a lattice point;
(D3) if $C \cap D \neq \varnothing$, then $(C \cap D)^i = C^i \cap D^i$;
(D4) for nonempty collections of cells $\mathcal{J} \subseteq \mathcal{I}$ in the same subdivision and every $A \subseteq \{0,1\}$,

$$\sum_{i\in A} \bigcap_{C\in\mathcal{I}} C^i \prec \sum_{i\in A} \bigcap_{C\in\mathcal{J}} C^i.$$

The empty polyhedron is allowed as a face; the empty index set A is tautological.

The pairwise consequence of (D4) is the one that does the work in Theorem 9.2: take $\mathcal{J} = \{\lambda\}$, $\mathcal{I} = \{\kappa, \lambda\}$ and $A = \{i\}$ when $\kappa \prec \lambda$. Together with (D3), this says that the decomposition induced on a face is a face of the decomposition of the cell above it. In particular, if a cell admits only homothetic summands, then so does every face of it.

Theorem 7.2 ([IV12, Thm. 4.4])**.** *A Minkowski decomposition of the slice complex in the sense of Definition 7.1 determines a flat family $\pi\colon \mathcal{P} \to B$ over a curve germ B, with $\mathcal{P}_0 = \mathbb{P}_\Delta$ and with T-action on every fibre.*

Definition 7.3. The family of Theorem 7.2, for a fixed primitive $R \in M$ and a fixed decomposition of the slice complex at R, is the *single-degree family at R*. Its fibre over $b \in B$ is denoted $\mathcal{P}_b$, and the base parameter is written b throughout, the letter t being reserved for edge dilations.

Theorem 7.4 ([IV12, Cor. 2.12])**.** *Suppose the polyhedral divisor at a germ has* affine locus*. Then the singularities of the general fibre at that germ are the cones over the summands C^0, C^1. In particular the germ is smoothed if and only if each summand cone is smooth; for the bounded summands used below, this means a point or a unimodular simplex, in particular a primitive segment in dimension one.*

Theorem 7.5 ([IV12, Rem. 2.13])**.** *If a polyhedral divisor has* complete locus *and its associated affine T-variety is singular, no T-deformation obtained by decomposing that polyhedral divisor is a smoothing. Thus the general fibre is singular; the remark does not assert that its analytic germ is unchanged.*

The following two consequences pass from a singular ambient fibre to its anticanonical hypersurfaces.

Proposition 7.6 (Relative anticanonical ampleness)**.** *Let $\pi\colon \mathcal{P} \to B$ be a single-degree family of $\mathbb{P}_\Delta$, with Δ reflexive and B a curve germ. After shrinking B, the morphism π is proper and Gorenstein of relative dimension 4, and*

$$\mathcal{L} := \omega_{\mathcal{P}/B}^{-1}$$

is π-ample and satisfies $\mathcal{L}|_{\mathcal{P}_b} \cong \omega_{\mathcal{P}_b}^{-1}$ for every $b \in B$.

Proof. The divisorial fan of the special fibre is complete, so [HI13, Thm. 3.2 and Rem. 3.3] makes π proper; flatness is Theorem 7.2. The special fibre $\mathbb{P}_\Delta$ is Gorenstein. The fibrewise Gorenstein locus of a flat morphism of finite presentation is open [Stacks, Tag 0C09]; it contains the whole proper fibre $\mathcal{P}_0$, so properness permits us to shrink B until every fibre is Gorenstein. Thus π is Gorenstein, and relative duality gives the asserted invertible dualizing sheaf and its base-change isomorphisms [Con00]. Finally, $\mathcal{L}_0 \cong \mathcal{O}_{\mathbb{P}_\Delta}(-K_{\mathbb{P}_\Delta})$ is ample because Δ is reflexive. Relative ampleness is open for a line bundle under a proper morphism, so another shrinking makes $\mathcal{L}$ relatively ample [Stacks, Tag 0D2S]. $\square$

Lemma 7.7 (Ample sections and singular curves)**.** *Let Z be a normal projective variety, let L be ample, and suppose* $\mathrm{Sing}(Z)$ *contains a complete curve C. Every effective Cartier divisor $D \in |L|$ is singular.*

Proof. Pass to the normalization $\widetilde{C}$ of an irreducible component of C. If the section defining D vanishes identically on C, then D contains C. Otherwise it restricts to a nonzero section of the positive-degree line bundle $L|_{\widetilde{C}}$, and hence has a zero. In either case choose a closed point $x \in C \cap D$.

Put $A = \mathcal{O}_{Z,x}$ and let $g \in A$ define D. Since D is Cartier, g is a nonzerodivisor. If $A/(g)$ were regular, then A would be regular: indeed, if $g \notin \mathfrak{m}_A^2$, both embedding dimension and dimension rise by one from $A/(g)$ to A; while if $g \in \mathfrak{m}_A^2$, regularity of $A/(g)$ would give $\operatorname{edim} A = \dim(A/(g)) = \dim A - 1$, impossible. This contradicts $x \in \mathrm{Sing}(Z)$, so D is singular at x. □

For a cone σ_G in the face fan the dichotomy is combinatorial. The locus is affine when $\langle R, \cdot\rangle$ is nonnegative or nonpositive on the rays of σ_G (we then say σ_G is *one-sided* at R), and complete when σ_G has rays strictly on both sides of $R^\perp$. If R vanishes on the whole cone, neither slice acts on its chart. The following lemma gives the resulting persistence statement.

8. Persistent ambient singularities and Minkowski rigidity

Throughout this section Δ is in the admissible framework and G is a singular 2-face.

8.1. Degrees taking both signs on a cone.

Lemma 8.1 (Persistence for degrees taking both signs)**.** *Let $R \in M$ be primitive. If σ_G has rays strictly on both sides of $R^\perp$, then the corresponding affine chart of the general fibre of every single-degree family at R has positive-dimensional singular locus.*

Proof. The rays of σ_G are the vertices of G. If $\langle R, \cdot\rangle$ takes both signs on them, then in the downgrade along R the polyhedral divisor attached to σ_G has nonempty coefficients at both 0 and ∞, hence complete locus. By Definition 6.4 the germ is a node or a del Pezzo cone, in particular singular, so Theorem 7.5 applies.

It remains to justify the asserted dimension. The tail cone $\tau = \sigma_G \cap R^\perp$ has dimension 2 in the rank-three acting lattice $N \cap R^\perp$. Choose a nonzero character u annihilating τ. Both u and $-u$ lie in $\tau^\vee$. For the proper polyhedral divisor $\mathcal{D}_b$ of the general fibre, semiampleness gives $\deg \mathcal{D}_b(u), \deg \mathcal{D}_b(-u) \geq 0$. Coefficient by coefficient,

$$\min_{v \in (\mathcal{D}_b)_y} \langle u, v\rangle + \min_{v \in (\mathcal{D}_b)_y} \langle -u, v\rangle \leq 0,$$

so both degrees are zero, equality holds at every coefficient, and $\mathcal{D}_b(-u) = -\mathcal{D}_b(u)$. After multiplying u to clear denominators, this degree-zero semiample divisor on $\mathbb{P}^1$ is principal. The coordinate ring therefore contains a nonconstant homogeneous unit $f\chi^u$ and its inverse. No point can be fixed by the acting torus while this unit is nonzero there. The singular locus is torus-invariant, so every one of its nonempty orbits has positive dimension. □

Corollary 8.2. *If for every primitive $R \in M$ some singular 2-face of Δ is two-sided at R, then no relative anticanonical effective Cartier divisor in any single-degree family has smooth general fibre.*

Proof. By Lemma 8.1, the singular locus of the general ambient fibre has a positive-dimensional irreducible component. Its closure in the projective fibre is a positive-dimensional projective subvariety contained in the singular locus, and hence contains a complete curve. Apply Proposition 7.6 and Lemma 7.7. ☐

Remark 8.3. The quantifier in Corollary 8.2 must range over *all* primitive degrees, not over the degrees $R = u_F$ dual to a facet F. With the normalisation $\langle u_F, v\rangle = -1$ these are exactly the vertices of the polar polytope Δ°, and we call them *vertex degrees* for that reason; at such an R the cell of F is the unique maximal bounded cell of the level-(-1) slice, and is F itself, which is why we call it the *lattice facet*. On Δ_{19} the vertex degrees leave at most 13 of the 15 singular 2-faces one-sided, while a general primitive degree, for instance $R = (-3, 0, 0, 1)$, leaves all 15 one-sided. Thus the sign condition in Lemma 8.1 does not obstruct X_{19}; its ambient obstruction follows instead from Minkowski rigidity, as proved in Theorem 9.2. A test restricted to vertex degrees would miss this distinction.

8.2. **Restriction to the local Gorenstein degree.** Fix a unit-edge pentagonal 2-face G and put

$$L_G = N \cap \operatorname{span}_{\mathbb{R}}(\sigma_G), \qquad Y_G = \mathrm{TV}_{L_G}(\sigma_G).$$

Then L_G is saturated of rank 3, Y_G is the isolated Gorenstein threefold cone over dP_7, and a choice of splitting gives

$$U_{\sigma_G} \cong Y_G \times \mathbb{G}_m.$$

Let $R^*_G \in L^\vee_G$ be the Gorenstein degree, characterized by $\langle R^*_G, v\rangle = 1$ on every ray of σ_G.

Definition 8.4. If $r \in M$ is nonnegative and nonzero on σ_G, its *crosscut* is

$$Q_G(r) := \sigma_G \cap [r = 1].$$

It is a polyhedron with tail $\sigma_G \cap r^\perp$. When r is positive on all five rays it is the rational polygon $\operatorname{conv}\{v/\langle r, v\rangle : v \in V(G)\}$.

Lemma 8.5 (Restriction to the Gorenstein degree)**.** *Let $R \in M$ be primitive and suppose σ_G is one-sided at R but $R|_{L_G} \neq 0$. Choose $\epsilon \in \{1, -1\}$ so that $r = \epsilon R$ is nonnegative on σ_G. For every single-degree family at R:*

1. *if $r|_{L_G} \neq R^*_G$, the induced deformation of U_{σ_G} is trivial;*
2. *if $r|_{L_G} = R^*_G$, then $\langle r, v\rangle = 1$ for all $v \in V(G)$ and every admissible induced decomposition of $G = Q_G(r)$ is a decomposition into lattice polygons. If that decomposition is trivial, the induced deformation of U_{σ_G} is trivial.*

Thus a one-sided single-degree family can move the transverse dP_7 *cone only on the affine lattice*

$$\mathcal{A}_G := \{r \in M : \langle r, v\rangle = 1 \text{ for every } v \in V(G)\}.$$

Proof. The deformation datum lives in the cone $\sigma_G \subset (L_G)_{\mathbb{R}}$. After choosing the splitting $U_{\sigma_G} = Y_G \times \mathbb{G}_m$, the restriction of the Ilten–Vollmert construction to this chart is the Laurent extension of the homogeneous Altmann deformation of Y_G associated with the same crosscut [IV12, §3]. Its Kodaira–Spencer class in $T^1(Y_G)$ has degree $-r|_{L_G}$ [Alt95, (5.3)]. Altmann's theorem for an isolated three-dimensional Gorenstein toric singularity says that $T^1(Y_G)$ is concentrated in degree $-R^*_G$ [Alt95, Thm. (6.5)]. Hence the Kodaira–Spencer map is zero unless $r|_{L_G} = R^*_G$.

Zero Kodaira–Spencer class is enough here, although it is not enough for an arbitrary deformation. Altmann's classification of the transverse admissible deformation datum with zero Kodaira–Spencer map gives only types A and B [Alt95, Prop. (5.6)]. Type A yields a trivial family [Alt95, (5.7)]; and a nontrivial type B datum does not exist when the toric variety is smooth in codimension two [Alt95, (5.8)]. The transverse variety Y_G is smooth in codimension two because its singularity is isolated. Thus its Altmann family is trivial, and so is its Laurent extension to U_{σ_G}. This proves (1).

In the remaining degree the crosscut is G itself. Altmann's Gorenstein specialization says precisely that admissibility is equivalent to all summands being lattice polytopes [Alt95, Thm. (6.6)]. A trivial decomposition is type A, so (5.7) again makes the induced family trivial. □

8.3. **Locking: a combinatorial rigidity criterion.** The dimension of a Minkowski summand cone depends on the chosen cross-section. We use a rigidity criterion depending only on the face lattice.

Definition 8.6. A positive-dimensional polytope C is *Minkowski-rigid* if every Minkowski summand is a homothet of C up to translation. Equivalently, its *Minkowski summand cone* has dimension one. This cone is the solution cone of the *edge-cycle system*: one variable $t_e \geq 0$ for each edge, and for each 2-face with cyclic edge vectors $e_1, \dots, e_k$ the closing condition

$$\sum_{i=1}^{k} t_i e_i = 0. \tag{63}$$

The all-ones vector represents homothety and satisfies every closing condition. Thus the summand cone has dimension at least one. Only bounded positive-dimensional cross-sections are used below.

Remark 8.7. The dimension of the Minkowski summand cone is *not* an invariant of the cone σ_F: it depends on the degree. On Δ_{19}, facet 12 has summand cone of dimension 2 at the lattice facet and dimension 1 at other cells of σ_{12}. The summand dimension at one degree therefore does not determine rigidity at all degrees. The following criterion applies uniformly to all bounded cross-sections.

The reason such a criterion exists is elementary.

Lemma 8.8 (Cross-section)**.** *Let $\sigma \subset N_{\mathbb{R}}$ be a pointed cone of dimension n and let $R \in \operatorname{int} \sigma^\vee$. Then $C_R := \sigma \cap [R = 1]$ is a polytope of dimension $n-1$, its faces are the cross-sections of the nonzero faces of σ, and its face lattice is that of σ with the apex removed. In particular, for any two $R, R' \in \operatorname{int} \sigma^\vee$ the polytopes C_R and $C_{R'}$ are combinatorially isomorphic, by the isomorphism matching cross-sections of the same face.*

Proof. R is positive on $\sigma \setminus \{0\}$, so every ray meets $[R = 1]$ in exactly one point and C_R is bounded; the correspondence $\sigma' \mapsto \sigma' \cap [R = 1]$ between nonzero faces of σ and faces of C_R is inclusion-preserving in both directions. □

The edge *directions* of C_R do change with R: the vertex of C_R on the ray through v is $v/\langle R, v\rangle$, so an edge from v to v' has direction $v'/\langle R, v'\rangle - v/\langle R, v\rangle$. The point is that the forcing rules below never use the directions, only the following two facts, both of which hold in every cross-section:

(a) the edge vectors of a bounded 2-face sum to zero around the cycle;
(b) two consecutive edges of a convex polygon are linearly independent.

Proposition 8.9 (The forcing rules)**.** *Let C be a polyhedron, f a bounded 2-face of C with edge vectors $e_1, \dots, e_k$ in cyclic order, and $t_1, \dots, t_k$ a solution of the edge-cycle system.*

(1) ***Triangle.*** *If $k = 3$ then $t_1 = t_2 = t_3$.*
(2) ***Single.*** *If $t_i = \lambda$ for every i except one, say $i = k$, then $t_k = \lambda$.*

(3) ***Adjacent.*** *If* $t_i = \lambda$ *for every* i *except two* consecutive *indices, say* $i = 1, 2$*, then* $t_1 = t_2 = \lambda$.

None of the three refers to the lattice, to the degree, or to the edge directions beyond (b).

Proof. (1) By (a), $e_3 = -e_1 - e_2$, so (63) reads $(t_1 - t_3)e_1 + (t_2 - t_3)e_2 = 0$, and e_1, e_2 are independent by (b). (2) By (a), $\sum_{i<k} e_i = -e_k$, so (63) reads $(t_k - \lambda)e_k = 0$ and $e_k \neq 0$. (3) By (a), $\sum_{i\geq 3} e_i = -(e_1 + e_2)$, so (63) reads $(t_1 - \lambda)e_1 + (t_2 - \lambda)e_2 = 0$, and e_1, e_2 are consecutive, hence independent by (b). ☐

Remark 8.10. The hypothesis that the edges in (3) are consecutive is necessary. Two opposite edges of a quadrilateral may be parallel, and then the same computation gives only one equation for two unknowns. Applying rule (3) to opposite pairs would incorrectly imply rigidity of the cube; the cube is the Minkowski sum of three segments and has Minkowski summand cone of dimension three.

Definition 8.11. Let F be a 3-polytope and S a set of vertices of F. Maintain a partition of the edge set of F, initially into singletons, and call two edges *tied* when they lie in a common class. Using only those 2-faces all of whose vertices lie in S, apply the following three rules until none applies.

(T) *Triangle.* Merge the three edges of a triangular 2-face into one class.

(S) *Single edge.* If all but one edge of a 2-face are tied to one another, merge the remaining edge into their class.

(A) *Adjacent pair.* If all but two edges of a 2-face are tied to one another and those two are consecutive, merge both into their class.

Each application either changes nothing or decreases the number of classes, so the propagation terminates. Its outcome does not depend on the order in which the rules are applied: each rule's hypothesis is preserved under coarsening the partition, so the three rules define a monotone closure operator and the terminal partition is its least fixed point. Say that a set E of edges is *locked in* (F, S) if at termination all of E lies in a single class; that a facet F is *locked* if its whole edge set is locked in $(F, V(F))$; and that a 2-face G lying in the facets F_1, F_2 is *locked* if both F_1 and F_2 are locked. This last is hypothesis (L) of Theorem 9.1, and it is what the census of Section 10 counts.

The requirement in (S) and (A) that the other edges be tied *to one another*, and not merely each already forced, is what makes the rules valid: what they propagate is the value of a common dilation parameter, and two edges carrying forced but unequal parameters license nothing. A partition, rather than a set of forced edges, is what records that.

The propagation needs a seed, and (T) is the only rule that fires from singletons, so a 3-polytope with no triangular 2-face is never locked.

Proposition 8.12. *Let* F *be a facet of* Δ *and* $R \in M$ *with* $\langle R, \cdot\rangle > 0$ *on every vertex of* F*. If* F *is locked then the cell* $\sigma_F \cap [R = 1]$ *is Minkowski-rigid.*

Proof. By Lemma 8.8 the cell is a polytope combinatorially isomorphic to F, with the same triangular 2-faces and the same edge-adjacency. By Proposition 8.9 every step of the propagation is a valid deduction about any solution of the edge-cycle system, so all t_e are equal and the Minkowski summand cone has dimension one. ☐

Corollary 8.13. *Let* G *be a unit-edge pentagonal* 2*-face, let* $r \in \mathcal{A}_G$*, and let* F *be a facet containing* G*. If* r *is positive on every vertex of* F *and* F *is locked, then every decomposition of the slice complex induces the trivial lattice Minkowski decomposition on* G*. Consequently the induced deformation of* U_{σ_G} *is trivial.*

Proof. Proposition 8.12 makes $C = \sigma_F \cap [r = 1]$ Minkowski-rigid, so in a decomposition $C = C^0 + C^1$ we have $C^i = \lambda_i C + t_i$ with $\lambda_0 + \lambda_1 = 1$. Condition (D3) and the pairwise consequence of (D4) then give $G^i = \lambda_i G + t_i'$. Lemma 8.5 says that both G^i are lattice polytopes. Every edge of G is primitive, so the lattice length of a corresponding edge of G^i is λ_i; hence $\lambda_i \in \mathbb{Z}_{\geq 0}$. Their sum is one, so one summand is a translate of G and the other is a point. This is the trivial decomposition, and the last assertion is Lemma 8.5(2). ☐

8.4. **The sign lemma.** The affine lattice $\mathcal{A}_G$ is a line. The following determines how the two facets containing G behave along it; it is a statement about reflexive polytopes and does not involve deformation theory.

Lemma 8.14 (Sign lemma)**.** *Let Δ be reflexive, G a 2-face, and F_1, F_2 the two facets containing G. The annihilator of* $\operatorname{span}(G)$ *in M has rank one; let d generate it, normalised so that $u_{F_1} - u_{F_2} = c\, d$ with $c > 0$. Then*

$$d(w) \leq -\tfrac{1}{c} < 0 \ \text{ for } w \in V(F_1) \smallsetminus V(G), \qquad d(w') \geq \tfrac{1}{c} > 0 \ \text{ for } w' \in V(F_2) \smallsetminus V(G).$$

Proof. $\operatorname{span}(G)$ is three-dimensional, so its annihilator has rank one. Both u_{F_1} and u_{F_2} take the value -1 on G, so their difference kills $\operatorname{span}(G)$ and is an integer multiple of d; this defines $c \neq 0$ and fixes the sign of d. For $w \in V(F_1) \smallsetminus V(G)$ we have $u_{F_1}(w) = -1$, and $w \notin F_2$ with Δ reflexive gives $u_{F_2}(w) \geq 0$, whence $c\, d(w) = -1 - u_{F_2}(w) \leq -1$. Symmetrically $c\, d(w') = u_{F_1}(w') + 1 \geq 1$. ☐

Corollary 8.15. *Fix a pairing vector on G and let $R_\tau = R_0 + \tau\, d$ be the corresponding line of degrees. Then the cell of F_1 is bounded exactly for $\tau < T_1$ and the cell of F_2 exactly for $\tau > T_2$, for thresholds T_1, T_2 computed from R_0 and d by finitely many divisions. In particular the sign pattern of R_τ on $V(F_1) \cup V(F_2)$ takes at most $\#(V(F_1) \cup V(F_2)) + 1$ values along the line.*

9. The obstruction: a rigidity criterion, and X_{19}

We can now state the obstruction for a class of polytopes rather than for one.

Theorem 9.1. *Let Δ be a reflexive 4-polytope in the admissible framework and let G be a unit-edge pentagonal 2-face, with containing facets F_1, F_2. Suppose that*

- (L) *both F_1 and F_2 are locked;*
- (C) *for every $r \in \mathcal{A}_G$, the functional r is positive on every vertex of F_1 or on every vertex of F_2.*

Then, for every primitive $R \in M$, the general ambient fibre of every single-degree family at R has positive-dimensional singular locus. After shrinking the base, no relative anticanonical effective Cartier divisor in such a family has smooth general fibre.

The hypothesis is decidable: $\mathcal{A}_G$ is an affine lattice line, and condition (C) *is a finite comparison of the thresholds in Corollary 8.15.*

Proof. Let R be primitive. If σ_G is two-sided at R, Lemma 8.1 already gives a singular curve in the corresponding chart. If R vanishes on L_G, neither slice changes U_{σ_G}, so its singular locus remains $\operatorname{Sing}(Y_G) \times \mathbb{G}_m$.

It remains to consider the one-sided case with nonzero restriction. Choose $r = R$ or $-R$ nonnegative on σ_G. If $r|_{L_G} \neq R_G^*$, Lemma 8.5(1) makes the local family trivial. If $r|_{L_G} = R_G^*$, then $r \in \mathcal{A}_G$; by (C), one containing facet is positive at r, and by (L) it is locked. Corollary 8.13 again makes the local family trivial. Thus in every one-sided case the general fibre contains the singular curve $\operatorname{Sing}(Y_G) \times \mathbb{G}_m$ in this chart.

Its closure in the proper ambient fibre is a complete curve contained in the singular locus in the one-sided case. In the two-sided case, Lemma 8.1 gives a positive-dimensional singular subvariety; its projective closure contains a complete curve in the singular locus, as in Corollary 8.2. Proposition 7.6 and Lemma 7.7 now give the assertion about anticanonical Cartier divisors. □

9.1. X_{19}**.** Let

$$\begin{aligned}\Delta_{19} = \operatorname{conv}\big\{&(1,0,0,0),\ (0,1,0,0),\ (0,0,1,0),\ (0,0,-1,0),\ (0,-1,0,0),\ (1,-1,-1,0),\\ &(0,0,0,1),\ (-1,1,1,-1),\ (0,-1,0,1),\ (0,0,-1,1),\ (-1,1,0,-1),\ (-1,0,1,-1),\\ &(-1,0,1,0),\ (-1,1,0,0),\ (0,-1,-1,0),\ (0,-1,-1,1),\ (-1,0,0,-1),\\ &(-2,1,1,-1),\ (-1,0,0,1)\big\},\end{aligned} \tag{64}$$

and label the displayed vertices $v_0, \ldots, v_{18}$ in that order. This is a reflexive polytope with 19 vertices, 27 facets and 64 edges, all of lattice length one. It lies in the admissible framework: its singular 2-faces are fourteen unit squares and one pentagon $P = \{v_{14}, v_{15}, v_{16}, v_{17}, v_{18}\}$ with one interior point, each with dual edge of length one. So X_{19} has fourteen nodes and one dP_7 cone point, all isolated. The pentagon lies in exactly two facets, which we call F_{21} and F_{24} after their zero-based position in the ordering the certificates use. Their supporting equations $\langle u, \cdot\rangle = 1$ and vertex sets are

$$\begin{aligned} F_{21}:\ u &= (-1,0,-1,0), \quad \{v_3, v_9, v_{10}, v_{13}, v_{14}, v_{15}, v_{16}, v_{17}, v_{18}\},\\ F_{24}:\ u &= (-1,-1,0,0), \quad \{v_4, v_8, v_{11}, v_{12}, v_{14}, v_{15}, v_{16}, v_{17}, v_{18}\},\end{aligned}$$

and the facet F_{12} of Remark 8.7 is $u = (0,1,1,1)$ on $\{v_1, v_2, v_6, v_7, v_{12}, v_{13}, v_{17}, v_{18}\}$.

Theorem 9.2. *Δ_{19} satisfies the hypothesis of Theorem 9.1 at P. Consequently, for every primitive $R \in M$ the general fibre of every single-degree family of $\mathbb{P}_{\Delta_{19}}$ at R has a singular curve. No relative anticanonical effective Cartier divisor in such a single-degree ambient family, with central fibre X_{19}, has smooth general fibre.*

Proof. Both facets are locked: each has 9 vertices, 16 edges and nine 2-faces, of degrees $(3,3,3,3,3,4,4,4,5)$ summing to $32 = 2 \cdot 16$, and the propagation of Definition 8.11 closes on all 16 edges.

It remains to check (C), and here there is no enumeration. Solving $\langle r, v_i\rangle = 1$ for $i = 14, \ldots, 18$ gives exactly

$$\mathcal{A}_P = \{r_s = (-1, s, -1-s, 0) : s \in \mathbb{Z}\}.$$

The five pentagon vertices pair to 1. Each of the other four vertices of F_{21} pairs to $s+1$, and each of the other four vertices of F_{24} pairs to $-s$. Hence r_s is positive on every vertex of F_{21} when $s \geq 0$ and on every vertex of F_{24} when $s \leq -1$. These two ranges cover every integer s, proving (C). Theorem 9.1 completes the proof. □

Corollary 9.3. *The full semiuniversal deformation space of a Δ_{19}-regular anticanonical hypersurface X_{19} is smooth of dimension* 30*. The threefold X_{19} has a convergent smoothing.*

Proof. We give the finite relation calculation explicitly. Order its nodal faces by the following vertex sets, using (64):

0	1	3	5	0	2	4	5	6	8	9	15
7	10	11	16	1	6	13	18	1	7	13	17
2	6	12	18	2	7	12	17	3	9	10	13
4	8	11	12	3	9	14	15	3	10	14	16
4	8	14	15	4	11	14	16				

For each set take the primitive relation on its four rays whose first nonzero coefficient is positive, and extend it by zero on the other rays. Let N be the resulting 14×19 matrix. The degree-seven root has divisor row

$$r = (0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,-1,1,-1).$$

Exact row reduction gives

$$\operatorname{rank} N = 12, \qquad \operatorname{rank} \begin{pmatrix} N \\ r \end{pmatrix} = 12.$$

A relation in the latter matrix, nonzero in all fifteen coordinates, is

$$(-2,2,-1,1,1,-3,-2,4,-2,2,3,1,-1,-3,-1).$$

Replace r by a basis of the full two-dimensional relation space on the pentagon rays $v_{14}, \dots, v_{18}$. The resulting 16×19 matrix has rank 13. Thus both the selected and the full local relation kernels have dimension three; they are equal. These matrices are reconstructed from the printed vertices and face sets in the exact computation listed in Appendix B.

The rows detect homology relations. Indeed, the correction term for non-toric divisor classes on a smooth crepant resolution vanishes: every singular two-face has a dual edge of lattice length one and every other two-face is a unimodular triangle. Toric divisors therefore span H^2 by Batyrev's formula [Bat94]. The inserted divisor over the pentagon pairs to zero with $K_E^{\perp}$. Consequently the full link kernel K is already K_S, and the displayed vector proves smoothing by Theorem 4.7.

For the dimension, the polar has 32 lattice points, no facet-interior lattice points, and zero correction term in Batyrev's $h^{2,1}$ formula, since the original edges are primitive. Hence $h^{2,1}(Y_0) = 32 - 5 = 27$. One also has

$$f_{0*}\Theta_{Y_0} = \Theta_{X_{19}}, \qquad R^1 f_{0*}\Theta_{Y_0} = 0.$$

Vector fields lift to the vertex blow-up of a cone and to a chosen nodal small resolution. For the higher direct image, formal functions reduces the degree-seven assertion to

$$H^1(E, \Theta_E \otimes \mathcal{O}_E(-kK_E)) = H^1(E, \mathcal{O}_E((1-k)K_E)) = 0 \qquad (k \geq 0).$$

The first equality follows from $\Theta_E \cong \Omega^1_E \otimes \mathcal{O}_E(-K_E)$ and toric Bott vanishing. The second follows from rationality and Serre duality for $k = 0, 1$, and Kodaira vanishing for $k \geq 2$. At a node the graded terms are copies of $\mathcal{O}(k+2)$ and $\mathcal{O}(k-1)$ on $\mathbb{P}^1$, whose first cohomology vanishes. Leray and contraction with the volume form give $h^1(X_{19}, \Theta_{X_{19}}) = 27$. Theorem 2.5 now gives $\dim \mathbb{T}^1_{X_{19}} = 27 + 3 = 30$.

The construction in the proof of Theorem 4.5 gives a smooth formal source whose blow-down differential covers V_S. Here $V_S = \mathbb{T}^1_{X_{19}}$ because $K_S = K$. The same independent-coordinate argument therefore makes the *full* deformation ring regular, not only its selected quotient. Regularity of the completed analytic ring implies smoothness of the analytic germ, with the dimension just computed. $\square$

Theorem 9.2 and Corollary 9.3 thus concern different deformation constructions: X_{19} is smoothable, while every single-degree ambient family in the stated class retains a singular anticanonical fibre.

Remark 9.4. The affine line in the proof contains every one-sided degree that can move the transverse cone. This follows from Altmann's concentration of T^1 in the Gorenstein degree and his classification of deformation data with zero Kodaira–Spencer class. Smoothness of cones over summands alone does not impose the admissibility condition (D2), and therefore does not yield this necessary degree restriction.

10. A census of the facet rigidity condition

We apply the combinatorial criterion to the Kreuzer–Skarke classification [KS00]. The broad census tests locking, condition (L) of Theorem 9.1; it does not test the degree coverage condition (C). It also records rigidity of the containing lattice facets when the locking criterion is inconclusive. These counts concern facet rigidity. The separate smoothability classification in Section 10.3 uses the homological obstruction of [Wue26b] and the explicit family constructed below.

10.1. **Enumeration and rigidity classes.** For each unit-edge pentagonal 2-face with one interior lattice point, we consider its two containing facets. Definition 8.11 tests rigidity by propagation of equalities among edge dilations. For either facet where that criterion is inconclusive, the edge-cycle system determines its Minkowski summand cone directly. This gives three disjoint classes of pentagonal faces:

(1) *locked*: both containing facets satisfy Definition 8.11;
(2) *rigid, not locked*: both containing lattice facets are Minkowski-rigid, but at least one does not satisfy that definition;
(3) *decomposable*: at least one containing lattice facet has a nonhomothetic Minkowski summand.

The first condition implies rigidity for all bounded cross-sections of both facet cones. The second asserts rigidity only at their lattice cross-sections.

Table 3. Facet rigidity in the Kreuzer–Skarke classification. Each pentagonal face is counted once. "Locked" means both facets satisfy the edge-propagation criterion; "rigid, not locked" means both lattice facets are rigid but the criterion is inconclusive; "decomposable" means at least one has a nonhomothetic summand.

vertices	polytopes	pentagons	locked	rigid, not locked	decomposable
5	1 561	0	n/a	n/a	n/a
6	24 189	0	n/a	n/a	n/a
7	177 446	30	30	0	0
8	834 638	882	763	0	119
9	2 867 955	11 596	8 673	37	2 886
total	3 905 789	12 508	9 466	37	3 005

The three classes exhaust the pentagonal faces: $8\,673+37+2\,886 = 11\,596$ at nine vertices, and $9\,466 + 37 + 3\,005 = 12\,508$ in total.

Exactly 9 466 of the 12 508 pentagonal faces satisfy (L). Including the 37 cases where the criterion is inconclusive, 9 503 have rigid containing lattice facets. The latter is a statement about those specific cross-sections: Remark 8.7 explains why it does not imply rigidity at every degree. Neither number is an all-degree immovability count, because the broad census does not impose (C). The 77 polytopes of Section 10.3 are counted separately.

Remark 10.1. Definition 8.11 records a partition of the edges because rules (S) and (A) require equality of the other dilation parameters. Knowing only which parameters are constrained does not suffice.

For a construction at a lattice facet degree, a decomposing facet is necessary but not sufficient, because by Definition 7.1(D4) what matters is the decomposition *induced* on the pentagon, and by (61) that must be segment + triangle with all dilation parameters nonnegative across the facet. Of the 119 at eight vertices: all 119 induce a nontrivial decomposition of the pentagon, 49 attain the segment-plus-triangle datum with nonnegative dilations, and 0 lie in a polytope whose whole germ inventory lies in the framework of Definition 6.4.

Remark 10.2. A filter that classified a 2-face by its numbers of vertices and interior points alone would return two admissible polytopes at eight vertices instead of none, because it misses the lattice smoothness condition. A triangle with no interior point need not be smooth: it is smooth only if *unimodular*, and one with an edge of lattice length two gives a non-isolated germ. Exactly two of the 49 at eight vertices carry such a face.

10.2. **Nine vertices, and Δ_9.** At nine vertices the database contains 11 596 pentagonal 2-faces. Of these, 63 lie in a polytope inside the admissible framework, and 2 886 lie in a facet that decomposes. Exactly one does both, and it is the pentagon of Δ_9. Section 11 constructs a smoothing from the compatible Minkowski decomposition of this polytope.

Across all five vertex counts the number of pentagons in a framework polytope is 77, and the number lying in a decomposing facet is 3 005; the single polytope in both is Δ_9. Thus this range contains one candidate satisfying both combinatorial conditions.

10.3. **The searched range, classified.** Restricting the enumeration to polytopes that lie in the admissible framework and carry a pentagonal 2-face returns exactly 77 of them, one pentagon apiece: one at seven vertices, thirteen at eight, sixty-three at nine. The negative classification in [Wue26b, Prop. 9.2], together with the smoothing below, decides all 77 cases.

Theorem 10.3. *Let Δ be a reflexive 4-polytope with at most nine vertices lying in the admissible framework and carrying a pentagonal 2-face. There are 77 such Δ, and:*

(1) *for 76 of them the criterion of* [Wue26b] *obstructs, so X admits no one-parameter smoothing* [Wue26b, Prop. 9.2] *and hence, by* [Wue26b, Cor. 6.7]*, is* not smoothable *over any base germ; 67 of these have a single singular point, always a lone* dP_7 *cone point, and 9 have one node in addition;*
(2) *for the remaining one, which is Δ_9, the criterion is silent, and X_9* is *smoothable by Theorem 11.8;*
(3) *exactly 76 of the 77 pentagons are locked, and the unlocked one is Δ_9's.*

Proof. The classification contains one framework polytope at seven vertices, thirteen at eight, and sixty-three at nine, each with one pentagonal face. The nonsmoothability of the 76 cases in part (1), including their division into 67 single-cone and 9 node-plus-cone cases, is [Wue26b, Prop. 9.2]. Its Corollary 6.7 gives the assertion over an arbitrary base germ. Part (2) follows from Proposition 11.2 and Theorem 11.8. The edge partitions of Definition 8.11 identify precisely those 76 pentagons as locked, proving part (3). The exact enumeration, relation matrices and partitions are reproduced by the programs listed in Appendix B. □

Remark 10.4. The 76 nonsmoothability assertions in part (1) are proved in [Wue26b, Prop. 9.2]. The explicit family constructed here treats the remaining polytope and completes the classification. Its abstract smoothability also follows from Theorem 4.7. Part (3) is a separate combinatorial observation: locking agrees with the negative classification in this range. The agreement does not extend to X_{19}, which is locked and smoothable by Corollary 9.3.

11. An ambient smoothing of X_9

11.1. **The polytope and its germs.** Let Δ_9 be the convex hull of the following vertices:

$$\begin{aligned} &v_0=(1,0,0,0), && v_1=(0,1,0,0), && v_2=(-1,-1,0,0),\\ &v_3=(0,0,1,0), && v_4=(0,0,0,1), && v_5=(-4,-2,-1,0),\\ &v_6=(-4,-2,0,-1), && v_7=(3,1,1,1), && v_8=(2,1,1,1). \end{aligned}$$

It is reflexive, with 12 facets and 27 two-faces, and it lies in the admissible framework.

Proposition 11.1. $\operatorname{Sing}(X_9)$ *consists of exactly three points: two ordinary double points, at the* 2*-faces* $\{v_2,v_3,v_6,v_7\}$ *and* $\{v_2,v_4,v_5,v_7\}$*, and one* dP_7 *cone point, at the pentagon* $\{v_3,v_4,v_5,v_6,v_8\}$.

Proof. Every vertex of Δ_9 is primitive, so every ray of Σ is a smooth cone. Every edge has lattice length one and unit greatest common divisor of its 2×2 minors, so every two-dimensional cone is smooth; hence $\mathbb{P}_{\Delta_9}$ is smooth along the orbits of all cones of dimension ≤ 2. Of the 27 two-faces exactly the three listed have a singular cone, and each has dual edge of lattice length one, so contributes exactly one point. Finally, $\#(\Delta_9^\circ\cap M)=162$ and this set contains every vertex of Δ_9°, so a generic anticanonical section is nonzero at every torus-fixed point and X_9 misses the zero-dimensional strata. □

Proposition 11.2. *The node relation system of* X_9 *has rank* 2 *and both nodes are coloops. The criterion of* [Wue26b] *is silent on* X_9*: at the* dP_7 *point the branch subspace is a single line, since* $K_E^\perp$ *has discriminant* 7 *and carries a unique* (-2)*-class up to sign, so the branch restriction there is one linear condition; the branch-restricted kernel is one-dimensional, and no coordinate vanishes identically on it.*

Proof. The two nodal rows are independent. Adding the degree-seven root row gives a matrix with one-dimensional relation kernel, generated by $(-2,-2,2)$, with entries indexed by the two nodes and the (-2)-line at the cone point. Every entry is nonzero, so the criterion of [Wue26b] imposes no forced-zero condition. The vertex and ray-relation computations are reproduced in Appendix B. □

Thus X_9 is genuinely mixed in the sense of Remark 6.5. The displayed relation and Theorem 4.7 prove that it is smoothable. We now construct a smoothing in an ambient family.

11.2. **The family.** Take the vertex degree

$$R = u_{11} = (0,1,-1,-1), \qquad \langle R,v\rangle=-1 \text{ for every } v\in F_{11},$$

dual to the facet $F_{11}=\{v_2,v_3,v_4,v_5,v_6,v_7,v_8\}$, which contains all three singular 2-faces. Downgrading along R, the bounded cell of the level-(-1) slice is the lattice facet F_{11} itself.

We use the following integral dilation vector; its two summands have lattice vertices, which will give admissibility.

The Minkowski summand cone of F_{11} has dimension 2. We take the decomposition given by the dilation vector

$$t = (0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0)$$

on the eleven edges of F_{11}, listed in lexicographic order of their vertex index pairs as

$$(2, 5),\ (2, 6),\ (2, 7),\ (3, 6),\ (3, 7),\ (3, 8),\ (4, 5),\ (4, 7),\ (4, 8),\ (5, 6),\ (7, 8),$$

so the dilation is 1 on the three edges v_2v_7, v_3v_6, v_4v_5 and 0 on the other eight. Figure 1 shows F_{11}, its three singular 2-faces and the three dilated edges.

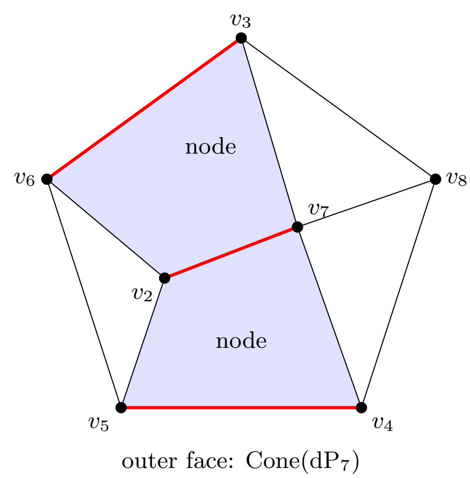

FIGURE 1. Schlegel diagram of the facet F_{11}, projected from beyond its pentagonal 2-face, which becomes the outer boundary. The two shaded faces are the squares $\{v_2, v_3, v_6, v_7\}$ and $\{v_2, v_4, v_5, v_7\}$ carrying the nodes; the third singular face is the pentagon $\{v_3, v_4, v_5, v_6, v_8\}$ carrying the dP$_7$ cone point, which the projection sends to the outer boundary. The three unshaded faces are triangles, and the germ there is smooth. The three heavy edges are those on which the dilation vector t is 1; it is 0 on the other eight.

A dilation vector determines a summand through the *displacement maps* φ_0, φ_1 on the vertices of F_{11}: fix the base vertex v_2, set $\varphi_0(v_2) = 0$, and along each edge v_av_b put

$$\varphi_0(v_b) = \varphi_0(v_a) + t_{ab}\,(v_b - v_a),$$

which is well defined precisely because t solves the edge-cycle system, that being what makes the value independent of the path from v_2. Put $\varphi_1(v_a) = (v_a - v_2) - \varphi_0(v_a)$. Then $\operatorname{conv}\varphi_0(V(F_{11})) + \operatorname{conv}\varphi_1(V(F_{11})) = F_{11} - v_2$, so the two maps exhibit the summands. Their restrictions to the vertices of each face, with its tail cone added, define the cellwise decomposition after the affine identification specified below. The three singular 2-faces inherit the following decompositions:

To specify the affine identifications and the gluing, use the saturated basis

$$(1, 0, 0, 0), \qquad (0, 1, 1, 0), \qquad (0, 1, 0, 1)$$

TABLE 4. The decomposition induced at each germ.

2-face	germ	induced decomposition
$\{v_2, v_3, v_6, v_7\}$	node	primitive segment + primitive segment
$\{v_2, v_4, v_5, v_7\}$	node	primitive segment + primitive segment
$\{v_3, v_4, v_5, v_6, v_8\}$	$\mathrm{Cone}(\mathrm{dP}_7)$	primitive segment + unimodular triangle

of $\ker R$, with splitting vector $w = v_1$. In these coordinates put

$$a_0 = (-1, 0, 0), \quad a_1 = (3, 1, 1),$$
$$b_0 = (-3, -1, 0), \quad b_1 = (-3, 0, -1), \quad b_2 = (-1, 0, 0), \quad b_3 = 0.$$

The two summands in the negative slice are the convex hulls of the images $A(v) = \varphi_0(v) + v_2 + w$ and $B(v) = \varphi_1(v)$, with the tail cone added. The vertex images, in the order $v_2, \ldots, v_8$, are

$$A: \ (a_0, a_1, a_1, a_0, a_0, a_1, a_1), \qquad B: \ (b_3, b_1, b_0, b_0, b_1, b_3, b_2).$$

The tail rays are

$$q_0 = (-4, -1, 0), \quad q_1 = (-4, 0, -1), \quad q_2 = (0, 0, 1),$$
$$q_3 = (0, 1, 0), \quad q_4 = (1, 0, 0), \quad q_5 = (2, 1, 1), \quad q_6 = (3, 1, 1).$$

Table 5 lists all maximal original charts. An index set in the A or B column means the convex hull of those vertices plus the indicated tail. The marking records complete locus. Write A_j, B_j for the summands in row j. All remaining cells are obtained from the nonzero face cones of the twelve listed facets by the same slice construction.

TABLE 5. The two displaced slices and their markings.

facet	vertices	tail rays	A	B	marked
0	0136	134	01	1	yes
1	0137	346	1	13	yes
2	0145	024	01	0	yes
3	0147	246	1	03	yes
4	0156	014	0	01	yes
5	02367	4	01	13	no
6	02457	4	01	03	no
7	0256	4	0	013	no
8	134568	01235	01	012	yes
9	1378	356	1	123	yes
10	1478	256	1	023	yes
11	2345678	$\varnothing$	01	0123	no

Proposition 11.3. *The displayed assignment is an admissible Minkowski decomposition of the two-sided slice complex in the sense of Definition 7.1.*

Proof. The summands in Table 5 have lattice vertices. Their minimizing faces therefore contain lattice points, proving (D2). Taking their sums and intersections verifies (D1) and (D3) on the 69 nonempty negative cells obtained from the displayed facet data.

For (D4), form the finite set of ordered pairs

$$\mathcal{I} = \left\{ \left(\bigcap_{C \in J} C^0, \bigcap_{C \in J} C^1 \right) : \varnothing \neq J \subseteq \mathcal{S}^- \right\}.$$

It is obtained from the 69 pairs (C^0, C^1) by closing under $(U, V) \mapsto (U \cap D^0, V \cap D^1)$ for each cell D. For every resulting pair and every D, the displayed rational polyhedra satisfy

$$U \cap D^0 \prec U, \qquad V \cap D^1 \prec V, \qquad (U \cap D^0) + (V \cap D^1) \prec U + V.$$

These three face relations suffice for arbitrary inclusions of collections by transitivity.

One case illustrates why disjoint original cells must be included. For facets 0 and 2, let $S = [a_0, a_1]$ and let $\ell(x, y, z) = y - z$. The functional is nonnegative on A_0 and nonpositive on A_2, with

$$A_0 \cap A_2 = S + \mathrm{pos}(q_4) = A_0 \cap \{\ell = 0\} = A_2 \cap \{\ell = 0\}.$$

On B_0 it is at least 1, whereas on B_2 it is at most -1. Thus $B_0 \cap B_2 = \varnothing$. The first intersection is a proper face and the other two required partial sums are empty faces, although the original cells are disjoint.

The positive slice uses $C = C + \mathrm{tail}(C)$; its tail summand contains zero in every minimizing face. The same face comparisons verify (D1)–(D4) there. The complete rational intersection enumeration is implemented in `slice_admissible.sage` from the vertex and dilation data printed here. It gives 86 distinct pairs and 5 934 extensions on the negative slice, and 29 pairs and 841 extensions on the positive slice. □

Corollary 11.4. *By Theorem 7.2 the decomposition determines a flat family $\pi\colon \mathcal{P} \to B$ with $\mathcal{P}_0 = \mathbb{P}_{\Delta_9}$, and by Theorem 7.4 the general fibre $\mathcal{P}_b$ is smooth at all three germs of X_9: the cones over a primitive segment and over a unimodular simplex are smooth.*

All three cones σ_G are one-sided at R with bounded cells, so Lemma 8.1 does not intervene and the affine-locus hypothesis of Theorem 7.4 is satisfied at each.

11.3. **The general ambient fibre.** The general fibre has a larger torus action than the one used to construct the family. We describe its fan explicitly; this also determines its singular locus.

Proposition 11.5. *For $b \neq 0$, the fibre $\mathcal{P}_b$ is the toric fourfold with the following primitive rays in $\mathbb{Z} \oplus \ker R$:*

$$\begin{aligned} &r_0 = (-1,-3,-1,0), && r_1 = (-1,-3,0,-1),\\ &r_2 = (-1,-1,0,0), && r_3 = (-1,0,0,0),\\ &r_4 = (0,1,0,0), && r_5 = (1,-1,0,0),\\ &r_6 = (1,3,1,1). \end{aligned} \tag{65}$$

Its maximal cones, written as sets of ray indices, are

$$\begin{matrix} 1456, & 1346, & 0456, & 0346, & 0145,\\ 0134, & 01256, & 1236, & 0236, & 0123. \end{matrix} \tag{66}$$

This is a Gorenstein Fano fourfold.

Proof. The vertex levels are $R(v_1) = 1$, $R(v_0) = 0$, and $R(v_i) = -1$ for $i \geq 2$. With splitting vector $w = v_1$, each nonempty positive coefficient is its tail cone, with vertex at zero. A face cone with empty positive coefficient has tail either zero or $\mathrm{pos}(v_0)$. Both already occur as positive coefficients, from $\mathrm{pos}(v_1)$ and $\mathrm{pos}(v_0, v_1)$ respectively. Thus the positive slice is the entire tail fan. Only the two displaced negative slices can be nontrivial. By [Sü08, Cor. 1.9], the three-dimensional torus action extends to a four-dimensional torus action.

Here is the fan, including the gluing data. Move the two nontrivial support points to 0 and ∞. For each marked row of Table 5, take the cone generated by $(1, A)$, $(-1, B)$ and $(0, \tau)$. For each unmarked row, take the two cones generated separately by $(1, A), (0, \tau)$ and $(-1, B), (0, \tau)$. Removing proper faces gives (65)–(66). The sections at first coordinate 1 and -1 recover the two subdivisions in the table. Their markings also agree: the first has eight maximal cells, all marked; the second has ten maximal cells, eight marked. The generic tail subdivision has eight maximal cells, all marked. Hence these are the marked subdivisions of the given fibre, which determine the variety [Sü08, Rem. 1.6 and Thm. 1.8].

For completeness, the anticanonical Cartier characters on the ten cones, in their listed order, are

$$\begin{gathered}(-2,-1,-2,6),\ (1,-1,-2,3),\ (-2,-1,6,-2),\ (1,-1,3,-2),\ (-2,-1,6,6),\\ (1,-1,3,3),\ (0,1,-2,-2),\ (1,0,-2,0),\ (1,0,0,-2),\ (1,0,0,0).\end{gathered} \tag{67}$$

Each pairs to -1 with the rays of its cone and strictly greater than -1 with every other ray. The seven inequalities $\langle m, r_i \rangle \geq -1$ define a bounded polytope with exactly these vertices. Consequently the displayed cones are its normal fan, and $-K$ is Cartier and ample. All the data needed for these finite comparisons are printed above and in Table 5; `toric_fibre.sage` reproduces the fan and marked subdivisions. □

Proposition 11.6. *The singular locus of $\mathcal{P}_b$, for $b \neq 0$, consists of one torus-fixed point.*

Proof. Nine of the maximal cones in (66) are unimodular. The remaining cone 01256 has five extremal rays in dimension four, with relation

$$r_0 + r_1 + r_6 = 2r_2 + r_5.$$

Its six facets have ray sets

$$012, \quad 015, \quad 026, \quad 126, \quad 056, \quad 156.$$

For each facet, the greatest common divisor of the maximal minors of its three primitive ray vectors is one. Thus every proper face is smooth, while the full cone is singular. The corresponding fixed point is therefore the only singular point of the fourfold. In the original cover, Table 5 has eight complete-locus charts and four affine-locus charts. The only singular full cone occurs in the complete-locus chart F_8; every lower-dimensional cone arising from an affine-locus chart is smooth as well. □

11.4. **The relative anticanonical system.** By Proposition 7.6, after shrinking B the family is proper and Gorenstein and $\mathcal{L} = \omega_{\mathcal{P}/B}^{-1}$ is relatively ample and commutes with base change. Constancy and generation of its sections follow from the special fibre.

Proposition 11.7. *After shrinking B, the sheaf $E = \pi_* \mathcal{L}$ is locally free of rank 162 and commutes with base change. The evaluation map $\pi^* E \to \mathcal{L}$ is surjective. In particular $h^0(\mathcal{P}_b, \mathcal{L}_b) = 162$ and $\mathcal{L}_b$ is globally generated for every b near zero.*

Proof. The line bundle $\mathcal{L}_0$ is ample Cartier on the complete toric variety $\mathbb{P}_{\Delta_9}$. Toric vanishing gives $H^i(\mathcal{P}_0, \mathcal{L}_0) = 0$ for $i > 0$ [CLS11, Thm. 9.2.3], and $h^0(\mathcal{P}_0, \mathcal{L}_0) = \#(\Delta_9^\circ \cap M) = 162$. Cohomology and base change therefore makes E a vector bundle of rank 162 commuting with base change [Stacks, Tag 07VK]. Its evaluation map is surjective on the special fibre, since $\mathcal{L}_0$ is globally generated. The cokernel has closed support whose proper image in B misses zero. Removing that image proves the assertion. After a further shrinking, trivializing E gives a common 162-dimensional space of sections surjecting onto every fibre. □

The general fan provides a direct check of the section count. Writing a character as (s, a, b, c), its anticanonical inequalities are

$$a \geq -1, \qquad a - 1 \leq s \leq \min(1, 1 - a), \qquad b, c \leq 1 - s - 3a, \qquad b + c \geq -1 - s - 3a.$$

Only $a = -1, 0, 1$ occur at lattice points. At fixed a, s there are $\binom{5-s-3a}{2}$ choices of (b, c). Summing gives

$$(45 + 36 + 28 + 21) + (15 + 10 + 6) + 1 = 130 + 31 + 1 = 162.$$

11.5. **The general anticanonical member.**

Theorem 11.8 (Explicit ambient smoothing of X_9)**.** *X_9 is smoothable. Precisely, let $g \in H^0(\mathcal{P}, \mathcal{L})$ be general and $\mathcal{X} := \{g = 0\}$. Then $\mathcal{X} \to B$ is flat, $\mathcal{X}_0$ is a generic anticanonical hypersurface of $\mathbb{P}_{\Delta_9}$, and $\mathcal{X}_b$ is a smooth Calabi–Yau threefold for general $b \in B$.*

Proof. Use the trivialisation of $\pi_*\mathcal{L}$ from Proposition 11.7, and choose g from a general constant vector in its 162-dimensional fibre. A nonzero constant vector restricts to a nonzero section g_b on every fibre. The fibres $\mathcal{P}_b$ are integral normal T-varieties, so g_b is a nonzero-divisor. The fibrewise regular-section criterion therefore makes $\mathcal{X}$ a relative effective Cartier divisor: locally, multiplication by g gives an exact sequence

$$0 \longrightarrow \mathcal{L}^{-1} \xrightarrow{g} \mathcal{O}_{\mathcal{P}} \longrightarrow \mathcal{O}_{\mathcal{X}} \longrightarrow 0$$

which remains exact after restriction to every fibre. Since the first two terms are flat over B, the local criterion for flatness shows that $\mathcal{O}_{\mathcal{X}}$ is flat over B.

$\mathcal{X}_0$ is a general member of $|-K_{\mathbb{P}_{\Delta_9}}|$: by Proposition 11.7 the restriction $H^0(\mathcal{P}, \mathcal{L}) \to H^0(\mathcal{P}_0, \mathcal{L}_0)$ is surjective, so a general g restricts to a general section on the special fibre. Hence $\mathcal{X}_0$ has the singularities of Proposition 11.1.

Fix one $b_1 \neq 0$. By Proposition 11.7 the system $\mathcal{L}_{b_1}$ is globally generated, so Bertini makes its general member smooth away from $\mathrm{Sing}(\mathcal{P}_{b_1})$. Proposition 11.6 says that this singular locus is finite; global generation also makes a general section nonzero at each of its points. Thus smooth sections form a nonempty open subset of $H^0(\mathcal{P}_{b_1}, \mathcal{L}_{b_1})$. Under the trivialisation of $\pi_*\mathcal{L}$, its inverse image in the constant section space $V \cong \mathbb{C}^{162}$ is open and nonempty. The sections whose restriction to $\mathcal{P}_0$ is generic form another nonempty open subset of the same irreducible space V, so a general g has both properties. Smoothness is open in the flat family $\mathcal{X} \to B$; hence $\mathcal{X}_b$ is smooth for b in a nonempty open subset of B.

Finally relative adjunction gives

$$\omega_{\mathcal{X}/B} \cong (\omega_{\mathcal{P}/B} \otimes \mathcal{L})|_{\mathcal{X}} \cong \mathcal{O}_{\mathcal{X}}.$$

On the special fibre the sequence is

$$0 \longrightarrow \omega_{\mathcal{P}_0} \longrightarrow \mathcal{O}_{\mathcal{P}_0} \longrightarrow \mathcal{O}_{\mathcal{X}_0} \longrightarrow 0.$$

A complete toric variety has $H^i(\mathcal{P}_0, \mathcal{O}_{\mathcal{P}_0}) = 0$ for $i > 0$; Serre duality gives the corresponding vanishings for $\omega_{\mathcal{P}_0}$ below degree 4. The long exact sequence therefore gives $H^1(\mathcal{X}_0, \mathcal{O}_{\mathcal{X}_0}) = H^2(\mathcal{X}_0, \mathcal{O}_{\mathcal{X}_0}) = 0$. Upper semicontinuity for the proper flat family $\mathcal{X} \to B$ gives the same vanishings on nearby fibres. Thus every smooth good fibre has trivial canonical bundle and $h^1(\mathcal{O}) = h^2(\mathcal{O}) = 0$, so it is a Calabi–Yau threefold. □

Remark 11.9. The deformation is small in a precise sense: of the cells of the slice complex, only seven change. Three are the cells of the singular 2-faces, where Table 4 and Corollary 11.4 show that the cones over the summands are smooth. The other four are higher-dimensional cells. Their orbit strata cannot be inferred from cell dimension in a complete-locus chart; their contribution to the singular locus is covered instead by the fan and singular-locus calculation in Proposition 11.6.

11.6. **The relation supplied by the ambient family.** The two nodal exceptional curves are independent when considered alone. Nevertheless the local tangents induced by the ambient family give a relation involving both nodes and the degree-seven point. The jump formula of [Wue26b, Prop. 8.2], applied to the dilation vector above, gives three classes whose unique relation has every coefficient nonzero. The degree-seven component lies on its (-2)-root line. Thus the construction realizes the relation required by Theorem 4.7 and moves all three germs simultaneously; a relation involving only the nodes cannot do so.

Appendix A. Other ambient deformation constructions

We compare two further ambient constructions with the examples above. The non-polynomial correction term vanishes throughout the toric framework used here. The global Minkowski construction has no nontrivial input on Δ_{19}.

A.1. **Mavlyutov's non-polynomial deformations.** For a smooth crepant Batyrev hypersurface, Mavlyutov [Mav04, Mav05] identifies deformations not induced by the ambient linear system. Their contribution to the dimension is

$$\sum_{\operatorname{codim}\Gamma=2} \ell^*(\Gamma)\,\ell^*(\Gamma^*), \tag{68}$$

where ℓ^* is the number of relative-interior lattice points and Γ^* runs over the dual faces, which for a codimension-two face Γ of Δ° is an *edge* of Δ.

Remark A.1. If every edge of Δ has lattice length one, in particular if Δ is in the admissible framework, then (68) vanishes: $\ell^*(\Gamma^*) = 0$ for every edge Γ^*. This is a sufficient condition for the vanishing, not a characterization. The formula concerns the smooth crepant hypersurface and does not classify all deformations of the singular X.

A.2. **Global Minkowski decomposition.** The construction in [Mav09, §7, Thm. 7.1] takes as input a Minkowski decomposition of Δ itself.

Proposition A.2. *Δ_{19} is Minkowski-indecomposable: its only weak Minkowski summands are its own homothets.*

Proof. The weak Minkowski summands of a polytope are the solutions of the edge-cycle system (63) imposed simultaneously on all of its 2-faces, one vector equation per 2-face in the edge dilations. Δ_{19} has f-vector $(19, 64, 72, 27)$, so the system has 64 unknowns and 72 vector equations, and its solution space is one-dimensional, spanned by the all-ones vector, which is the homothety direction. A polytope whose summand space is a single ray is indecomposable. The rank computation is exact over $\mathbb{Q}$ in `global_decomp.py`, which also verifies the f-vector against Euler's relation. □

Thus the global Minkowski construction has no nontrivial input on Δ_{19}.

A.3. **Why one cannot simply refine to a simplicial ambient.** It is natural to try to avoid the non-simplicial ambient by passing to a maximal projective crepant partial (MPCP) desingularisation, which cones off every boundary lattice point of Δ. The specified Minkowski decomposition need not descend to the resulting subdivision.

Over a singular 2-face G, an MPCP subdivision refines σ_G using the interior lattice points of G; at a vertex degree the slice cell *is* G, so the cell is cut into triangles. A triangle has only homothetic Minkowski summands. For a unimodular triangle, lattice admissibility forces any such decomposition to be trivial: its primitive edges cannot split into two nonzero integral dilations. On Δ_9 a full MPCP triangulation also triangulates the two node squares, although they have no interior lattice points. The pentagon has one interior point and its star subdivision consists of five triangles. The original segment-plus-triangle datum is therefore no longer a decomposition of a cell in the refined complex.

This observation concerns the specified Minkowski datum on the slice complex. The partial-resolution method of Section 4 uses deformations of a different space and subsequent blow-down; it does not require that datum to survive on an MPCP refinement.

Appendix B. Exact computations and reproducibility

The following programs reproduce the finite equations, lattice data and relation matrices used in the proofs. Their inputs are the stated lattice and local-equation data, together with the Kreuzer–Skarke database for the enumeration. The table records which mathematical assertions each program checks.

statement	script	checks
Rem. 8.3	`one_facet.py`	7
Prop. 8.9, Def. 8.11, Rem. 8.7	`locking.py`	13
Thm. 9.2, the line $\mathcal{A}_P$, its two ranges, and cell rigidity	`slice_rigidity.py`	49
independent reimplementation	`slice_rigidity.sage`	24
face compatibility in Corollary 8.13	`def41_check.sage`	16
Prop. A.2	`global_decomp.py`	12
Section 10, the enumeration	`dp7_facet_sweep.py`	n/a
induced pentagon decompositions	`candidates.sage`	n/a
Appendix A.3	`refine.py`	5
Props. 11.1, 11.2	`v09_candidate.py`, `sing_locus.py`	6
Prop. 11.3, Cor. 11.4	`slice_admissible.sage`	36
Cor. 11.4, completeness of the general fibre	`general_fibre.sage`	6
Prop. 11.7; independent semiampleness diagnostic	`hzero.sage`	n/a
Prop. 11.5, 11.6	`toric_fibre.sage`	n/a
Prop. 11.6	`charts.sage`	3
Rem. 11.9	`final_check.sage`	3
Local branch tangents correspond to $\mathcal{R}_p$	`branch_locus.py`	17
comparison with cyclic period coordinates	`smoothing_locus.py`	13
Thm. 10.3, and [Wue26b, Prop. 9.2]	`classification.py`†	10
the relation classes of X_9 and X_{19}	`relation_class.py`†	36
from Theorem 11.8's family	`theoremB_class.py`	10

The degree restriction is proved in Lemma 8.5. Its finite input is the affine degree line and the two pairing formulas, reproduced by `slice_rigidity.py`. Smoothness of cones over summands alone does not impose admissibility (D2) and is not used as a necessary degree test.

The two marked † certify statements of [Wue26b] as well as statements here, and live in that paper's certificate directory of the shared repository, from which the scripts of this paper already import their exact toric modules. Two further scripts there, `kernel_equality.py` and `lone_germ.py`, certify [Wue26b, Prop. 8.3] and [Wue26b, §9.2], used in the companion computations.

`slice_rigidity.py` and `slice_rigidity.sage` implement the same statement in different systems, pure Python with integer edge equalities and Sage with convex hulls over $\mathbb{Q}$ and then over $\mathbb{Q}(s)$, and are compared against each other and against an independent facet computation.

`hzero.sage` implements the invariant-divisor formula of [PS11] and is validated on that source's worked example, reproducing the printed value $h^0(-K_{\mathbb{P}(\Omega_{\mathbb{P}^2})}) = 27$, and then cross-checked on the special fibre against the toric lattice-point count.

On the special fibre, `charts.sage` recovers the positive-dimensional singular loci in the four charts containing singular 2-faces. On the general fibre, it verifies the smoothness of every cone corresponding to a positive-dimensional orbit.

The jump formula of [Wue26b, Prop. 8.2], applied to the Minkowski data of Theorem 11.8, gives the three local tangent classes and their relation with every coefficient nonzero. The file `theoremB_class.py` reconstructs this relation and checks that the degree-seven component lies on the (-2)-line.

The summand cone of facet 12 of Δ_{19} has dimension two at the lattice cross-section and one at the other cross-sections in Remark 8.7. The file `locking.py` verifies these dimensions and the necessity of consecutive edges in Proposition 8.9. Its edge partitions agree with those computed by `dp7_facet_sweep.py` on all 27 facets of Δ_{19}.

B.1. **Certificates for the smoothing criterion and deformation spaces.** The following additional computations use exact arithmetic. Their geometric implications are proved in the corresponding sections; the scripts verify the finite local equations, face data, and relation matrices used there.

- `a2_simultaneous_partial_resolution.py` checks the incidence charts, nodal Hessians, and inverse maps of Proposition 3.8 (33 assertions).
- `dp6_nodal_partial_resolutions.py` checks the two degree-six subdivisions, normal bundles, root classes, and local parameter maps of Proposition 3.7 (43 assertions).
- `a2_branch_selection_counterexample.py` reconstructs the 25-vertex polytope, its faces, divisor rows, four relation kernels, and three obstruction identities in Section 5 (272 assertions).
- `branch_smoothness_inputs.py` verifies the Hodge data, degree-six vanishing inputs, nonzero local projections, and the relation-space comparisons under nodal replacement (52 assertions).
- `counterexample_deformation_germ.py` checks the full relation matrix, cone image, component tangent ideals, their intersection, and the resulting dimensions (57 assertions).
- `a2_counterexample_rational_replay.py` independently repeats the counterexample's rational linear algebra and Hodge input with Python's standard-library rational arithmetic (50 assertions).

- `d19_quadric_partial_resolution.py` checks the X_{19} partial fan, local quadric-cone cohomology, Hodge input, and the matrices printed in Corollary 9.3.

The Sage-dependent files run with `sage -python`; the rational replay runs with `python3`. They resolve their inputs relative to the repository. The finite computations do not replace the analytic lifting, blow-down, or necessity proofs.

The additional files in `figures/code/` run with Python's standard library. The file `compute01_degree5_local.py` verifies the five conic-pencil classes and their A_4 coordinates, the Cremona permutation, the Grassmannian Pieri integrals and Chern classes, and the two exterior-product ranks used in Section 3.1. The file `compute01_degree8_local.py` verifies the Cayley lattice, projection monomials, and link lattice in the quadric-cone proof.

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